PRODUCT MANAGEMENT & ANALYTICS GUIDE - 2023

BizTech | Product management club of IIM Lucknow



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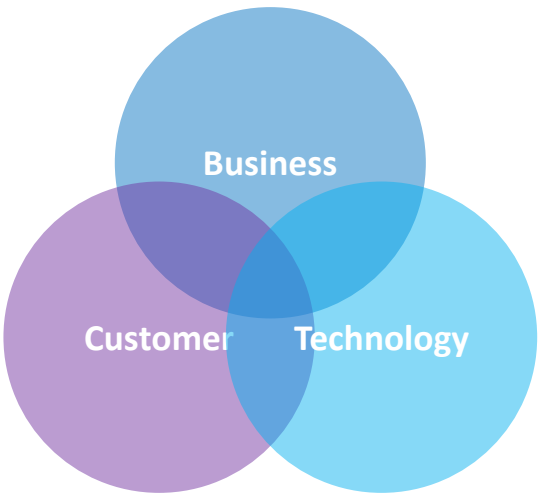
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What is Product Management?





What is product management?

When we think of a company, and we think of its products, we can think of certain functions straight off the bat, like, Engineers are the ones who build the product, Marketing makes sure that customers know about the product, and Sales makes sure that the product actually gets sold. So, where exactly does a Product Manager fit in? And ultimately, what is Product Management?

What do Product Managers do?

In very simple words, Product Manager represents the customer. Elaborating on this, let's first understand why does the customer buy a product. It's certainly not to give money to the company. Customers buy a product in order to fulfil a need gap that they have, i.e., a need which is not getting fulfilled. The product that the company thus makes, fulfils this gap and if done right, it helps the customer be awesome, and as the Product Manager is the one representing the customer, they are ultimately helping the customer be awesome.

In formal words, Product management is an organizational function that guides every step of a product's lifecycle — from development to positioning and pricing — by focusing on the product and its customers first and foremost. To build the best version of the product, Product Managers represent the voice of the

customer within the company, and makes sure that the product gets built the way it is being demanded in the market. Some of the responsibilities that a Product Manager has could be the following:

1. Conducting Research (Primary and Secondary Research to understand the exact need and how the product should be designed)
2. Setting Product Vision: After getting an understanding of the need gap, the Product Manager then translates this into a vision for the product that they are building.
3. Designing the features of the product
4. Building out a product prototype according to the design
5. Testing the prototype
6. Collect user feedback and usability data on the prototype in multiple iterations
7. Developing the final product, incorporating the feedback
8. Maintaining the product

These are just some of the responsibilities, but they could vary because, for almost every company, what the role of Product Management means, is different. Sometimes, Product Managers are referred to as the "CEOs of the product".

The task involves lots of **planning** and **decision-making**, and the decision includes:

a. What are the features that my product must have?
b. How do I segregate and prioritize my features?
c. What are the metrics I should track to evaluate my product's performance?

A Product Manager has to answer these questions with a proper understanding, and not all of them may have a concrete answer initially. It would also require **highly structured thinking** and **good problem-solving skills** to come up with a good plan of action. To answer these questions well, the PM has to talk to customers, which would require them to have good **communication skills**. It would also require good **research skills** to understand what and how to ask. Then, they would also require **Marketing skills**, as the customer might not know what they need and the PM has to make them aware of the need. They would also need to have a good **understanding of Technology** to translate the need of customer into the product, and also understand the limitations that there might be.

These are not all though, as the PM would

also have to produce the product for a company, and every company will have some constraint that the PM needs to take care of. There will either be time constraint, or budget constraint, or tech constraint, or resource constraint.

These gives us a pretty good idea about the Product Management role, and what a Product Manager does. Product Management role is famously referred to as an intersection of Business, User Experience and Technology.

Types of Product Manager

There are a few types when it comes to product management roles, and the expectations and job description varies vividly based on the type. Here, we will try to classify the Product Managers, which could be broadly mentioned as:

- **Tech Product Manager:** They try to solve the engineering problems, and are usually pretty good at it. They define the need for the Tech product, defining the WHY and WHAT of the product, but leaving out the HOW part to the engineering team. They take care of all the activities throughout the tech product's lifecycle. One of the pitfalls of this role is that they can end up being the Engineering Manager for the team, rather than the Product Manager.
- **Design Product Manager:** They have an understanding about what a Good Product looks like, and how to define the strategy and design for the product. They understand the user needs, and they figure out how to improve the features to meet the needs properly, as well as sense the future needs and how to cater for it. However, they do not have experience with code and haven't been anywhere near the engineering team or how they work.
- **Business Product Manager:** They work on the bigger picture for a Product, how the future looks like in terms of Finance, Marketing and Operations. Usually, they are well versed in the Business terminologies and workings, and are good at communicating as well as understanding people around them. However, they are not well equipped with the tech or design, which could lead to conflicts and also, it takes time to build credibility with the tech/design teams.
- **Growth Product Manager:** They

work on scaling a business, identifying any problems in the product to bridge the gap between supply and demand. This role is a rather undefined role, and varies a great deal from one organization to another. These PMs are usually strong with data, as well as with communication. They focus a lot on Metrics, and the right ones can lead you in the right direction of solving a growth problem, but, being too focused on metrics sometimes leads to losing sight of the bigger picture and a lack of quick experimentation and right prioritization framework may lead to low impact work.

Types of products a PM might encounter

- **Mass Consumer Apps:** These are apps that find use for the whole mass and are built keeping this in mind. B2C product managers are often the most visible face of a Tech Company, and they build product appealing to the masses which is also functional.
- **B2B or SaaS products:** These products are usually built keeping in mind the requirements or needs of other businesses. The products are so built that they cater to needs of multiple businesses, and these products usually bring in larger profits.
- **Additive Features:** For products with multiple significant features, there are multiple Product Managers for the different features. Sometimes, certain features become the USP of the product and for larger products, there are multiple PMs manning these features or in charge of new features that could be added.
- **Early-stage products:** While launching a new product, usually a Minimum Viable Product (MVP) is first launched, which is the early-stage product. Based on the feedback and product performance analysis, this version is worked upon additively to make the final product. The emphasis here is on quick development and proper prioritization of the most essential features.
- **Marketplace platforms:** These products are built with the goal of reducing transaction costs as an important criterion. Awareness of supply-demand, and ability to understand client psychology are helpful attributes, more important than tech awareness.
- **IoT, Machine Learning and Physical Products:** These products are built to

- implement the technologies, which are IoT and Machine Learning. The goal here is not just the correct implementation of these technologies, but also ensuring that other parts of the product work in conjunction and provide information to improve these technologies.
- **Transformation of Existing Products:** Here, usually the Traditional products are taken and transformed. These would include the digital transformation of various businesses or business models. The push for digital transformation has been very high since pandemic. Here, an understanding of risk that come with decisions and the planning for the product are important, as these are usually established businesses. Some such examples would be Newspapers, banking systems, even Netflix.

Product Manager's Career Path

The positions could have different names at different organizations, but the roles and responsibilities remain somewhat similar for the stages. For a better understanding, lets take a closer look at each of the roles and what they entail.



Associate Product Manager: Here, one has a subset of the responsibilities of a PM. So, one might not be setting the product strategy or product roadmap, but they could be doing the same for a feature or working on Prioritization and shipping of a particular feature.

Product Manager: Here, one owns the product and is responsible for shipping the product.

Senior Product Manager: Here, one is responsible to look at a problem and create a strategy to get past the problem.

Director of Product: Other than doing thorough market research for their product and setting the future roadmap, the other responsibilities include guiding a person and helping them level up through the career path.

VP of Product: Here, one looks to budget for the products, ensures product strategy aligns with business objectives and protects the team from infighting and internal politics.

Chief Product Officer: Here, one looks at the Top-line for the current product portfolio and looks for new product lines, create product strategies and communicate it, spearhead product development teams, lead, manage and mentor PMs and other leadership duties.

How to build your profile and prepare for PM role?

Case Study/Design Competitions

There are multiple case study competitions which are conducted around the year. Participating and winning in these competitions go a long way in creating meaningful domain related impact to a person's profile. As a bonus point, a lot of times, people are able to secure PPIs as a prize for winning or being finalists in certain competitions.

Live Projects and Internships

Working on internships and live projects of Product Management domain also adds a lot of value to one's profile and provides good experience about the work. They also provide good, relevant pointers for resume.

PPTs of visiting companies

Several companies that come to campus for placement, organize PPTs. While these PPTs may not always be very useful, more often than not, they talk about some new technology or project the company is working on currently, and it is good to be aware of the recent developments within the company. They also make for better conversation and preparation topics, and sometimes, direct questions are also asked in the interview from these PPT topics.

PM cases to land interviews

Most PM companies provide a case study as the entry-level criteria for the interview round, and cracking these case studies provide you with the opportunity to appear for the interviews. A good way to approach these case studies is to provide structured answers, following the frameworks, and then adding some Wireframes/mock-ups and properly prioritizing the solution using some of the frameworks add a lot of value to the Case solutions that you provide.

Forming groups to prepare for interviews and GD

Many of us prepared for interviews by

forming groups among ourselves. Groups of 3 or 4 are a good practice to have, with these folks also preparing with Product Management as their primary domain focus. It is a good choice to make groups with people who are at similar levels of preparation, so that the discussion can be of richer value. There are some advantages to preparing in groups:

- Discussing and preparing with a group gives you a lot of diverse views on one topic, and you get better ideas along lines which you may not have been thinking along. Some of the solutions may be more efficient and/or optimized than the one you had thought of.
- You are exposed to past experiences of your group members, and the different situations they faced in their past employment and how they acted in those situations. All these experiences are useful, as you might be given a situation in interview which you may not have encountered, but thanks to the experiences you heard from your group, you can mould and answer the question in a justifiable manner.
- Group members can aid each other's preparation, by dividing some common tasks within the group. For example, someone can prepare some tech repository to be regularly discussed, while another can make a repo of company details and products, and another person can collect the types of interview experiences for different companies visiting campus.
- The journey of preparation for interviews is a gruelling one, and leaves you exhausted and crest-fallen a lot of times, often due to external factors. At such times, your group helps you pick yourself up and keep the preparation levels up.

When preparing in a group, it is a good practice to meet once a day, and practice various PM topics, cases, discuss tech news. When trying to simulate case-solving scenarios, one person can be Interviewer, another can be interviewee and the others can act as shadows/spectators and they can discuss solutions at the end. The group members can rotate and interchange between these roles. A good practice is to solve ~2 cases per day, or 1 case and 1 guesstimate/RCA question.

Another good practice is to research, critique and discuss popular apps and also

your favourite apps. This helps develop a product acumen, and comes in handy later. It is also a very good practice to write down the frameworks on a whiteboard, or some similar place and keep practicing them regularly, as these frameworks provide a structure to your PM solutions. Over time, you will get a hang of these frameworks and be better prepared to answer questions in interviews.

Last but not the least, you should know your CV in detail. You should have a good story/explanation behind each sentence that you have in your CV, and you should be able to justify the numbers that you have mentioned. You should also prepare and be ready for some HR questions, as most companies have such a round to see culture fit. You should have good stories from your past, which you can use for multiple situations with some tweaks here and there.

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PM Concepts



Artificial Intelligence

Artificial intelligence (AI) and its subset, machine learning is indisputably a rising star on the world's technology stage. AI initiatives are taking root everywhere & Product managers (PMs) of every stripe need to understand the powerful influence of AI to both stay competitive in their product role and also ensure that their companies maintain a competitive advantage.

"*A recent survey of global business leaders indicated that 70% have started AI initiatives. With the proliferation of AI into business, it is easy to see applications to B2C and B2B products and services: Google Search / Photos / Translate, Alexa, Amazon Recommendations, Stitch Fix. Nest, Tesla Autopilot. Machine Learning (and data) is the common thread among all these. Of course, there are many more products powered by ML behind the scenes*."

-Babar Bhatti, co-founder of Dallas AI

AI product management focuses on using artificial intelligence, deep learning, or machine learning to enhance, improve, create, and shape products.

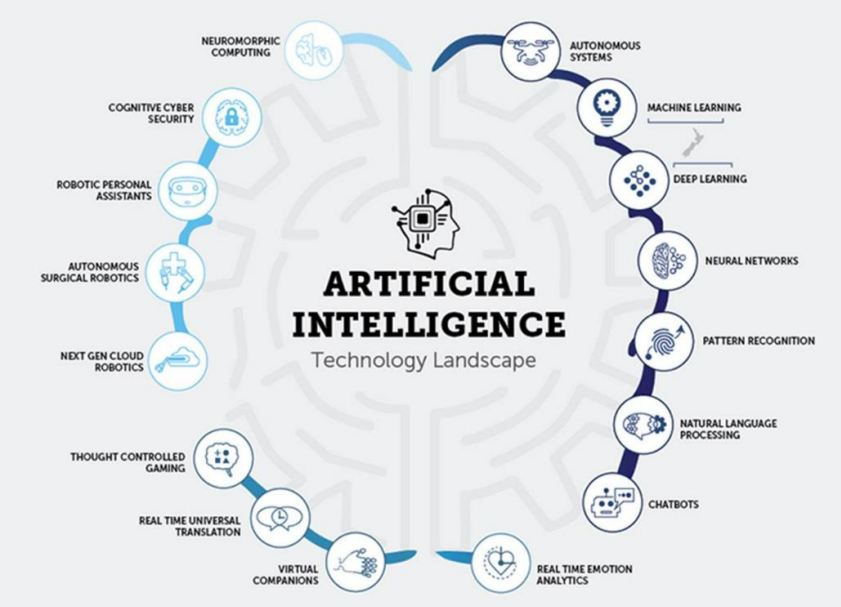
Let us begin with understanding what is AI.

DEFINITION:

It Is the ability of a digital machine to perform tasks that is commonly associated with intelligent beings. The simplest human behaviour is ascribed to intelligence.

Some examples of Artificial intelligence is computers that play chess and self-driving cars. Each

of these machines must weigh the consequences of any action they take, as each action will impact the end result.



Research in AI has focused chiefly on the following components of intelligence: learning, reasoning, problem solving, perception, and using language.

Learning:

There are multiple learning forms that can be applied to Artificial Intelligence. Simplest of all is trial and error. One simple example of this kind is solving check-mate in chess. The machine will try multiple moves at random for mate, and will store the information to implement next time in same situation. This simple memorizing of individual items and procedures is known as rote learning and is relatively easy to implement on a computer.

Reasoning:

The motive is to draw inferences appropriate to the situation, which can be either deductive or inductive. An example of deductive reasoning can be “Ria will buy either Apple or a Kiwi”. Ria didn’t buy Apple therefore she bought a Kiwi. An example of Inductive reasoning can be “These kinds of accidents was caused by instrument failure in past; therefore this accident was

caused by instrument failure". Inductive reasoning is common in science, where data are collected and tentative models are developed to describe and predict future behaviour whereas Deductive reasoning is more common in mathematics and logic, where elaborate structures of irrefutable theorems are built up from a small set of basic axioms and rules.

Problem Solving:

Problem Solving in AI is either special purpose or general purpose.

A special-purpose method is tailor-made for a particular type of problem and goes into depth of very specific features of the situation at hand whereas a general purpose technique means end analysis. A simple program will select the actions from PICKUP, PUTDOWN, MOVEFORWARD, MOVEBACK, MOVELEFT, and MOVERIGHT to reach the goal.

Language:

Language in AI is not confined to spoken words but it is a system of symbols having certain conventional meanings. For instance, the clouds symbol means rain. Although none of these programs actually understands language, they may, in principle, reach the point where their command of a language is indistinguishable from that of a normal human.

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Machine Learning

Machine Learning for Product Managers

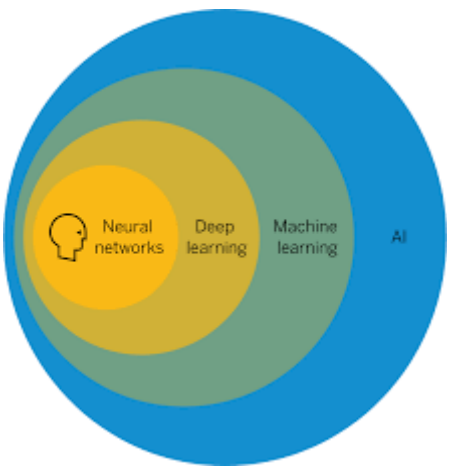
It seems that, regardless of whether it makes good use of the technology or not, practically every new software product aims to use ML to satisfy consumers' needs more effectively. There are many reasons to become familiar with the technology and its fundamentals so that you can assess how it will affect your own products

What is Machine Learning

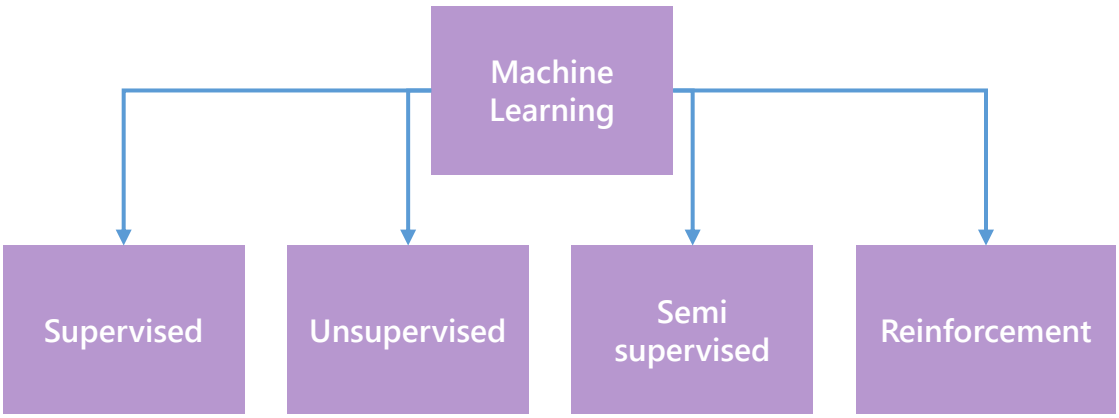
"*A computer program is said to learn from experience (E) to some class of tasks (T) and performance measure (P), if its performance at tasks in T, as measured by P, improves with experience E.*"

- Tom Mitchell

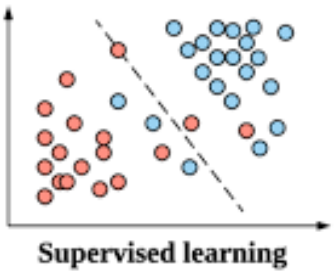
Machine learning is a subset of Artificial intelligence that gives machines(computers) the ability to learn and take decision without being explicitly programmed. It is basically a process of training a piece of software to make useful predictions from a given data. This piece of software is known as algorithm or a model.



There are four basic approaches: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning. The type of algorithm data scientists chooses to use depends on what type of data they want to predict.



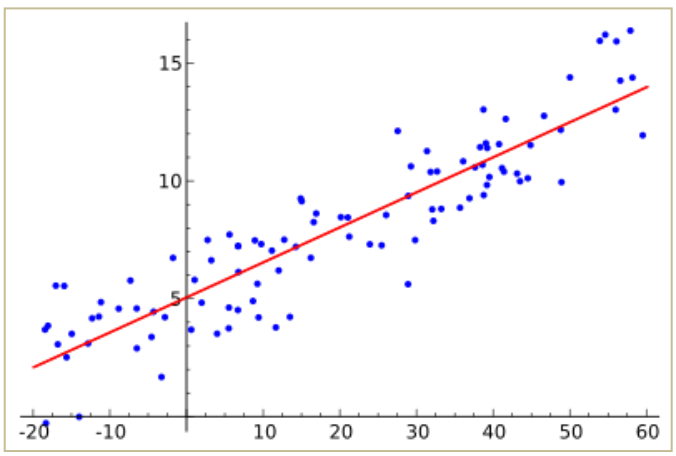
Supervised Learning

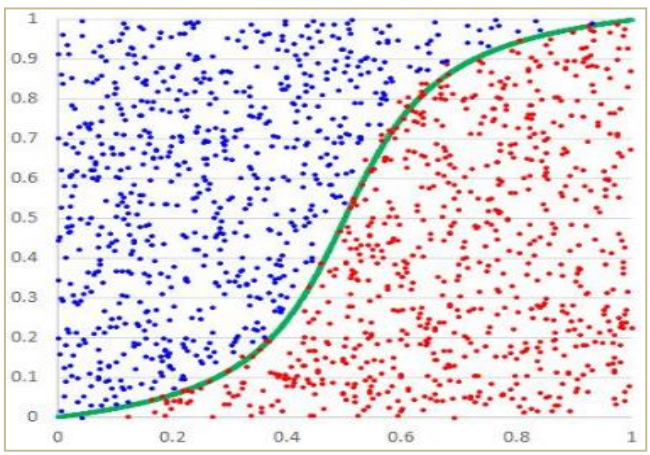


Supervised learning is the machine learning task that learn by example. In the training stage, the algorithm is fed with "training data" which consists of inputs paired with the correct outputs. During training, the algorithm attempts to figure out patterns in the data that correlate with the desired output. The efficacy of the algorithm can be measured by making predictions on data of which the output is not known and subsequently comparing the prediction with actual values.

Few of the commonly used supervised learning models are: Linear Regression, Classification, Logistic Regression, K-nearest neighbour.

Following is a brief explanation of some of the prevalent supervised algorithms



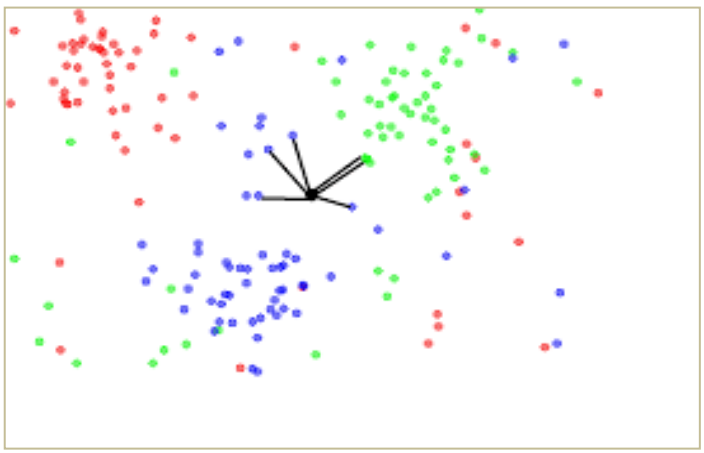


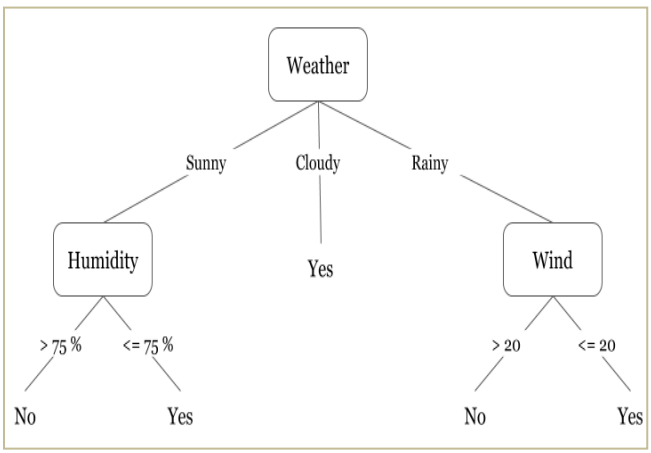
Linear Regression

- Used to find **linear relationship** between independent variables (attributes) and the output (often referred to as dependent variable and is **continuous**). The idea is to obtain a line that best fits the data (has low prediction errors)
- Application: Predicting price of house in an area

Logistic Regression

- Used when the dependent variable is **categorical** in nature. Finds extensive use when the number of levels of dependent variable is 2 (e.g. "spam" or "no spam", "rain" or "no rain"). The output is provided in the form of probability.
- Application: Predicting if a tumor is benign or not





K-Nearest neighbours

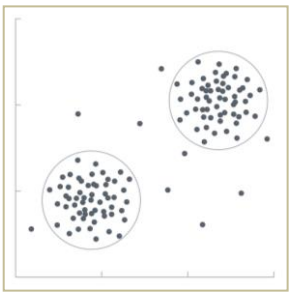
- For every new observation, k-nearest observations from training set (Euclidean distances) are chosen and the algorithm predicts based on the output values of the neighbours
- Application: Search applications in which users are trying to find similar items

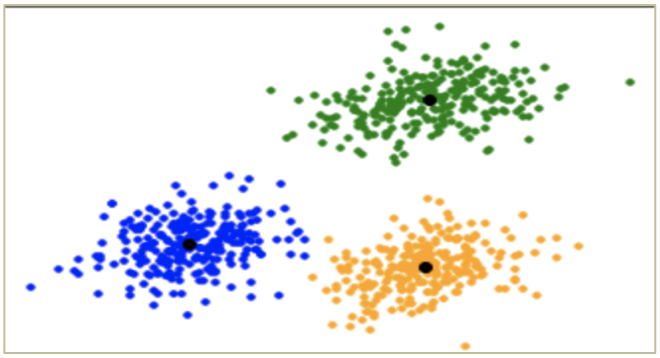
Decision tree

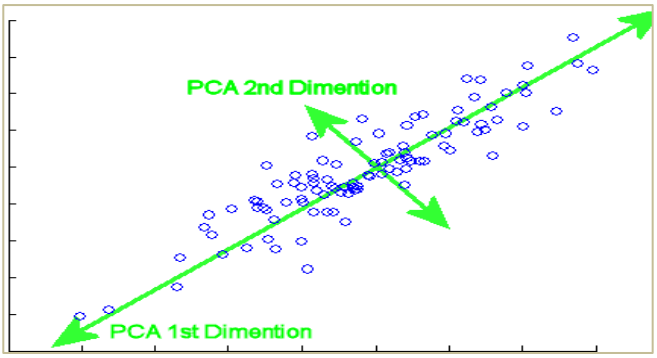
- Represents a tree like structure where each branch represents a decision point and the end nodes (leaf nodes) represent the label.
- Popularly used in classification problems
- Application: Lenders use this to predict if the customer will default on loan

Unsupervised Learning

Unsupervised algorithms belong to a class of algorithms that are used to derive inferences from datasets which consists of input data which is not labeled. The most commonly used unsupervised learning algorithm used is clustering algorithm which is implemented in exploratory analysis to identify hidden patterns.







K-means clustering

- The process starts with algorithm selecting random centroids, and then iteratively calculations to optimize position of centroids
- Application: Google news clustering news on the basis of topics and keywords in news

Principal Component Analysis (PCA)

- Main idea of PCA is to reduce dimensionality of a dataset containing many variables that are correlated with each other
- Application: Used to filter noisy datasets, image compression

Semi-supervised Learning

The algorithm represents an intermediate ground between Supervised and Unsupervised learning algorithms. It uses a combination of labelled and unlabeled datasets during the training period. Semi-supervised learning uses pseudo labeling to train the model with less labeled training data than supervised learning.

The whole working of semi-supervised learning is explained in the below points

- Firstly, it trains the model with less amount of training data similar to the supervised

- learning models. The training continues until the model gives accurate results.
- The algorithms use the unlabeled dataset with pseudo labels in the next step, and now the result may not be accurate.
- Now, the labels from labeled training data and pseudo labels data are linked together.
- The input data in labeled training data and unlabeled training data are also linked.
- In the end, again train the model with the new combined input as did in the first step. It will reduce errors and improve the accuracy of the model.

Reinforcement Learning

It trains a model to return an optimum solution for a problem by taking a sequence of decisions by itself. Reinforcement Learning uses a feedback mechanism from its own actions and experiences that enables an agent to learn in an interactive environment by trial and error.

Important Terms in Reinforcement Learning

- Agent: Agent is the model that is being trained via reinforcement learning
- Environment: The training situation that the model must optimize to is called its environment
- Action: All possible steps that can be taken by the model
- State: The current position/ condition returned by the model
- Reward: To help the model move in the right direction, it is rewarded/points are given to it to appraise some action
- Policy: Policy determines how an agent will behave at any time. It acts as a mapping between Action and present State

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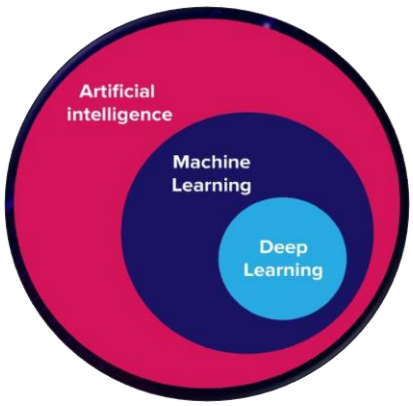
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Artificial Intelligence vs Machine Learning vs Deep Learning



Succinctly put, AI is the ability of computer program to function like a human

AI includes algorithms that imitate the intelligence exhibited by humans, and are able to solve problems in ways that are considered "smart".

E.g. Amazon's accurate AI that provides one of the most accurate of predictions

Machine Learning

Machine Learning is considered as a subset of AI. This comprises of algorithms that parse data, "learn" from it and have the ability to apply what the algorithms have learned to make informed decisions. The machines are able to learn from the data with the help of statistical techniques

E.g. Email filtering, fraud detection and dynamic pricing

Deep Learning

Further subset of Machine Learning is Deep Learning. The objective of these techniques is to achieve a goal or an artificial intelligence power that teaches computers to task and ability to understand anything. The algorithms permit software to train itself to perform tasks, like speech and image recognition.

E.g. Application of Deep Learning to differentiate between dialects, autonomous vehicles using deep learning to identify pedestrians

Introduction to Analytics concepts

What is Analytics?

Analytics is the science of analyzing raw data to derive actionable insights and conclusions. The techniques and processes may involve complex algorithms and machine learning concepts.

Data Analytics plays a pivotal role in helping businesses identify patterns, customer trends and accordingly take decisions.

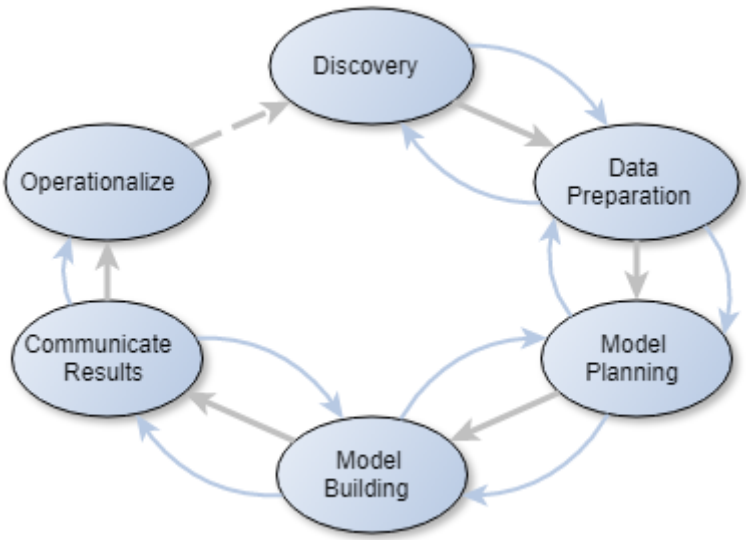
Types of Analytics






Descriptive Analytics	Diagnostic Analytics	Predictive Analytics	Prespective Analytics
Answers the question of "What has happened?". The objective is to analyze historical trends and figure out relevant patterns and to gain insights	Diagnostic analytics helps answer the question of "Why it happened?". It involves techniques such as correlation, drill down, data mining.	Predictive Analytics uses statistical modeling, machine learning to predict the probability of a future event. It answers the question of "What could happen?"	It answers the question of "What should happen?" Prescriptive analytics is used to determine the most appropriate step to be taken based on data.

Analytics Lifecycle



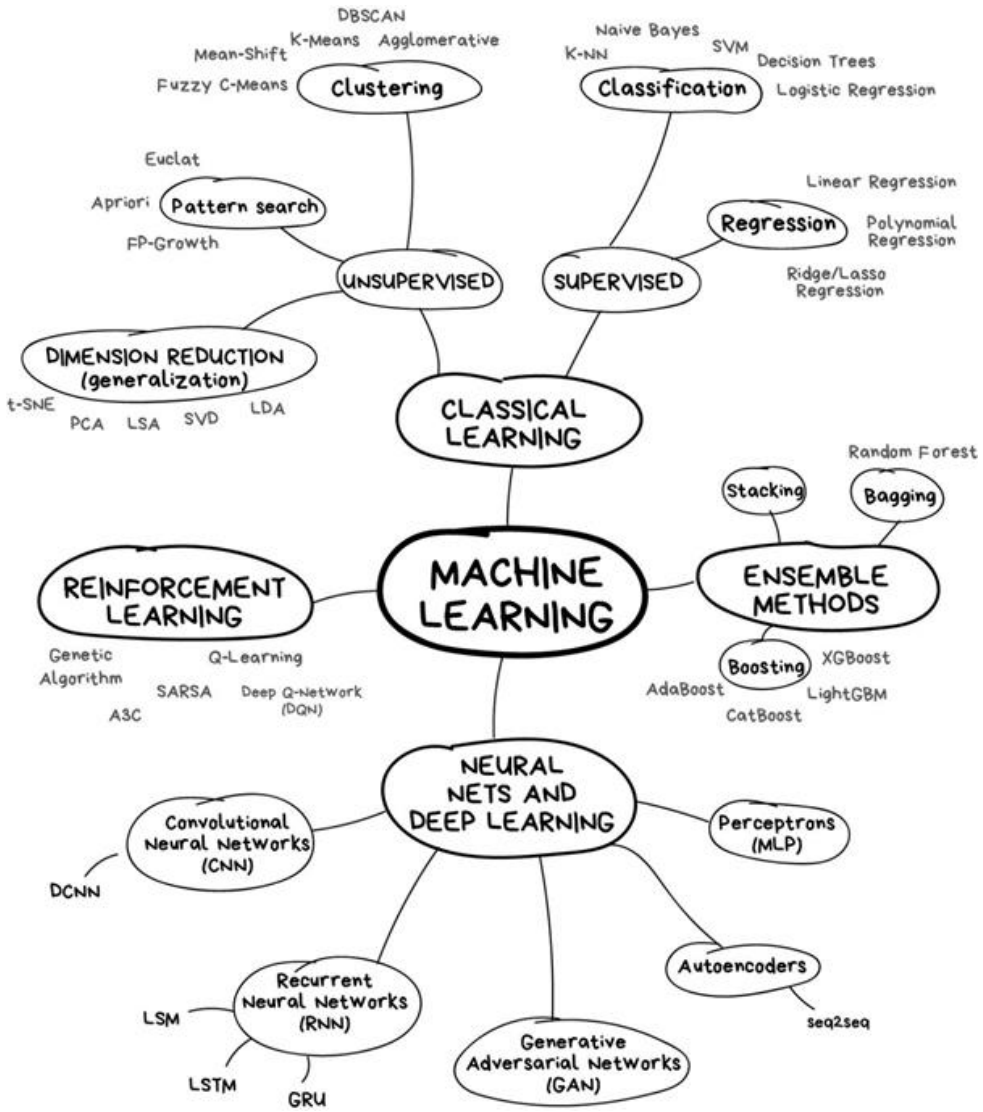
1. Discovery:
    1. Objective is to understand the business domain and know the available resources (people, time, technology, data)
    2. Developing initial hypothesis to test with the data
2. Data preparation:
    1. Understanding the data in detail
    2. Performing ETL: Extract → Transform → Load
    3. Data visualization to understand nuances, trends and overview of the data
3. Model Planning:
    1. Understand relationships between different variables and select the most appropriate and relevant variables
    2. Determining causal relationships between input and output, if it exists
    3. Shortlisting possible techniques that can be employed: Classification, Clustering, etc.
4. Model Selection:
    1. Develop and fit models on training data and evaluate on test data

    2. Determine if the model is robust enough to be scaled to a larger dataset

1. Communicate results:
    1. Comparing the outcomes with the defined criteria for success and failure
    2. Checking if the results of the model are statistically significant
2. Operationalize:
    1. Deploying the model to production environment
    2. Monitoring of ongoing model deployment

"How vast can Machine Learning really be? I mean isn't it just a few algori....

Oh okay"



Augmented reality & Virtual Reality

Augmented reality is adding virtual elements into real world for an enhanced experience. These digital elements could be digital visual elements, sound, or other sensory stimuli and delivered via technology. Augmented Reality is a growing trend majorly in mobile computing companies. Augmented reality continues to develop and become more pervasive among a wide range of applications

Examples of Augmented Reality

Store catalogue apps, that allows consumers to visualize what different products would look like in different settings. For example, while shopping for furniture for a better understanding in terms of looks, shoppers point the camera to the appropriate room and the product will appear in the foreground.

How AR works:

Augmented reality starts with a camera-equipped device such as a smartphone, a tablet, or smart glasses with AR software. By using a computer to decrypt, or by identifying & recognizing live video stream from the camera the computer renders the relevant 3D graphics and superimpose it on the live camera feed to add the virtual element into the picture. As the user moves, the size and orientation of the AR display automatically adjusts. The 3D model is usually created using CAD(computer-aided design). Based on the type, AR can use different sensors, accelerometers, cameras, gyroscopes, and light sensors to collect data on the user's surroundings. The parameters to measure can be the distance to the objects, speed of the motion, direction, and angle, and overall orientation in space. The data is then processed to CAD to show animation.

Virtual Reality

Virtual reality is creating a simulation. Instead of viewing a screen in front of them, users are immersed in and interact with 3D world. Major players in Virtual Reality are HTC Vive, Oculus Rift and PlayStation VR (PSVR).

VR Categories:

The three types of VR are non-immersive, semi-immersive, full immersive.

Non-immersive Virtual reality: Video games are a prime example of non-immersive VR where the user simultaneously remains aware of the virtual environment and controlled by their physical environment.

Semi-Immersive Virtual Reality: The VR experience in this category is partially based in virtual environment. This type can be widely used in training purposes such as flight simulators for pilot trainees.

Fully Immersive Virtual Reality: Currently there are no completely immersive VR technologies but with the advancements it might be right around the corner. This category generates most realistic simulation experience, from sight to sound to sometimes even olfactory sensations. In racing games an immersive virtual reality can give the user the sensation of speed and driving skills.

Developed for gaming and other entertainment purposes, VR use in other sectors is increasing.

HOW VR works?

Virtual Reality uses hardware and software to create that immersive experience. Hardware supports sensory stimulation and simulation such as sounds, touch, smell or heat intensity, while software creates the rendered virtual environment.

Hardware used by VR

Virtual reality hardware includes sensory accessories such as headsets, controllers, 3D cameras, hand trackers, and treadmills.

Software used by VR:

Various software is used by developers to build VR. There are VR software development kits, visualization software, content management, game engines, social platforms, and training simulators

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Agile

Agile is the ability to create and respond to change

In software development, Agile is an approach that revolves around incremental and iterative steps to completing projects. Agile methodologies aim to deliver the right product, with incremental and frequent delivery of small chunks of functionality, through small cross-functional self-organizing teams, enabling frequent customer feedback and course correction as needed.

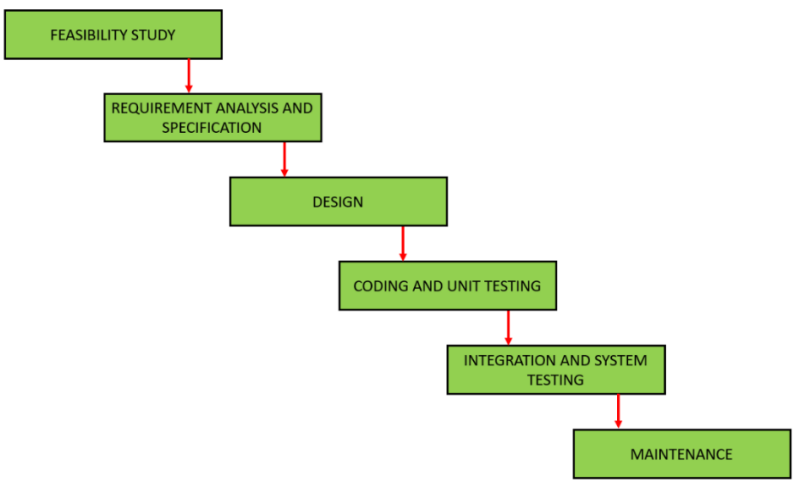
In doing so, Agile aims to right the challenges faced by the traditional “waterfall” approaches of delivering large products.

We will first look into the waterfall model:

Waterfall Model:

The classical waterfall model It is very simple but idealistic. Earlier this model was very popular but nowadays it is not used. But it is very important because all the other software development life cycle models are based on the classical waterfall model.

The classical waterfall model divides the life cycle into a set of phases. This model considers that one phase can be started after the completion of the previous phase. That is the output of one phase will be the input to the next phase. Thus, the development process can be considered as a sequential flow in the waterfall. Here the phases do not overlap with each other



By doing this Agile breaks up the traditionally long delivery cycle (typical of the legacy "waterfall methods") into shorter periods, called sprints or iterations.

Reasons for failure of waterfall model

The traditional waterfall model suffers from various shortcomings.

No way back:

In the waterfall model after being done with any phase there's no way to go back.

Sequential:

The waterfall model recommends that a new phase can start only after the completion of the previous phase. But in real projects, it is neither possible nor is it very efficient.

Lack of Feedback.

The waterfall model has no feedback path. In the traditional waterfall model evolution of software from one phase to another phase is like a waterfall. The waterfall model assumes that no error is ever committed by developers during any phase. Hence, it does not incorporate any mechanism for error correction.

Inflexible:

In a waterfall, it is almost impossible to accommodate change The waterfall model assumes that all the customer requirements can be completely and correctly defined at the beginning of the project, but actually, customers' requirements keep on changing with time. After the requirements specification phase is completed difficult to accommodate any change requests.

To overcome these drawbacks companies practices Agile software development

Agile Model:

As mentioned in the beginning the Agile concepts revolves around iterative development that is the requirements and solutions evolve through collaboration across cross-functional teams. The ultimate objective agile practice tries to achieve is greater aptitude to respond to change at any stage.

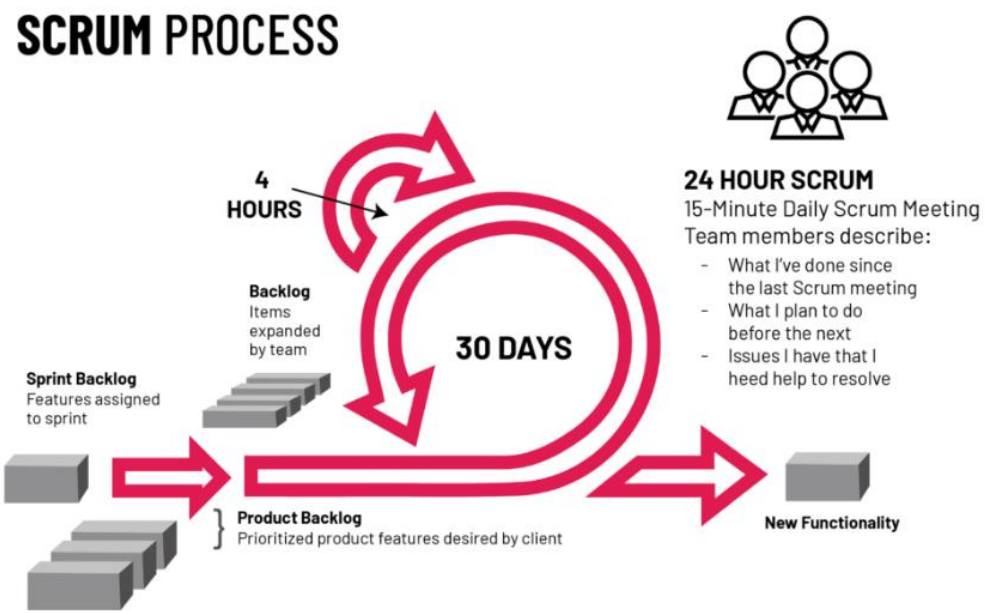


It helps the teams to deliver value faster, with greater quality and predictability. Agile project management is not a singular framework — rather, it can be used as an umbrella term to include many different frameworks. Agile project management can refer to terms including Scrum, Kanban, Extreme Programming (XP), and Adaptive Project Framework (APF).

One of the widely used Agile methodologies is Scrum.

SCRUM

Scrum is a way to get work done as a team in small pieces at a time, with continuous experimentation and feedback loops along the way to learn and improve. The Scrum Team consists of a Product Owner, a Scrum Master, and Developers, each of which has specific accountabilities. The Three pillars of scrum are transparency, inspection and adaptation which supports the concept of working iteratively.



Scrum Events:

Sprint - short cycles of one month or less, during which the work is done; the Sprint contains all the other Scrum events; a new Sprint starts immediately after the conclusion of the previous

Sprint.

Sprint Planning – an event dedicated to planning out the work that will take place during the Sprint

Daily Scrum – an event held every day where the Developers inspect the progress toward the Sprint Goal, uncover anything that may be getting in their way and adapt accordingly

Sprint Review – an event held at the end of the Sprint where the Scrum Team and key stakeholders review what was accomplished in the Sprint and what has changed in their environment; next, attendees collaborate on what to do next

Sprint Retrospective - the Scrum Team gets together during this event to talk about how the last Sprint went and identify the most helpful changes to improve their effectiveness

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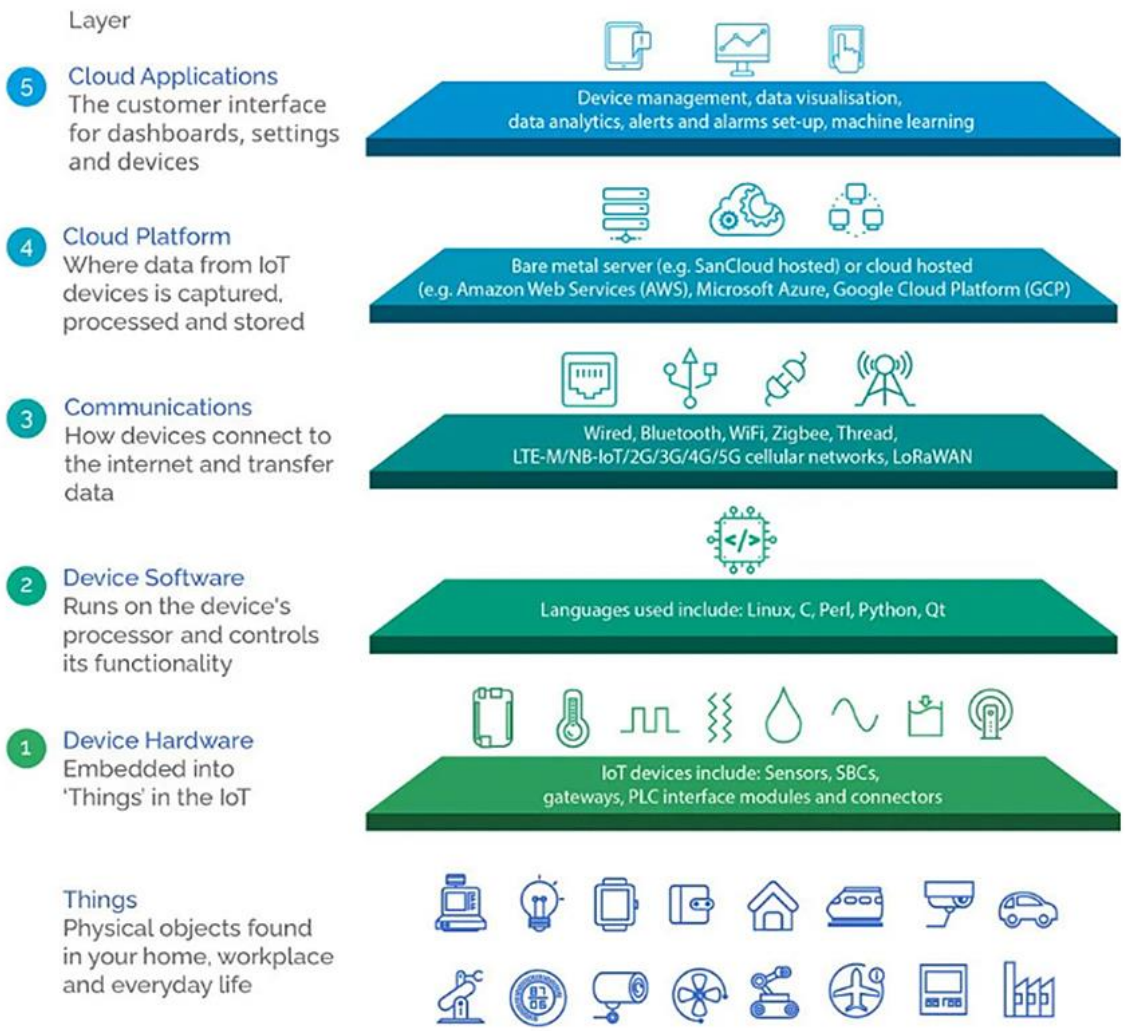
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IoT – Internet Of Things

The Internet of Things is a network of various devices, software, and sensors connected to each other that can transfer data via the internet without any human interaction. It is widely regarded as the fourth stage of the industrial revolution.

IoT Tech Stack



Device hardware: Bottom most layer of the tech-stack and is composed of sensors and processors. Raw data collected through sensors is processed to generate an output that works on another system. Microprocessors form the basis for crunching the high-velocity data that are generated by the sensors and use it for processing in other stages.

Device software: It is primarily the operating system and the set of applications that connect

with upper layers of the tech stack.

Communication layer: Connects the various devices that form the IoT system based on Bluetooth, radio, wi-fi, cellular or satellite technologies and differs mainly in the frequency and range supported by them.

Data and Analytics: From the information that is collected, relevant data needs to be extracted and structured before being processed. With the help of many cloud APIs, insights can be generated to make sense out of this data using sophisticated data analytics techniques.

Applications: It is typically the layer that the end-user interacts with and sees. These can take different forms like web-based or app-based depending on the requirement.

Use cases:

1. Smart Cities: IoT systems have the capability to sync up different traffic lights across cities as per real-time data, which can help in redirecting and synchronizing the city traffic. Another use case includes the sensors in buildings like CO2 sensors, smoke detectors, cameras etc. that can help in optimizing building's energy usage.
2. Healthcare: Along with the use of health trackers and wearables, IoT devices are also used in collecting critical patient information and can send it to another other site quickly. Additionally, it can help in monitoring heart rate, blood pressure, and glucose for patients who require regular attention. Caregivers can also monitor and provide medication using IoT devices.

Autonomous Vehicles: Drivers can gather information regarding the state of the road using cameras, LIDAR, and other onboard sensors and take preventative driving measures. IoT is playing a part in making "vehicle-to-everything" or "V2X" technology a reality.

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Cloud computing

Cloud computing is the delivery of computing services—including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet ("the cloud") to offer faster innovation, flexible resources, and economies of scale. Cloud service providers provide their customers with all the core services that are involved in a traditional on-premise IT infrastructure with added services and benefits in terms of pricing, availability of services, scaling up or down, security, maintenance, and much more. The customers usually pay for what they use at very reasonable costs and just have to worry about the kind of infrastructure they want and its deployment all virtually. Everything is accessible over the internet through some interface provided by the cloud service provider and users log in and simply deploy the kind of infrastructure they want.

Major Players:

- Amazon (AWS) with a 60% market share in 2020
- Microsoft (Azure) with 31% market share in 2020
- Google (GCP) with 9% in 2020

Types of cloud services



- **Infrastructure as a Service (IaaS):**

  Users can choose and rent IT infrastructure like servers and virtual machines (VMs), databases, networks, operating systems etc. from a cloud provider. Example : AWS, Microsoft Azure, Google Cloud Platform(GCP)

- **Platform as a Service (PaaS):** Allows users to build, manage and deploy the applications without having to manage the underlying infrastructure such as networking, servers, databases, Example: OpenShift, Github, Gitlab, Docker, Kubernetes

- **Software as a Service (SaaS):** It's a complete service provided and managed completely by the service provided. Users just have to create accounts, configure and use the service. examples include Google Apps (Gmail), Dropbox, Zoom.

Benefits:

Pay on demand: Customers can use it as and when required and pay for only what's used which helps in reducing TCO and Operational Expenses.

Provision the right type of resources: Cloud provides the customers an opportunity to choose the right type of resources in terms of power, storage, speed, network, etc for meeting their business needs.

Metered and self-service: Payment is based on the basis of resource usage. Users need not rely on 3rd party for any services as they can directly use it from the service provider.

Flexibility and Scalability: Cloud environment is very flexible and elastic. Users can easily switch between different kinds of resources to have more processing power or to increase/decrease the number of nodes provisioned depending on the type of workload they face.

High availability and fault tolerance: cloud is based on a global infrastructure and hence provides high availability and fault tolerance as resources are automatically provisioned across multiple data centres.

Global Deployment: customers can easily deploy their entire infrastructure within minutes across the globe and have a global presence.

No operational or maintenance costs: Cloud turns out to be cost-effective for customers as they don't have to worry about any costs relating to staffing, maintenance, rents, power, cooling, hardware, licensing, etc.

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Blockchain

Blockchain is sequence of blocks or list of records that are chained together and distributed among the users. It's a ledger that keeps on growing as more data is added and are immutable in nature. There are 3 elements in each block – a cryptographic hash function that points to the previous block, transaction data of the current block and its timestamp and a cryptographic hash function pointing to the current block.

1. Blockchain has become popular due to the following properties:

1. Immutable: Any record that has been validated and added to the ledger is final and irreversible. In order to change any block, every block preceding it must be changed, which is computationally impossible.
2. Distributed: Every user has its own copy of the transactions and hashed blocks, and they spread the information of any new transaction to the entire network.
3. Decentralized: There is no central governing authority that will responsible for all the decisions. Rather a group of nodes makes and maintain the network. Each and every node in the blockchain network has the same copy of the ledger.
4. Secure: All the blocks are encrypted using cryptographic hash functions. All the blocks contain a unique hash of their own and the hash of the previous block. Due to this property, the blocks are cryptographically linked with each other.
5. Consensus: Ensures that the next block is the only version of the truth. Consensus is a decision-making algorithm for the group of nodes active on the network to reach an agreement quickly and faster and for the smooth functioning of the system. Nodes might not trust each other but they can trust the algorithm that runs at the core of the network to make decisions.
6. Faster Settlement: Blockchain can allow the quicker settlement of trades. It does not take a lengthy process for verification, settlement, and clearance. It is due to the availability of a single version of data that has been agreed upon by all stakeholders.

Types of blockchain:

Public Blockchain: The public blockchain is an open-source, distributed, and decentralised public ledger that anyone can access and review. They can read, write, and audit the blockchain. It is considered to be Permissionless blockchain.

Private Blockchain: A Private blockchain is a private property that belongs to an individual or an organization. It is managed by a single organisation, which controls access rights to read and submit transactions, and take part in consensus. It is considered to be permissioned blockchain.

Consortium or federated blockchain: A pre-selected group, such as a collection of businesses or representative persons, is in charge of a consensus process. These pre-selected group is coming together and making decisions for the best benefit of the whole network. Such groups are also called consortiums or a federation and hence the name consortium or federated blockchain.

Business Applications of blockchain:

a. Record keeping: Availing feature of immutability and easy accessibility, along with a high level of security has potential uses in keeping records for land titles, patents, government databases like IDs and social security numbers.

b. Dynamic transactions registry: Timestamping makes blockchain useful for storing and updating real-time tradable information.

c. Blockchain-augmented supply chain: Currently, only limited information about the product in the supply chain is recorded, with little to no visibility of the same to multiple players in the chain. Blockchain helps bring about transparency in the functioning of the supply chain, improving operational efficiency as well.

d. Government/civic-tech applications: Blockchain's trust-based record-keeping and transaction mechanisms can be used to ensure that ensuring that the government's record-keeping facilities are tamper-free and transparent.

e. Banking and finance: need for peer-to-peer transactions to facilitate cross-border payments and trade without any intermediary banks and financial institutions.

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SDLC

Software Development Life Cycle (SDLC) is a framework that defines the steps involved in the development of software at each phase. Proper planning and execution are the key components of a successful software development process. It ensures that the end product is able to meet the customer's expectations and fits in the overall budget. The entire software development process includes 6 stages. These stages are:

1. Requirement gathering and analysis: In this stage, all the business requirements are gathered from the customer/client to develop a product as per their expectation. Meetings are held with management, stakeholders, and users to define the needs, such as: Who will use the system? How will they employ the apparatus? What information has to be added to the system? What information should the system output? After further analysis to determine whether the product's development is feasible, these requirements are included in a document known as the Software Requirement Specification (SRS).
2. Design: The requirements gathered in the SRS document are utilised as an input during this phase, and software architecture that is used for implementing development as well as how each and every feature in the product should work and every component is derived.
3. Implementation: On receiving system design documents, the work is divided into modules or units, and the development team begins the actual coding process in accordance with the design discussed in the previous stage.
4. Testing: As soon as the coding is completed, modules are made available for testing. The software is rigorously tested in this phase, and any defects detected are assigned to developers for fixing. In this phase, non-functional testing as well as all forms of functional testing, including unit testing, integration testing, system testing, and acceptance testing, are performed.
5. Deployment: Post successful testing the product is delivered / deployed to the customer for their use. Customers are recommended to conduct a beta testing as soon as the product is made available to them. Any required changes or discovered bugs will be

- reported to the engineering team. If the problem is severe, the development team will address it right away; if it is not, it will wait until the next version release.
- Maintenance: Customers using the developed system might eventually encounter issues that needs to be fixed from time to time. In order to maintain operational effectiveness, such as to improve performance, strengthen security features, this phase also involves making modifications to hardware and software in accordance with the customer's requirements.

SDLC Models:

- Waterfall model: It follows a linear sequential approach. The outcome of one phase is the input for the next phase and the development of the next phase starts only when the previous phase is complete.
- Prototype model: Software prototypes are built prior to the actual software to get valuable feedback from the customer. Feedbacks are implemented and the prototype is again reviewed by the customer for any change. This process goes on until the model is accepted by the customer.
- Incremental Model: It is necessarily a series of waterfall cycles. The requirements are divided into groups at the start of the project. For each group, the SDLC model is followed to develop software. The
- SDLC process is repeated, with each release adding more functionality until all requirements are met.
- Iterative Model: Iterative process starts with a simple implementation of a subset of the software requirements and iteratively enhances the evolving versions until the full system is implemented. At each iteration, design modifications are made and new functional capabilities are added.
- Agile Model: Agile SDLC model is a combination of iterative and incremental process models with focus on process adaptability and customer satisfaction by rapid delivery of working software product. Agile methods break the product into small incremental builds. These builds are provided in iterations. Each iteration typically lasts from about two to

four weeks. Every iteration involves cross functional teams working simultaneously on various areas like designing, coding and testing.

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Product Design Frameworks



One of the most common types of question in a Product Management interview is a Product Design question, or a Product Improvement question. When answering these questions, a very critical thing to do is to look at the customer/user when answering. These questions are asked to evaluate your creativity, ability to empathize with a user or customer, ability to provide solution keeping business goals in mind, ability to prioritize among the features and solutions. The products you encounter for design could be both physical or digital, or a mix of both.

A few example questions:

Design an ATM machine to cater for the blind.

How would you improve WhatsApp?

You are a PM at Gaana. How would you increase User Engagement?

There are some common mistakes which people commit when attempting these cases, like

Jumping to features directly: It is a common habit where we try to provide a solution as soon as we see a problem or pain point. However, doing that disturbs the flow and distracts us from identification of pain points efficiently. As we are looking at the problems from a customer's viewpoint, it is important to understand all the issues that there might be, before we move on to solve them. This gives a holistic view of the problem statement at hand.

Unstructured answers: When answering these questions, if you don't gather your thoughts before speaking, the answers are all over the place and makes it a very painful experience for the listener. It is better to take a few pauses at various stages, gather your thoughts and then answer.

Assuming things, and not clarifying: A very important aspect of solving these questions is to

ask really good clarifying questions. This is an important step, as in real world, you would have to scope the requirements when speaking to different stakeholders. During the interview, not asking the right questions and moving ahead with assumptions more often than not lead you to the wrong solution, something entirely different from what the interviewer was expecting to discuss. A good set of clarifying questions show clarity of thought.

To answer these questions, it is better to follow a framework. The framework provides you with a structured way of going through the problem. It is essentially a checklist of things to keep in mind when attempting a design/improvement question in interview.

CIRCLES method

C – Comprehend the situation

I – Identify the customer

R – Report the Customer's needs

C – Cut through prioritization

L – List the solutions

E – Evaluate the trade-offs

S – Summarize your recommendations

	**Comprehend**	• What needs to be done? • Who will be impacted? • How to achieve?
	**Identify customer**	• Consider different personas • If particular use case to be considered, opt for 1-2 user personas
	**Report Customer Needs**	• Drill down the problem statement • Identify absolute requirement
	**Cut through prioritization**	• If multiple requirements are at hand, prioritize the one with most impact • May use ROI estimates
	**List solutions**	Before coming to solution, identify multiple routes
	**Evaluate Tradeoffs**	Amongst identified solutions, evaluate the different pros and cons analytically & objectively
	**Summarize**	Few things to reiterate before concluding • What & Why • Brief recap for sanity check

Comprehend the situation

A product manager seeks to gather as much information, and scope a problem and that is exactly what you do in this phase. Here, you get as much clarity about your problem statement as you can. Questions include the type of product you are building, scope of your solution, constraints that you might have and also, clarify your understanding of the problem statement. It is better to sound out what you understood before proceeding.

Identify the Customer

In this phase, you look for the possible customers for the product. Here, you first categorise the segment of customer base, for whom you will be building your product. This is important, so that you can solve for the targeted segment and not make a product for everyone, which ends up not catering to anyone's interest.

Report the Customer's needs

After you have identified your customer segment, you move on to identifying the needs for this customer segment. Here, you try to find the current pain points that the customer is facing, or you look at the needs that the customer might have. Identifying good pain points is really important, as this shows how you can empathize with your customer. A good way to frame the needs is:

As a <role>, I want <goal/need> so that <benefit>.

Cut through Prioritization

In the previous section, one can identify a lot of pain points based on how much one can think of. As it is a time-constrained interview, it is not possible to go through all the solutions at once. You have to prioritize the pain points you would be solving for, and there are a few ways of prioritizing. One can prioritize based on how acute and important the problem is, or what is the value the solution is to the customer. You choose the solution which has maximum impact for the customer, with the lowest or moderate effort to build.

List the solutions

After you have prioritized pain points you will be solving for, you now provide your solution. This is the portion to get creative and show your passion for product management. Here, you

showcase your understanding of the problem, and also the business case where you provide solutions keeping both the customer and the business in mind. Provide realistic solutions but still try to think of creative ways to solve a problem. If you chose more than one pain point, provide solutions for all of them

After you have provided solutions, you can prioritize solutions based on popular frameworks, such as RICE, Kano model, ICE, MoSCoW method, etc.

Evaluate trade-offs

In this step, you self-evaluate your solutions, and provide the pros and cons that there might be. This gives you an edge in the interview, as it shows that you can think of the criticisms for your solutions, and it would be better if you justify or a future way of resolving them.

Summarize your recommendation

Here, you provide a summary of the recommended solution, in a 20-30 second span. Provide a recap of the problem, and feature you recommended and why.

Feature Prioritization Frameworks



The practise of ranking and organising features in a product according to consumer value, business objectives, the amount of time and money required, and technological viability is known as feature prioritisation in product management. For stakeholders to develop an effective roadmap, structure is essential. This enables product managers to concentrate on the most crucial features and choose the things that should be developed and released next on the roadmap more wisely.

- **RICE**: Each feature or initiative is scored according to the four factors: reach, impact, confidence, and effort. Here's a breakdown of what each factor stands for and how it should be quantified:

REACH	IMPACT	CONFIDENCE	EFFORT
How many people will this feature affect within a given time period? **Example:** customers per quarter, transactions per month	How much will this impact individual users? Use a multiple choice scale: **3** = massive impact **2** = high impact **1** = medium impact **0.5** = low impact **0.25** = minimal impact **Example:** How much will this feature affect conversion rates?	How confident are we about the impact and reach scores? How much data do we have to back up those estimates? Use a % score where: **100%** = high confidence **80%** = medium confidence **50%** = low confidence	How much of a time investment will this initiative require from product, design and development? Measure as persons per month (how much work one team member can do in a month).

Then, a formula is used to combine all of those individual numbers into a single score. With the help of this formula, product teams may generate a standard number that can be used for every kind of initiative that has to be included to the roadmap.

$$\frac{\text{REACH} \times \text{IMPACT} \times \text{CONFIDENCE}}{\text{EFFORT}} = \text{RICE SCORE}$$

User will receive a final RICE score after applying this formula to each feature. The order in which user will address each idea, initiative, or feature can then be determined using the final score.

- **Kano Model**: It operates on two sliding scales: how much consumer satisfaction this will provide vs how it will enhance functionality. Product teams may clearly see which features will have the greatest impact on user satisfaction and functionality by putting them on this chart. Also, which features should be avoided because they're simply a waste of time and resources and won't deliver customer satisfaction

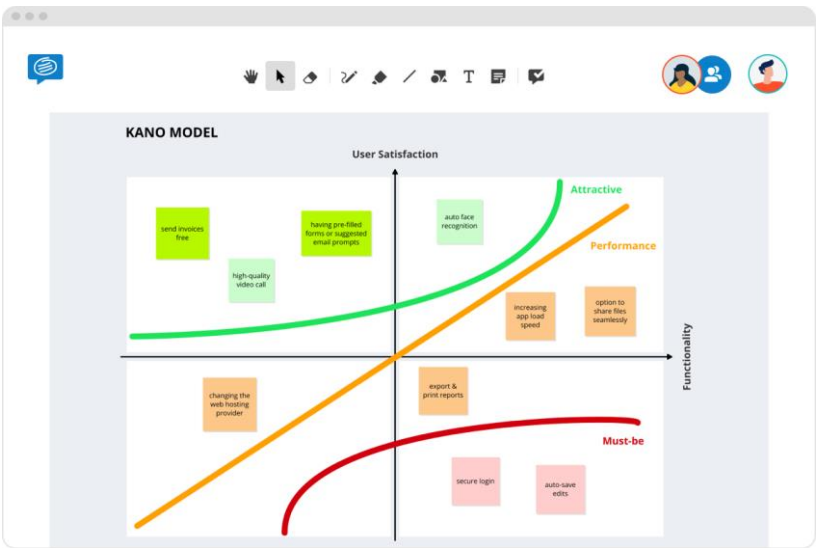
These features can be classified into three buckets:

a. Must-haves or basic features: Customers won't even consider your product as a solution to their problem if it lacks these features

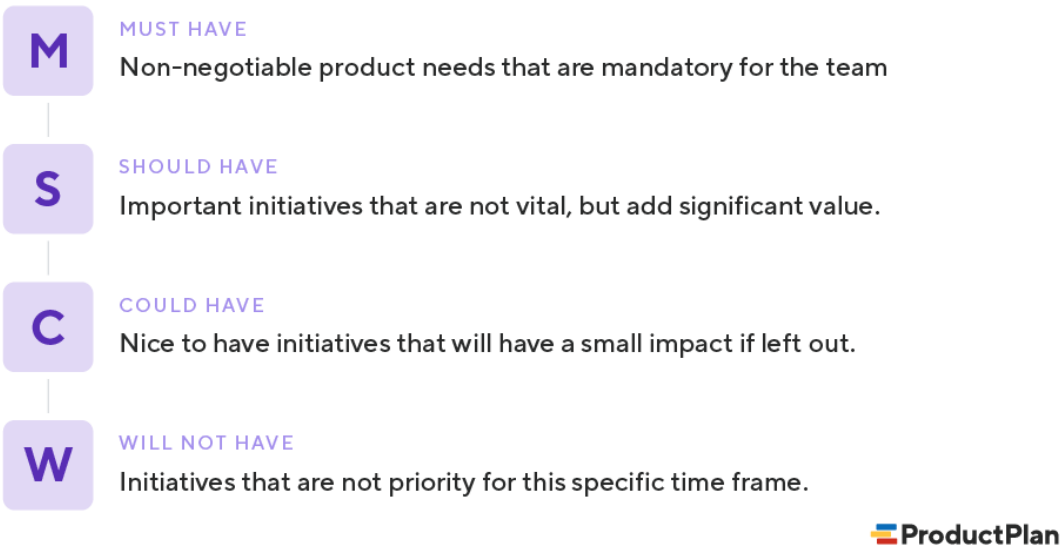
b. Performance features: The more you spend on these, the more satisfied customers you'll have.

c. Delighters or excitement features: These are the delightful surprises that customers don't anticipate, but when offered, produces a delighted response.

An example can be:



- **MoSCoW**: By grouping features into four priority buckets, the MoSCoW technique enables you to determine what matters most to your stakeholders and customers.



  a. Must-Have: These are the features that have to be present for the product to be functional at all. They're non-negotiable, essential and most time-sensitive of all the buckets.

  b. Should-Have: Although they are not time-sensitive, these requirements are crucial to deliver.

  c. Could-Have: This is a feature that is neither necessary nor urgent to deliver by the deadline. They are bonuses that, if included, would significantly increase consumer satisfaction but have little effect otherwise.

  d. Won't-Have: These are the least important criteria, tasks, or features (and the first to go when there are resource constraints). Future releases will take into account these features.

- **ICE**: In order to decide on a feature's priority, ICE considers impact, confidence, and ease of implementation of features. To get the ICE score for a specific feature, these elements are typically assessed on a scale of 1 to 10, with the sum being averaged. An example of ICE prioritising is:

Project idea	Impact	Confidence	Ease	Ice Score
Billing System	7	1	5	35
Community tab	7	2	8	112
Update receipt design	5	5	3	75

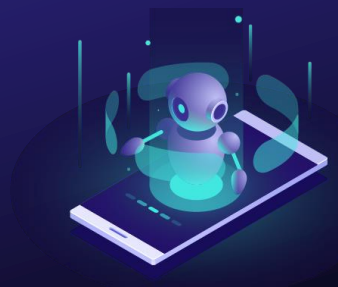
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7 Key Mobile app metrics to track



Acquisition

Number of **new downloads** in a given time (daily/weekly/monthly)

Download Attribution: The channels your new users come from. Use this data to optimize marketing spend and improve ROI/customer LTV

You should be tracking downloads daily to get clearer insights into the impact of your marketing campaigns. Knowing exactly where your users are coming from tells you where to invest your marketing resources to ramp up new user growth.

Activation

Activation Rate: Percentage of downloads that launched the app

Metric Range: If your activation rate is 85% or above, you're doing well. Anything lower calls for a deeper analysis to understand and fix this issue.

For most growing mobile apps, the ratio of first-time app launch to total app launches over a rolling 30-day period is between 5% and 15%.

Many seemingly trivial factors can influence decline, from UI typos to lengthy user registrations, but this is the first drop off point you should examine.

Retention

Simply put, **retention = revenue**. The longer you keep users, the more valuable they are to

your business.

By tracking user retention metrics and identifying exactly where in the user journey people are dropping off, apps can be improvised for users' attention and engagement.

Use first time app launch to create acquisition cohorts. You can then track how long each cohort stays active in your app and find out which in-app behaviours have a positive or negative effect on retention.

Engagement

Daily Active Users (DAU) and Monthly Active Users (MAU): The number of active app users on a given day or month

Average Session Length & Frequency: How often users are launching your app, and how long an average session lasts

Not all apps are meant to be used daily, so you'll need to define exactly what an "active user" means for your app. Is it launching the app, logging in, or completing a key action?

Uninstalls

Number of Uninstalls: How many users uninstall your app on a daily or weekly basis

Churn Rate: Users at the start of the period – users at the end of the period / users at the start of the period

Compare daily uninstalls with the number of daily downloads and daily activations to get a net result of your mobile app growth.

If the major churn problem on your hands, look at your app's performance to rule out bugs, crashes, or latency issues. 80% of users ditch buggy or slow-loading apps after just a few uses.

Drop Offs

Conversion Rate: Users who performed a key action / user who could have taken the action

Cost Per Conversion: Total cost of user acquisition / number of conversions

CAC-to-Conversion: How much it's costing you to acquire new users who actually pay money for your product or service. Plus, it's a metric that you can directly improve. By lowering this rate, you get better ROI. And because conversion doesn't stop when a user installs your app, you can continue to upsell, cross-sell, and encourage users to make repeat purchases or renew subscriptions.

Ideally, you need to identify 1 to 3 key flows within your mobile app that you want your users to take.

Then use simple conversion funnels to track those flows over time, discover the drop off points, and zero in on areas that will have the greatest impact on growth

Reachability

Click-Through Rates: Percentage of users who tap on push notifications, in-app notifications, and email links

Opt-In Rate: Percentage of users who opt-in to receive notifications or allow location access

Your ability to connect with users via different channels is an important mobile app metric to track over time.

Monitoring reachability per channel over time, and comparing those trends to your daily app installs and activations, is critical to be able to engage your users and create sticky experiences.

(**Source**: https://clevertap.com/blog/are-you-tracking-these-7-key-mobile-app-metrics/)

Interview Tips for product design cases



Tip Number 1: Be structured in your approach. You do not need to follow CIRCLES to the letter. However, broadly you should follow a structure of Comprehend situation - Create customer persona and identify their pain points, and then how you solve customer problems

Tip Number 2: Articulate a product vision that covers:

1. The value proposition that the product feature offers to the customers
2. What sets it apart from competitors

For example, for a hypothetical Facebook Movies product, the product vision can be "Enable users to share their life stories in a memorable manner with friends and families on Facebook"

For Facebook Messenger Rooms, the product vision can be:
"Enable friends on Facebook to hang out casually in easily accessible rooms on Facebook"

Tip Number 3: Be creative in

1. The customer personas that you visualize
2. Identifying the pain points that they face
3. Suggesting creative solutions rather than mimicking existing solutions from competitors

Tip Number 4: Be Simple and Concrete - When mentioning pain points, tie them to a specific customer persona, even when it is applicable to a broader set of users. Tying to a specific customer persona would help you ideate how to solve the problem for that customer persona.

Tip Number 5: When designing the MVP, articulate the Key Design Principles that you will follow.

Product Design / Improvement cases



Problem Statement: Improve Whatsapp

Clarifying the broad area of improvement - what is the goal

Initially asked if there was a goal in terms of acquisition, engagement and monetization. Interviewer mentions that monetization is not a priority currently. Then interviewee starts explaining what he means by acquisition and engagement as goals

a) Acquisition - How do we onboard more new users to the platform. Is there a customer segment we are not targeting?

b) Engagement - There are a billion users who are already using WhatsApp. How do we make sure that users are having a good user experience, and How do we ensure that users are spending more time, and starting more conversations through WhatsApp?

WhatsApp already has a large market share. Focus should be on delivering a much more positive user experience.

Customer segmentation and use cases

- School-children for whom access to social networks needs to be moderated
- Elderly, for whom communication through Whatsapp may not be so intuitive

- Millennials/ young people who want a fun medium in which to express themselves
- Police offers, who require a secure and easy-to-use medium to update news

Prioritization

Interviewer asked him to focus on the Elderly segment, and so the interviewee started detailing the pain points better -

a) Elderly may not be able to interpret messages sent on Whatsapp (for example emojis are commonly-used lingo),

b) They may not know how to respond appropriately

Solution

Once the pain point was identified, the interviewee started listing solution - primarily a chat assistant to enable elderly to interpret messages with helpful recommendations (Think of Gmail's automatic reply)

Exponent: https://www.youtube.com/watch?v=e0Nj_eYDj2s

Problem Statement: Improve Adobe Scan to onboard utility users towards Adobe Document Cloud

Stakeholders	User scanning the document User with whom pic is being shared **Additional details** Users are often pressed for time when performing this task - they don't want to spend too much time scanning on the app - and want to just scan and share the image
Problem Identification	User wants to share a printed/ handwritten document using phone camera
User Story	• As a student, I want to submit hand-written assignments so that they can be evaluated • As a student, I want to share my notes with peers so that they can help them • As an office manager, I would like to scan office documents so that they can be shared with other business teams • **As a regular user, I want to scan documents and share with others** • **As a regular user, I want to scan documents and share with my PC so that I can work on it further on my PC** • As a regular user, I want to scan documents **quickly** so I can use them further

Prioritization

- Is this problem use case big enough?
- Does solving the user story encourage users to use Adobe's Suite of offerings for all their requirements
- Is the core document scanning product reliable enough?
- Which user feature would improve customer experience better?
- How much development effort is required in either solutions?
    - Improving document scanning product may mean
        - Improving ML algorithms
        - Making customer UX more intuitive
    - Improving shareability may mean
        - Making it easier for customer to scan from Adobe Scan app and access it immediately on their Desktop
        - Making it easier for customers to share with their friends through any platforms

User Journey on App

- User opens the App
- User clicks a button to start scanning

Scanning of Image

- User places phone camera over scanned material - Positive experience
- User clicks photo of document - Negative experience
- User evaluates and processes scanned image

If more pages

- User clicks on button to click new image, and goes back to Scanning of Image

If scanning done

- User clicks on button to confirm completion of scanning
- User saves the image to the mobile or

User shares the image with somebody

How currently the App makes customer Journey better

- When the user opens the App, initially it shows home page, which within 5 seconds opens into the camera, so that user can immediately start scanning. This shows an understanding that the customer opens the app for document scanning. The home screen is shown to highlight that there are other use cases as well. However, this may confuse the user
- Currently an ML system automatically scans the image once the document is in focus
- The ML system also automatically determines the document edges
- Once the image is clicked, users can opt to either - Retake, Continue, or Finish. However, finish button is not so clear, as you have to click on the small preview of Finished pages in bottom right (for first-time users' journey is not so intuitive)
- Provides users options for use cases such as for scanning
    - Forms (which they can then fill using Adobe Acrobat)

Business Cards (which they can save as Contacts)

Key metrics to

- Number of save to file vs Number of file shares - The more the

Track success of features suggested

- number of file shares, the more confident customers are about the final quality of the image
- Amount of time it takes camera to focus on and click photo (speed of ML-based capturing)
- Number of camera retakes (success of ML algorithm to capture image)
- Number of backs and forths from one page to another (may highlight that user is confused about the product use)

Competitor Analysis

- **Camscanner** has wide user base, and also applies a watermark - the watermark gives an incentive to users to pay to remove the watermark, while doing organic publicity for the Camscanner app
- **WhatsApp** - has wide user base across mobile phones and can be used for sharing documents
- Google Drive is one of the most popular file sharing service
- Gmail - Native integration of mobile app with desktop client means that clients may often transfer files

While the above three apps do not perform scanning of image, they do play a role in sharing of image

Feature suggestions

#1 Get user to create an account on Adobe Document Cloud

- Notifications on mobile app to prompt user to login
- Native integration with G-Suite for one-click Authorization
- User can login to Document Cloud from PC, and then download from there

#2 Create an automatic temporary URL where file is stored, from which user can download

- Integrate with Google Chrome browser, so that the link gets saved in the browser itself, and can be accessed from the PC then, and so user can download image directly

#3 Improvements in Document Scanning product

- Currently it is not clear for the user how to finish scanning the image. The finish button should be more prominent

#4 Promote utilization for Form scanning - Onboard institutional clients to utilize Adobe Scan for Forms feature

- Auto-convert images that are forms for use in Adobe Acrobat, driving use for Adobe Cloud

#5 Faster Optical Character recognition

Problem Statement: Design a product to encourage voting

Clarifying Question	Region - United States Physical or Digital Product - No constraints, but preferably digital product - on the voter's side
Process as a whole	Candidates are nominated by their political party **Candidates campaign and promote themselves, and voters become aware of the campaigns** Voters decide who to vote for, and turn up for voting in the polling booth Results are declared
Stakeholders	• Voters • Candidates • Election commission overseeing election
Customers - voters	Voters should be able to influence the choice of candidates that political parties nominate **Voters should have reliable and complete information about candidates** Results should be transparent; Voters should be shown the importance of their vote Voting should be made more convenient on the final voting day
Product vision	Reliable and easy to understand information sourcve

MVP

Acquisition strategy - In-app or as a featured page - Like LinkedIn - Feature on LinkedIn as an acquisition strategy

Can have an app for greater control over content

Content Creation - 3 sides, and hybrid model should be there

- Candidates themselves
- General public
- Editors

Visual Design elements

- History of public service & professional service
- Political views - Personalization element by comparing with views of individual voter
- Images of candidate

Exponent: https://www.youtube.com/watch?v=jQhdz4j29fU

Problem Statement: Design a smart shoe

Clarifying Questions

Q : What is a smart shoe?

A: Something that involves technological upgrades

Q. Specific customer segment?

A: free to make assumptions

Q. Any specific kind of shoes like Shoes, boots, rain boots etc

A. Normal shoes

Customer segment

- Users who care about fashion, want great looking shoes, particularly children who may open to new ideas
  - Augment great looking shoes with tech for eg. Led lights on shoes or some simple screens that display graphics/logos
  - They can program their own little logos and can have an educational experience of learning programming and designing
- Athletes: elite runners who are really focused on a great running performance.
- Elderly, especially if they live alone and are not assisted too often. An elderly with Alzheimer's, for instance will still always remember to put on shoes but maybe not a watch or a life alert badge.
  - A smart shoe that could change temperature, comfort and/or a GPS tracker to assist with directions

Prioritizing customer groups

- Fashion users: too much variability/subjectivity in fashion
- Elderly

- might not end up buying shoes because they already have it
- High barrier to entry: there might be constraints around these shoes e.g. you can't wash them, have to be connected to Wi-Fi at all times etc

Therefore, picking the segment of runners for this design!

Product Vision

Helping runners run faster and better

Design

Hardware needs to be light, comfortable etc that is essential for running

Software features:

- Tracking and giving useful analytics to the runner:
  - Tracking the runner's form based on how the shoe tilts/lands etc that tell us if the runner is able to run effectively
- Based on tracker analytics, recommendations for making adjustments to shoes and running form
- If the runner runs again in the same track (for e.g. a second round of a circular track), we can compare running performances of the two sets.
  The way to do this is:
  - The smart shoe can know when the runners are starting on the track based on a period of inactivity followed by a period of high intensity activity
  - Shoe can differentiate between these two laps- track the steps and pace and relate it to variables like weather conditions, diet etc

	○ Overtime, we can build a machine learning model that can tell based on a variety of factors like weather, diet etc what can be the increases in running performance and what can be done to make running better Encourage healthy competition by allowing comparison of running performance between two runners on the synced app
Scaling the use	• Olympians • Marathon/triathlon runners
Concerns	• Too many sensors might make the shoe big and bulky The technology needs to be precise to really make the shoe helpful and achieve the product vision

Problem Statement: Design a bicycle renting app for tourists

Comprehend

- Mobile app for commuting between tourist places in a specific city (say Agra).
- By bicycle we mean normal bicycle & e-bikes with batteries which do not require Driving licenses**.**

Identify customers & stakeholders

- Tourists (Solo traveller/group of friends/family) - This is the main user group and their objectives are to explore the popular places in the city (museums, local markets, monuments, sightseeing)
- Guides/Travel Agency - They may want to include rented bikes to their customers and offer them this optional service included in the guide package.

Reporting Customer needs

Before delving into the user journey, first understand some of the key requirements from a customer's point of view:

- As a tourist, I wish to spend less on travel for short distances.
- Apart from all the famous tourist destinations, I would like to explore some hidden gems too, which are usually not accessible by motorised vehicles.
- As an aware global citizen, I would prefer a mode that is environment friendly.
- If I am staying at a place for more time, I would also like to explore the city culture and gel up with locals, which can't be usually done if I am in a cab.
- Apart from tourist destinations, it would be great if I get pre-emptive suggestions for food, recreational activities etc. nearby

rental service user's perspective:

- Native language of the tourists
- Locate a cycle
- Get an idea of the cost
- Start my ride & ride safely
- Deposit the cycle after completing the journey
- Make easy payment for the trip
- Get suggestions of popular places & restaurants during my journey

Cut through prioritization

After identifying customer needs, it's important to prioritize and analyse the MVP (Minimum viable product) for first draft or design and subsequent development.

Here are the key components to be included can be prioritized as follows-

- Locating the bike (GPS enabled system)
- Starting the journey (QR code enabled)
- Cost display and payment (mode familiar to international tourists like PayPal)
- Native language of tourists
- Depositing kiosks

Suggestions of popular places based on proximity and user rating

List Solutions

- **Linguistics** - Have a multilingual app which prompts users to select language after installation (English, Spanish, German, French etc.). Have an option to switch language in settings option
- **Locate bikes nearby** - Each bike will have a GPS system installed and bikes will be displayed to the user in a map view (he can view which bikes are near him) and go to the respective location to pick-up the bike.
- **Cost** - Users can select bike types and it will display the cost (per hour, per day basis).
- **Start my ride**- To start the ride there are the following prerequisites
- **Setup payment** - Setup a payment wallet (Paypal, GPay etc.), Credit card, Debit Card
- **Unlock the bicycle** - Scan QR code to unlock the bike which will start the trip automatically and time will be recorded and displayed to the rider
- **Safety** - Each bike will have an attached helmet to it
- **Deposit the bike** - Parking spots will be displayed to the users on app to ensure bikes are parked in high density areas of the city. (users will be provided suggestions of parking spots so that the other riders can find them easily for subsequent trips).

Once a user ends his/her trip, an invoice/receipt will be displayed and the amount will be deducted from the wallet.

Evaluate

Feature↓/Factor→	Freq of usage	Impact on user	Impact on business	Development time
Linguistics	Medium	Medium	Medium	Low
Locate cycles nearby	High	High	High	Low
Cycle type & associated cost	High	Medium	High	Low
Locking/unlocking & trip payment	High	High	High	High
Deposit (Parking spots)	Medium	Low	Medium	High

Summarize

To summarize, we want to build a bicycle renting mobile app for tourists with a goal of helping them travel the city using an efficient & clean energy solution that meets their needs.

Following points constitute our MVP and are achievable in lesser amount of time:

- Linguistics
- Locating cycles nearby
- Bike type & associated cost
- Locking/unlocking & payments (Most crucial & probably maximum development time)
- Parking spot (may not necessarily be a part of MVP)

Problem Statement: Design news platform for younger generation

Clarifying Questions

Q: Is the platform a mobile application or desktop application or both?

A: It's supposed to be a mobile app.

Q: By younger generation do you mean any specific age group or school/college-going student?

A: Consider school-going students from class 6th to 12th.

Q: Where will the students use this app at home, school, or anywhere else?

A: Make required assumptions.

Provide structure

- Identify different users.
- List down needs/use cases.
- Identify weak spots of the current product in the market for these use cases.
- Brainstorm features to improve the weak spots
- Prioritize features
- Define basic metrics to measure success
- Wrap it up

(In online interviews, make it clear to your interviewer when you are transitioning from one step to the next.)

Identify Users

Primary users -

1. School-going children from the age of 13 - 18.
2. They will use the app to surf news mostly related to their interest

1. i.e. sports, academics, entertainment, etc.

Secondary users:-

1. These can be parents, guardians, or school teachers. They will help the primary user in logging into the app
2. They will moderate the content to filter relevant news according to the age
3. They will pay for ethe app (if it is monetized)

User needs

- Curated Categories - School-going children need different news categories than adults.
- User experience - Teenagers love variety and experiment a lot with the look and feel of the apps. They also prefer graphic and video content.
- News deep dive - Children are curious and would like to know more about the news or related history.

Engage with fellow users:- Teenagers love to share and know what their friends are doing.

Gaps in the current products in the market

- Current apps focus more on news preferred by adults such as politics. They also lack/deprioritize children-friendly categories such as social sciences, music, etc.
- Most news applications are text-based.
- At present, not many options are given to experiment with the app display. Most apps just have variety in terms of selecting the night mode.

- Most applications just have the article/news snippet without any option to delve deeper into the topic.
- Current apps have the option to share the news article. However, all the platforms do not give proper visibility into the most read articles/categories and the ability to interact with fellow users.

Features improving weak spots

- Categories - Students will prefer news with a focus on the following categories:-
  - Academic - Science, Social science, Literature, Business, etc.
  - Non-Academic -
    - Extra-Curricular - Sports, Entertainment - Movies & Music, etc.
    - Current Affairs - Country, World

  Users will have the option to select relevant categories according to the interest.
- User experience - Graphics and video snippets can be used to convey news. Also, students will be given more options to change the display in terms of font type, text size, page colour.
- News deep dive - Users can explore more about the news using relevant redirects such as links to Wikipedia pages.

  Ex. - For the news - "American poet Louise Glück wins the Nobel Prize for Literature", link to the Wikipedia page on 'Nobel in Literature' can be attached with the news
- Engagement with fellow users:- Following features will make the app more interesting for the primary user group:-
  - Adding friends to the network

- Share news with friends
- Visibility to the most read news and categories among the similar age group

Prioritization

S. No.	Features	Impact	Effort
1	Identifying news categories for the younger generation	High	Low
2	Using video snippets and graphics for news	High	High
3	Giving the option of font type, size, page colour, etc.	Low	Low
4	Including relevant link	Low	High
5	Engagement features	High	Low

Key metrics

- **Total Users** - Daily active users (DAU) or Monthly active users (MAU)
- **New Users** - Number of new users added in a month
- **Retention Rate** - # of people using the app within a set period of time /# of people using the app within a previously set time
- **Session length -** Average time per session

If the app is monetized:-

- **Cost per Acquisition**(CPA)
- **Average revenue per user**(ARPU)

Miscellaneous Cases



Go-to-market case: You are the product manager at Facebook India. Facebook is planning to launch a fitness app. Give us a Go-to-market strategy for Facebook

Clarifying Questions

Candidate: What is the goal of this new App?

Interviewer: Add new users to FB

Candidate: Since this will target a new customer segment, will we be looking to increase revenue from advertisements as well?

Interviewer: Facebook's mission is to engage people so revenue generation will be secondary.

Candidate: Will this app be an additional feature of the existing Facebook app

Interviewer: Yes

Structure

Since there are already a lot of players in the Fitness market, the feature adoption of our app should be done very uniquely. Here Facebook can leverage its understanding of user's behaviour to provide a better recommendation system to users and by aligning with Facebook's mission of connecting people, Our Fitness app can provide a peer-to-peer platform that will allow the users to connect

among themselves which will be an additional motivating factor for new user to sign up.

In addition to that our application will also provide Fitness influencers to further expand their reach.

These can be the USPs for our product which should be strategically marketed.

Strategy

The marketing activities can be categorized into three buckets. Pre-launch, Launch, and Post-launch. Let's go into each one of them.

Pre-launch Facebook should ideally start by testing the app in a few regions within India. Probably start with metro cities where the fitness culture is more progressive. By releasing the teasers, and advertising small clips to highlight the unique features of the product Facebook should first create the hype.

Launch – The product can be launched by creating a virtual/ live event. All the competitors in the market are kind of established, so Facebook should seek to make an impression on its users.

We should create blogs with influencers to create a positive image around the product.

Post-launch – This is the stage where we should do the health check of our product by measuring the metrics. Also Facebook should actively keep marketing the product and educate the users about how this platform is different from its competitor. They can use existing products like Facebook and Messenger to market this product

Pricing case: You want to launch a subscription-based product like Netflix. How would you price this product?

Clarifying Questions

1. To confirm, it's a subscription-based product to stream content.

   Yes. We will be competing against Netflix, Hotstar, Prime etc.

1. Any specific geography?
   India
2. What types of contents will be streamed? Will it have live sports or UGC or is it very similar to Netflix?

   Can consider as similar to Netflix. Right mix of syndicated content and our own originals but won't be doing live sports.

1. Will it be available on different devices like websites, TV app, phone app, etc.? Yes
2. Any particular target audience? Anyone who consumes online content in India
3. What will be our business goal?

   Test the market, grow membership and become #1 in the long term

Customer segments

- Kids & Toddlers -Assuming we will have kid's section. Watches cartoons. Parents will be the deciding factor and will watch on whatever devices they provide.
- High school kids – Educational/historical and super heroic. Again, will watch on whatever devices parents have.
- College going students – Consumes variety of content. Will get pocket money to pay and will either use laptop or phone to watch.
- Mid/Low-income adults – More preference to entertainment and latest contents. No particular preference for devices and will mostly be using phones

	• Affluent Family/Adults – Seeks value and requires good quality content. Limited time to watch (late night/travelling). Would watch through many devices but preference would be TV
Pricing Options	1. Cost-based pricing – Aim is not to turn profitable and hence doesn't fit 2. Value-based pricing – Fits as quality of the content matters. Analysis on competitors pricing shows that Netflix – Rs.649/month, Hotstar- Rs.299/month, Prime- Rs.179/month 3. Should be less than 649rs/month or else will price ourselves out.
Strategy	1. Based on type of content and no. of devices used for each segment, can follow mix of value based and competitive pricing. 2. Since idea to test - Launch all plans with limited duration of free trial. This helps in figuring out the segments of people interested in the product and content. Conversion rate from free to paid service can be used as a metric to check profit/revenue point of view.
Pricing recommendations	1. Mobile only & Kids Only – Less than Rs.149/month. Can start with free trail + Rs.99/month. No of devices to be 1-2. 2. Mobile only, one device all content – Free trail + Rs. 129/month. 3. Multiple devices (except TV), all content – Free trial + Rs.199/month. Can have 3-4 devices. 4. Multiple Devices including TV & all content – Free trail. Starts

from Rs.499/month. Can have separate tiers within the tier for various video qualities like HD,2k,4k etc. Max devices limit can be 5-6.

Pricing case: You are a PM at Amazon and you are asked to change the pricing for Prime membership. How would you go about it?

Clarifying Questions

1. To confirm, Prime is a membership program that provides access to the OTT platform as well as some extra benefits on the Amazon e-commerce platform.

   Yes. The benefits include Free delivery, Prime Video, prime music, audio books, podcasts, amazon mom, amazon pantry, etc.

1. Is the plan to increase or decrease the price?
   Increase the price as Amazon is already providing users with good deals and offers
2. Any specific geography?
   India
3. Have our competitors changes their pricing recently?
   No
4. What is the business goal? Is it to increase profitability or customer base or sales/revenue? Can assume it's to increase revenue.

Customer segments

1. Segment 1: Interested in online video/ audio streaming services. Consists of Millennials/Teenagers aged between 16-21years old and prefers to share their accounts among group of friends. May not be willing to pay extra for the membership.
2. Segment 2: Interested in both online video/audio services as well as benefits like- free delivery, exclusive deals etc. Composed of Working Professionals who are financially independent and

1. hence should be are willing to pay for quality content.
2. Segment 3: Interested in deals, offers, shipping and delivery benefits rather than the online streaming services. May/may not be willing to pay extra for the services.

Pricing Options

1. Cost based: Costs like shipping costs, original content creation, the cost of buying the media from other production houses, etc. to be incurred. Against these, Amazon gets more loyalty from the customers which improves CLTV in the long term. Above costs to be estimated to decide what value can be charged.
2. Value based: Add a new set of benefits like-streaming online sports, access to more tv shows, movies and original contents.
3. Competitive based: Main competitors for would be Hotstar and Netflix. Prime has an edge over these as it also provides the users with e-commerce services.

Strategy

Mix of value based and competitive pricing

Pricing & recommendations

<u>Pricing & recommendations</u>:

1. Add a new set of benefits like-streaming online sports, access to more tv shows, movies and original contents
2. Exiting customers can choose whether or not to upgrade.

Current users in India – 10M

Current price – 1499/year

Assume segment 2 covers 50% of users i.e. 5M

New increase in price - Rs 300 and assume 50%-60% of the

users upgrades i.e. 2.5M

There would be a profit of 300*2.5 = 750M -800 M; 7-8% revenue increase.

RCA Cases



Problem statement: Decline of 8-9 % of user bookings in Urban Company (formerly known as UrbanClap). Diagnose the issue.

Clarifying Questions

Q1) What kind of services are being provided?
A1: Beauty, Spa, Grooming, Home Repairs are some of the services provided to users

Q2) What is the timeframe of the issue?
A2: Timeframe of last 4 days

Q3) Which platform is the issue occurring on Desktop app/ Mobile App?
A3: Mobile app

Decide on framework

. Internal factors- These are actions have been taken within the company that have resulted in changes

Q1) Any historical pattern observed?
A1: Has happened before but not recently

Q2) Happening for existing or new users?
A2: Spread across user segments no particular segment affected

Q3) Any update released recently?
A3: Yes update to improve the UI

2. Analyse the entire user journey-For existing user:
Launches app-> Browses Homepage-> Selects specific category-> Searches for specific service type-> Selects the service-> Taps on 'Add' button
3.Deep Dive into product funnel-
Q1) Is the Category List populating fine?
A1) Yes
Q2) Selects service: Any changes done here?
A2: Some information about professional was added
Q3) Has the 'Add' button pushed down?
A3: Yes
We can ask Analytics team if the user is going back without scrolling all the way to the 'Add' button

Consider all cases

External factors- These are things that happen, regardless if you want them to or not. Examples include competitive price changes, shifts in user behaviour, government policy changes

Q1)Has there been any campaign initiated by competitors in the last one week?
A1: No
Q2) Any significant change in app uninstall number, user feedback,

app install numbers?

A2: No (this is asked to analyse any shift in user behaviour)

Q3)Has there been any significant change introduced by the government impacting the sector?

A3: No

Summarise and final recommendation

Internal factors- Recent update in the mobile app has caused a drop. We can do further testing & launch to company internal users & specific set of super users to get their feedback on the new update.

External factors- Can also be caused due to government guidelines issued in regards to Covid. This is out of our control. From the app's end, stronger marketing can be done to be totally Covid safe.

PM School: https://youtu.be/OiE5vvbnryI

Exponent: https://youtu.be/DSV-vuvmIro

Problem statement: Number of Ola cancellations have been increased by 7% in India

Clarifying Questions

- Which user base is reporting cancellations- riders, drivers or both?

  Riders
- What are the time-frames of these cancellations?

  Past one week
- Has there been change in metric definitions for rider cancellations?

  No change
- Were cancellations observed predominantly on Ola app or its lite version?

  Both apps
- What platforms are experiencing these cancellations - Android, iOS or both?

  Both platforms

Demographic segment and user segments

- Were there specific locations wherein the cancellations have increased?

  No (Spread across locations)
- Have these changes been reflected among new/ old app users?

  Both segments
- Has this change been observed by users of a specific profile? (ex- corporate profile)

  No specific profile
- Were cancellations observed in a particular ride option Daily, Intercity, Rental?

  No particular ride (such as Mini, Prime, Sedan etc.)

- Has the average waiting time for cabs increased over the past week?

  Minimal change

External factors and competitors

- Have there been any recent rider protests which gained media traction?

  No
- Were there Covid-19 restrictions on cab services in specific regions across country?

  No restrictions observed
- Did competitors launch a new feature to counter Ola's cab booking success?

No significant change observed

App update

- Has the application been recently updated in the past month?

  Android app updated in phases over 14 days, iOS one week back
- Did the app report changes in crash free sessions?

  Unchanged
- Did the application update improve the user journey?

  No change
- Does the application provide updated Covid-19 safety protocols across platforms?

  Yes, up to date
- Has there been a change in the push notification/ messaging services post booking?
- OTP removed from in-app notification due to security issue
- OTP service primarily sent via SMS, push notification via vendor

service

- Has there been a change in OTP services provided by third party vendor?

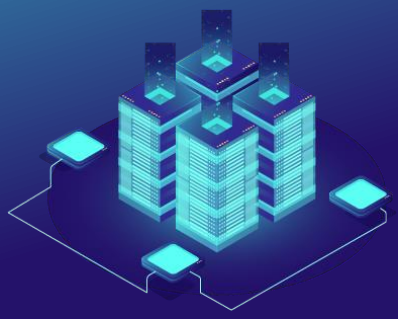
  Yes, OTP channels have been updated

Conclusion

- The updated OTP channels are a probable cause for concern
- SMS sent through vendor services are delayed due to channel update
- Users who remove push-notification services are unable to fetch OTP
- Therefore an increase in Rider cancellations

Source: Lyft Product Manager Interview: Driver Cancels - YouTube

Key Tech Products



Critical to being a Product Manager is having an eye for detail in Products.

Facebook, WhatsApp, Netflix and YouTube seemingly deliver a smooth experience to users.

As Product Managers, you need to develop an eye for the combination of various features and design principles that these apps use. These features are carefully optimized to achieve the desired user behaviour on the app.

Key Components include

- Value Proposition
- Users
- Some features & what they enable
- Improvements that can be made

Let's look at a couple of the examples for this.

Facebook

Value Proposition - Social media platform for people to engage with friends, share moments from their lives, and to engage with content creators

Users

- Individuals, as primary content consumers & creators
- Businesses - As Advertisers
- Publishers - Includes influencers and businesses that seek to organically generate audience and viewership for content

Features

- **News Feed** - Enable users to engage with content from friends and from publishers;

- Create New Post encourages you to post Live Videos, Photos
- **Stories** - Enable users to engage with ephemeral content from friends and family, and from businesses
- **Timeline** - Chronicles your posts over time, enabling it to be a single source of the history of your activities on Facebook
- **Facebook Rooms** - video calling experience integrated into Facebook
- Facebook Watch
- Portal - hardware device for video calling
- Marketplace - Discovering deals around you, currently used by sellers primarily to develop leads

WhatsApp

Value Proposition - Social media platform for individuals & groups to communicate instantaneously, and in a secure and privacy-centric manner

Users -

- Individuals, in Person to person communication
- Business users - Individuals who manage businesses, and register business accounts
- API-level users - Authorized to use WhatsApp APIs to send automated messages to users transacting (such as BookMyShow, Redbus)

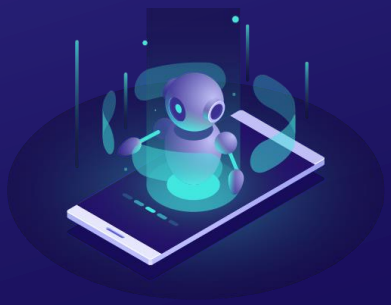
Features

- Stories
- Chats
- Individual Chats
- Group Chat
- WhatsApp for Web - enables users to communicate on PCs, so they can work on their systems & simultaneously engage in chat

- Reply feature
- Single Tick, Double tick, Blue Ticks
- “Forwarded” tick when forwarding a message

You can do a similar analysis for applications/products like Amazon, Netflix, YouTube, LinkedIn, iPhone, Swiggy, Amazon Alexa, FireTV stick, Echo, Kindle, Prime Video, Microsoft Bing, etc..

App Analysis



Kindle

App Choice	Kindle for desktop
Identify what problem the app solves for the user	• Cheaper and lighter substitute to physical books • Reducing time to access books once they have been identified online • Making e- reading and learning interactive • You do not necessarily purchase books - You can access large variety of free books - Integration of books downloaded from non-kindle sources
How does the app solve the problem? List three attributes / features that stand out	• Books from Kindle Store and PDFs/Mobi version of ebooks can be downloaded • In a single click, Kindle Store delivers the book within seconds to the app to be further downloaded in minimal time • Interactive and accessible learning ○ Colour coded highlights, flashcards/note taking feature- can be referred to, through the notebook feature

- changeable fonts, bg colours (3), viewing mode
- Notebook can be exported up to a limited % of the book contents

User Story – customer requirement

- As a student, I download course books as prescribed by school to study
- As a student, I want to highlight parts and take notes in the book itself to make sure I internalize what I'm reading
- As a student, I want to read further on specific topics mentioned in e-book because
  - E-book is not very clear on that topic
  - For in-depth understanding

  So, I google search the specifics
- As a student, I want to know what others find important in the book and exchange notes/flashcards with them
- As a regular reader, I want my ebooks to automatically appear at a single platform whenever I download/purchase it
- As a regular reader, ebooks should provide me a smooth reading experience so I can focus on reading only without having to adjust viewing mode

Customer journey – from opening to goal completion

- User Opens the kindle app
- User searches and clicks on the relevant title (from grid/list view)
- If it's a new book, the very start of the book is shown and if reading is in progress then, the page that is read the farthest

- appears
- User toggles through the contents and chooses the chapter to be read
- The user may want to tweak the viewing mode (full screen/ fit to window/fit to width) depending on the content she may be consuming. Eg. Fit to width for an accounting book and full screen for a marketing book that shows a diagram/process flow
    - For some types of books, there's an additional feature of column fit (single/double/auto) that is similar to EPUB format and enables better readability
- User highlights content using four colour options, that she finds important to go back again (prominence)
- While reading, she may also stumble upon examples from her own life/news that she would like to note down against the highlighted part and uses the 'Add note' feature to write it down
- For jargons that don't have a description in the book, user selects the term and clicks on search web/Wikipedia that takes her to the page in the browser
- User quits after reading for a while

Compare with competitors / alternatives

Calibre- eBook management

Pros

-Free, Open source ebook reader that supports 40 ebook formats and enables conversion to 18 output types

-Can edit the ebook. For eg. Table of Contents can be customized and text -content size, type, etc can be tweaked

-RSS feeds synchronized-> select newspapers can be auto downloaded and converted into epub file formats
-Well labelled controls buttons with tips that can be viewed once you hover over
-Books can be saved to disk
-Compatible with phones via wireless transfer of books

Cons

-There's no store/provision of book recommendations
-While there's a greater customization of fonts, colours and viewing mode- it is not intuitive- requires a lot of mental effort at the user's end to make changes.

- Adobe Digital - very easy to understand, simplest tool

	Key Decisions	Metrics
Acquisition – how to acquire new users	Collaboration with Bschools/and institutes	Increase in number of downloads
Engagement	Including a more 'visible' Flashcards feature - with a help guide on how to use it Enabling notebook share with other kindle users and peers	Number of minutes per session Number of minutes per day A/B Testing to see if there's any rise in usage of this feature If above is true, is there an increase in active time spent

Retention	Want people to return to Kindle app many times a day - positive reinforcement notifications e.g. You read 1 hour today/ Good job, you read 5 percent of Thinking Fast and Slow today! Blogs on how to read better auto downloaded in kindle app	Number of times you open Kindle app per day
Revenue – How to minimize costs	Recommendations on Kindle store Kindle unlimited adverts Audible purchase Advertising - Data analytics - the value of user data	Number of clicks on recommendations shown Number of actual purchases via recommendations Monetizing user's data to show

Problems	How new features solves this problem
In virtual learning mode, we refer to content in isolation rather than with a group of friends. Can learning be made as a social experience again	Problem 1: 1. Enabling social sharing of notes/highlighted content especially in a virtual academic setting where division of labour is difficult while preparing for notes/exams. 2. Provision of Audio Recording (by self/exchange with peers) of the summary at the end of each

	chapter/topics. In case of non-academic books, provision of hearing pre-recorded book reviews of friends before taking a call on whether or not to read it or just to understand others' interpretation of the book!
Bringing together different sources such as newspapers, magazines & books at one place for students	**Problem 2:** **Subscriptions:** Newspaper and journal subscriptions that can be auto downloaded at any given time in the app. For e.g. Calibre has RSS feeds synchronized with its platform and any given newspaper is auto downloaded and converted into epub file format
Enforceability: Is the kindle app motivating me to read not just during assignments but all year round?	Regular quizzes/reminders based on flashcards and highlighted material Collaborating with the author- can provide a basic quiz material which Kindle can use to shoot out questions time to time.

Duolingo

App Choice	Duolingo
Identify what problem the app solves for the user	• Need for a **free platform** to learn a new language • Need to keep up **motivation** while learning a new language • Need to make the learning process **easy and fun**
How does the app solve the problem? List three attributes / features that stand out	• **Free platform**: The app is **completely free** with some ads **for the entire length of the use journey.** Premium features just make the usability better but don't act as obstacles for completing the user journey. • **Motivation**: As soon as you achieve your first daily goal, Duolingo pushes you to commit to a 7-day streak goal. If you commit and miss your streak, you'd lose 50 gems, but if you maintain you'd get 100 gems back. A subtle use of **Endowment Effect/Loss Aversion** helps to ensure that the user engages and builds the habit of daily learning. • **Easy and Fun**: ◦ Duolingo uses **multiple stimuli** i.e. visuals, audio and writing to enhance the learning experience. It also gives a completely gamified experience with **variable rewards and leaderboards** which makes learning fun. Duolingo has divided the curriculum into skills, skills into levels and levels into **bite-sized chapters** that can be finished in a small sitting.

Customer journey – from opening to goal completion

1. Customer opens the app.
2. Experiences the core product offering without creating an account.
3. Creates an account.
4. Selects the languages he/she wants to learn and other attributes like daily practice goals, etc.
5. Customer keeps completing the lessons. Might opt for a premium model in case he/she feels the need to avoid ads and get offline access to the learning experience.
6. Customer completes all lessons.
7. Customer takes a certification test.

How could the goal completion be performed better

- Having actual conversations is a very important part of learning a new language which is missing from the app. Interactive sessions with native speakers can be added as a feature. One can earn one interactive session with a native speaker on completing one interactive session as a native speaker or with the use of gems.

Currently the certification test is charged at $20. Proficiency level wise certification tests can also be added to help authenticate progress till date in between the user journey. A new lower level certification feature without the test can also be added for free to make the goal completion look more authentic on completing all lessons.

Compare with competitors /

- Most competing apps are more expensive than duolingo as they don’t have a freemium model which gives as many features as

- duolingo for free. Even the premium features in Duolingo don't act as barriers for the user journey to be completed. The USP of duolingo therefore lies in it being cheap which it should continue maintaining.
- But competing apps like Babble, Mango Languages and Lingvist are better at support and better at meeting requirements. Duolingo can think of redesigning certains aspects of the learning journey or add some additional features to make the learning outcome on the app more effective.
- Duolingo supports more languages (34) than competitors like Babel (14)

	Key Decisions	Metrics
Acquisition – how to acquire new users	The first screen you see has a very crisp value proposition saying "Learn a language for free. Forever". A clear call-to-action helps the user get interested. An improvement to this could be to add statistics on how many people have mastered languages on the app and what is the average time required for the same which would imbibe greater confidence in the user towards the app. The app allows the user to explore the core product offering after 5 clicks while	Conversion ratio of number of app downloads to profile creation

	delaying the signup process. Duolingo excels at this. Taking a user to the core product as soon as possible is very critical, and Duolingo excels at it.	
Engagement	• The watching ads part makes the user engagement experience poor. This could be improved by showing ads in the language that the user is learning. • A section of curated content on interesting videos, blogs and songs in the language being learnt can be added. • A quiz battle feature can be added where you can play games with friends or other people online. • Allow a leader board creation feature with friends. Currently leader boards are level wise and created with random users.	Average amount of time spent per user on the app in a day
Retention	• Duolingo can ask the user to customise a specific time of the day for practice on the app, making it a part of a routine for	Number of inactive and discontinued users. Number of uninstalls.

	• the user. This way duolingo can improve timing of the reminder notifications being sent to the user. • The notifications can also be customised for every user and have interesting data and trivia which would help catch the attention of the user better. • There can be a flashcard section that auto-updates each time a user is introduced to a new word. Something for users to go back to and review standalone vocabulary without having to retake lessons or exams or stories, thus improving the usability.	
Revenue	• **Selling merchandise**: Let points earned be used for buying merchandise at discounted rates. • **Sell a market trends newsletter:** Survey the users monthly (or at least the ones who opt to be part of this program) and compile the results into a report which is then marketed on a subscription basis to those interested in the views of language learners.	Revenue generated from these new revenue streams

Amazon Alexa

Customer value proposition	Smart speakers connected to intelligent Voice Assistant Alexa, providing voice interaction and various audio services such as music, podcasts & news
Features	• Functionalities can be extended through Alexa skills provided by third party vendors • Can be activated by wake words (such as Alexa or Amazon) • Speech recognition service
Customer personas	• Elderly users in metro cities, living in joint families - looking to connect with extended family, and various services around them • Adults with young families, with different music preferences for every member • Fitness enthusiasts, using Alexa to moderate fitness activities
Pain points	Elderly users • Difficult to remember words for different skills • May not be able to troubleshoot if there is interpretation error • May need quick call functionality for family & caregivers in close proximity for any urgent help • Would have similar set of activities that they need to perform every time

Suggested Improvements

- Enable SOS feature on Echoes connected inside a home, to enable elderly to call out for urgent help
- Skills names should be prefixed Alexa and then the task - for example, to listen to News, the wakeword should be *Alexa News*
- Can have a feature that tells users about skills they can use when they just say Hello Alexa - kind of like a Tip of the Day
- Have a Stop word that Elderly Users can use to undo the previous Alexa request

Google Fit

Customer value proposition	Google Fit is a fitness tracking app that tracks movements to provide users a gist of their activity throughout the day and the calories they burnt. The app can track from your phone or your watch.
User personas & pain points	Working professionals who want to maintain or reduce weight, but are short on time **Pain Points:** -Needs regular reminders to put in maintenance amount of workout -Can't keep track of progress -Set Goals to put in the minimal amount of workout for a day Young population, looking to build muscle and get fit **Pain Points:** -Need the total output summary of workout -Need a summary of his activities throughout the day -Sleep and Calorie nutrient tracker -Needs to track his progress throughout his fitness journey Homemaker who wants to reduce weight, and track his health **Pain Points:** -Tracker to update regular health details, like Blood Sugar, Blood Pressure etc. -Track sleep and outdoor activity to maintain a regular amount of physical activity -Prefer digital alternatives of newspapers
Features	**1) Activity Summary-** -Basic overview of Activity for a day is available on the home screen

itself

-Users get to see their total step, distance covered, calories burnt and total movement time in one place.

-A dashboard indicating how the user has performed on the goals they set

-Detailed summary of each overview component is available on a day-to-day basis on the homepage.

2) Activity Journal-

-Detail of all activity throughout a day is available. Clicking each activity shows you the break-up of what happened for each activity

-Journal has entries for all the other days, so that one can track where they lagged behind for a particular day or check how much calorie was burnt doing a certain activity

3) Track Workout and Log Activity-

-Option to log an activity after working out or doing an outdoor activity.

-The details while tracking are gathered from either a connected smartwatch or from the type of activity and phone's sensors like gyroscope, accelerometer etc.

4) Track Vitals, Sleep, Body Measurements, Nutrition-

-Users can track their vitals from a connected device, or manually provide the data

-Tracking regularly gives a better picture of improvement while using the app

Product Improvement

a) App does not have food recipes, and more importantly, a catalogue of calories consumed per item.

Solution:

Database of calories for food items and dishes, as well as range of measurements- Providing an option to choose the recipe/item consumed and accordingly suggesting the calories consumed keeps track of calories more accurately. A unit of measurement, like bowl, gram, cups, etc to cater to the amount consumed, and identify total calories.

b) Set a specific goal, like Weight Loss, Muscle Gain, and suggest accordingly

Solution:

Set Goals- An option to set specific goal which user wants to attain.

c) Recommendations and alerts based on goal set by the user

Solution:

Customized Recommendation- Based on the goal set by the user, and the body measurements and vitals data provided, the app should suggest the optimum calorie one should eat, how much calorie one should burn, water consumption alerts and other related stuff.

d) Counting macros when following a particular diet plan

Solution:

Detailed breakdown of nutrients consumed- A detailed list of macros in a particular food can be provided. This will be particularly helpful when a person is following a diet plan, like a Keto diet, or a diet chart based on Macros consumed.

e) Unable to follow and compete against people/friends with same

goal. Leads to a drop in motivation.
Solution:

Add Friends – Allow adding friends from Google contacts, to track progress.
Leaderboard feature- Show a leaderboard among friends to indicate progress and weekly leaderboard to induce a healthy competition to stay fit.
Objective: Improved User Experience and User Engagement

f) Can't find appropriate gym or at-home workouts when targeting a particular muscle group
Solution:

Workout suggestions- When user enters a certain muscle group they want to target, the app can suggest workouts to follow with different variations and a short clip demonstrating the workout.

g) New engagements to inspire people to use the app more and be more active. Host Branded events.
Solution:

We can organize events related to some physical activities, organized by brands. This will motivate users to participate in order to compete.
Objective: Revenue generation and User Engagement

Prioritization Matrix

Name of the Feature	Impact	Effort	Priority
Database of calories for food items and dishes, as well as range of measurements	High	Low	P1
Set Goals	High	Low	P1
Customized Recommendation	High	Medium	P1
Detailed breakdown of nutrients consumed	High	Medium	P1
Add Friends & Leaderboard Feature	Medium	Medium	P2
Workout suggestions	Low	Medium	P3
Branded event hosting	Medium	Medium	P2

Microsoft Teams

Customer value proposition

- Hub for teamwork
- Brings together everything a team needs - chats, video conferencing, content collaboration
- Integration with Microsoft 365 applications,
- Ability to create & integrate apps and workflows used by businesses

Features

- Chat
- Video conferencing
- File sharing
- Integration with Microsoft365 - High customization

Customer personas

- *Manager at MNC based out Mumbai* - Office worker - integration with Office365 users - extensively use Office applications for professional collaboration
- *Student at a premier B-School* - Education -
    - Competitive context- Currently largely using Google Suite,

File sharing prefer Google Drive - Docs, Sheets & Slides for their ease of use

Suggested Improvements

Product Vision - Making Microsoft Teams the destination where students at B-Schools collaborate with each other

For Education users -

- Team Collaboration - Combining power of customization of

Office365 with ease-of-use and templates available in Teams

- Connecting - text chat
  - Ability to form groups for Placements, Studies & for classes
- Connecting – video

Enable users to create Video Rooms for themselves that they can use anytime, instead of having to generate meeting rooms each time; Privacy should be configurable

LinkedIn

Customer value proposition	• Build your professional brand • Grow and manage your network of professionals • Share information that can help others in the network
Features	• job recommendations to job seekers/professionals looking for a switch and talent search tool for recruiters • Sales navigator tool for lead generation for sales professionals • Ability to create articles and in essence have your own blog on LinkedIn itself
Customer personas	• Social Networker: Professionals looking to network, maintain business relationships seek jobs etc • Salesperson: Sales professionals looking for new sales opportunities • Passive content consumer: They just like to scroll through updates from network, news items etc • LinkedIn Recruiters: they connect with professionals who might have relevant skillsset etc
Problem	Roughly 62.5% of LinkedIn's revenue is based on the premise that users keep their profile updated. If companies are posting jobs and LinkedIn's users are not updating, LinkedIn could face customer problems

Suggested Improvements

For social networker:

User Story#1: As a LinkedIn User, I want to meet new people so that I can increase my number of connections.

- Group chats for networking with, say conference group
    - The user registers as a user and logged-in to access LinkedIn Groups.
    - The user picks a LinkedIn Group to join.
    - The user must be approved to join the group by a group moderator (someone who started the group in the first place).
    - Once approved, the user will have access to the group and its various channels.
    - Channels are set up by the moderator, the user can choose which ones sound interesting. For example, having a #news channel or #the general channel will segment out content in their respective areas.
    - The user selects a few channels to join and starts to listen to new people and their stories.
    - The user can add these people within the group as personal connections.
    - The user can directly message their new connection once connected.

For LinkedIn premium: Weekly analytics

Zerodha

App Choice	• Zerodha • Offers 3 main products - Zerodha Kite, Zerodha Coin, Zerodha Varsity
Vision	To create a brokerage-free world & benefit the retail investors when it comes to financial markets and to provide the best possible customer experience through highly personalized service and support.
Tech stack used	HTML 5, Elk, Kafka, VueJs, Java, Docker, APIs
Identify what problem the app solves for the user	• Convenience - Displays all investments and their performance in a **single interface**. Acts as a Link between the investor and stock market for giving a **simple buying and selling experience** to its users. • Cost effective & faster transaction- Charges **low commission** from the traders, has various methods of online payments that facilitate the **near-instant transfer of funds** between accounts ultimately reducing the trade barriers. • Better understanding - Helps new traders and investors to **learn basics**. • Great control over investments - Investors can **review all their options and monitor their investments on their own,** without waiting for a broker to tell them what their best recourse is. Since investors can make decisions without external interference, they gain more control of their investments.

How does the app solve the problem? List 3 attributes / features to stand out

- Kite allows customers to trade from anywhere, gives deep insights and provides AI recommendations. With full market watch, extensive charts with 100+ indicators and advance order type, Kite is loaded with features required by traders.
- Kite Connect enables users to gain programmatic access to data such as funds information, positions, live quotes so users can build risk models, stock selection models, and do back testing.
- Zerodha Varsity provides stock market education to enthusiastic investors. It is multilingual and is available in 11 regional languages.
- Kite web has a responsive design which makes it work nicely on a mobile web browser. Fund transfer and withdrawal is done in a single click.
- The feature Good Till Triggered (GTT) allows placing an order to be sent to the exchange only when the price condition is met.

User Story – Customer requirement

- As a user, I want to keep up with the real time changes without spending much time following all stock markets.
- As a user, I want to have greater control on my portfolio and customize whenever needed.
- As a user, I would like to learn more through hands-on practical learning.
- As a user, I want to know the current performance of my investments.
- As a user, I want to get favourable recommendations for my portfolio supported with facts and news articles.

Customer journey from registration to goal completion

- User opens the Zerodha Kite app and creates an account.
- Zerodha has 2-factor authentication first the user logins with username, password then second with a 6 digit pin/face recognition/fingerprint.
- User selects the market watch option to search and add required stocks
- In market watch, users can also sort the stocks based on exchange, alphabet or last traded price.
- There is a snapshot option where the user can click on the required stock and search for vital information about the stock. Snapshot feature also has options like fundamentals (with news, events, financial ratios regarding the app), Alerts (for users to set rule based events), technical (gives users the bullish, bearish probability of stocks), Stock reports +(gives comprehensive report on stocks, markets, sectors).
- After getting all the information about the stock, the user selects the order option to choose quantity, order type to buy/sell the stock.
- To complete the payment, the user goes to funds in the account section and adds an amount either through UPI, Gpay, Netbanking. User then uses these funds to place and order.
- Users can now see the portfolio with holdings and Console (the backend app) which has breakdown of all trades, profit and loss, tax reports.
- The user exits the application when they are done placing a deal, and checks the portfolio.

Compare with competitors / alternatives

Upstox

Pros

- Zerodha is safe for investment because it is a zero debt company and doesn't do proprietary trading with client funds.
- It provides an in-depth learning experience with excellent support forums, self-serving portals and education initiatives.
- Provides the facility to make direct IPO, FPO investments.

Cons

- Zerodha backend that is console is not integrated with its trading platform. The data in the back office gets updated overnight.
- Some customers face delays and service down during trading hours.
- Does not have free demat account opening and good deals like free brokerage credit like in Upstox.
- Guest login feature is not available. (available in upstox).
- Only 5 watchlists with maximum up to 50 stocks in each watchlist. (unlimited in upstox).

Suggested improvements

- Can improve the login page by adding guest login for new users to check the platform.
- Solved the problem of accessibility and availability, next need to find a way to help win markets with best strategies available to users.
- Covers only Indian market, can extend to other markets also

	Key Decisions	Metrics
Acquisition – how to acquire new users	It markets using word of mouth and partner referral	New users added in this month /Total users till the previous month Percentage of referrals used
Engagement – how to provide a great experience to customers	• Offers products like Kite, Coin, Varsity and is very agile, fast and smart with user friendly portal. • Provides stock market education to enthusiastic investors (both video and text format). • It is multilingual and is available in 11 regional languages.	• Number of trades per day • Number of users opting for regional linguistic support • Average time spent by users on app • Average time spent on subset products • No. of clicks and time spent on educational material
Retention – How to bring the users back repeatedly to use the app	• It has the best customer support, and is safe for investment because it is a zero debt company and doesn't do proprietary trading with client funds.	• Number of logins by registered users in a month • User logins during new recommendations/ Total logins in the period of time

	• Upskill and reskill users on the stock market domain, get updated with the latest market information.	
Revenue – How to make more money from users	• Discount broking, low commission charges - **Low margin and high-volume model** • Zerodha offers low-cost trading services. • The low brokerage brings down the break-even point for trades. This increases the frequency of trading and overall revenue. • Zerodha offers all its services online, with nearly no physical presence. This reduces the operational cost and helps them scale-up the business easily.	• Revenue earned per day • Revenue from brokerage charges/Total revenue • Total revenue/Total number of customers

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Behavioural Questions



Behavioural Questions

These are situational questions that you face in more or less in all interviews, but it is one of the most neglected parts of the prep. However, this should not be the case, as they are an integral section in interviews. You should prepare for these questions, and to answer these questions effectively, we have a few frameworks that we can follow, namely:

STAR Framework

S – Situation

T – Task

A – Action

R – Result

Using this framework, you can effectively communicate some of your real-life experiences in an effective manner. Let's look at an example of how we can utilise this in our answers:

Q: Tell me about the biggest challenge you faced, and how did you overcome it.

Ans:

- **Lay out the 'situation'**

  When asked a question, lay out the situation you are referring to, without including unnecessary details. Say, you want to refer to a difficult client, it is not needed to include how you onboarded the client. Elaborate about how you got stuck in the execution of client request, why you were unable to proceed.
- **Highlight the 'task'**

Next, you move on to show what was your goal/task for the particular example you took. Here, you answer questions like, what was your responsibility? What was the challenge you were facing? How were you planning to solve them? Were there any trade-offs for the plan of action? Was there any alternative solution?

- **Share how you took 'action'**

  Now that the interviewer knows the situation and what your role was, it is time to show what you did in the situation. Here, you should not '*globe',* and rather, you should answer it directly, like 'I started by doing ..'. Here you show your execution of what you planned, and if something unexpected came up.

- **Share the 'result'**

  After having gone through the whole situation, now is the time to share the outcome of the action you took. Here, you can also a few questions from the interviewer regarding some alternate plan of action. Think ahead for any such scenario, so that you are not caught off guard.

There are a few other frameworks to answer these questions, like:

SOAR framework

S – Situation

O – Obstacles

A – Actions

R – Results

Here, like STAR framework, you have almost the same criterion, except that you list the obstacles instead of task. You can choose this framework when you are faced with a situation where you had multiple obstacles to overcome.

CAR framework

C – Challenge

A – Action

R – Results

It's a similar framework, where you start by listing the situation and what were the challenges in the situation, before going on to show the Action and Result of your action. All of these frameworks are extremely useful when answering behavioural questions, like "Tell me a situation when..", "Tell me about a time when..".

Examples of some Behavioural Questions:

1. What kind of conflicts have you faced in the past?
2. What kind of changes have you installed in your organization?
3. Tell me about a time you had a conflict with someone. How did you resolve it and what did you learn?
4. Tell me about a time you had to decide to make short-term sacrifices for long-term gains. (https://www.tryexponent.com/questions/653/time-make-decision-short-term-sacrifices-long-term-gains)
5. How do you sell an idea to senior management? If you use slides, what would the content include.
6. What was your biggest failure in life?

"**Mistake**" questions are all about:

1. Showing your self-awareness
2. Willingness to learn and receive feedback
3. Explain how these learnings affected your future behaviours

Additional Readings



What is Ad Tech?

Ad tech (short for advertising technology) is the umbrella term for the software and tools that help agencies and brands target, deliver, and analyse their digital advertising efforts. Ad tech is designed to help advertisers make better use of their budgets. Ad tech methodologies deliver the right content at the right time to the right consumers, so there's less wasteful spending.

Why Ad Tech needs to be data driven?

- No real user feedback about ads
- Ad inventory is perishable, if a user visits a webpage and the ad agency fails to show an ad for whatever reason, that impression is gone. A lost impression can never be monetized again.
- Trade-offs are simple, for example trade-off between revenue and performance, an algorithm which increases revenue but at the same time reduce performance. However, supply chains used in the Ad Tech ecosystem are complex in nature. They involve a lot of stakeholders.
- Measuring all outcomes is difficult and challenging
- Involves a lot of customizations and lack on standards
- Full of risks, uncertainties and ambiguities about outcomes, metrics and priorities

Example:

Problem statement: Imagine that you are playing a lottery and if you win, the prize is another

lottery ticket. Consider the problem to be a chain of such lottery tickets and as you iterate, you unearth new lotteries whose outcomes are not clear.

What will you find in such situations?

1. Rationalizing the problem statement at hand is extremely hard.
2. At times, the behaviour of different metrics or measures is unpredictable and non-linear in nature.
3. Since there are multiple agents in play, their behaviour is not dictated by the system. They are self-motivated to work towards their own incentives.
4. Also, all the agents in the supply chain are extremely intelligent and adaptive. They learn as they gain experience and change behaviour over time

With respect to several ambiguities like which features to build, which metrics to chase, how do PMs usually take product decisions:

1. Opinions given by senior management/ peers
2. Based on intuition, lacking validation
3. Based on what a few customers say
4. Looking at a high-level aggregated metric or data set
5. Actionable insights derived out of large sets of data

Let us look at two simple examples to understand how data can help the PM make the right decision:

Example 1: Ambiguity about Metrics and Priorities

Let's say there's an e-learning platform where the platform success metric is 'Courses per year'. To drive this success metric, you have been given 2 workflows:

- Workflow A: Signup Flow, where you are getting new user signup and enrol to courses. Key Metric here is 'Signup Conversion Rate' which is currently around 20%
- Workflow B: Course Completion Flow, where you get the existing user to subscribe or

- enrol to new courses post completion of a course. Key Metric here is 'Conversion rate post course completion' which is currently around 30%

Which workflow will you prioritize:

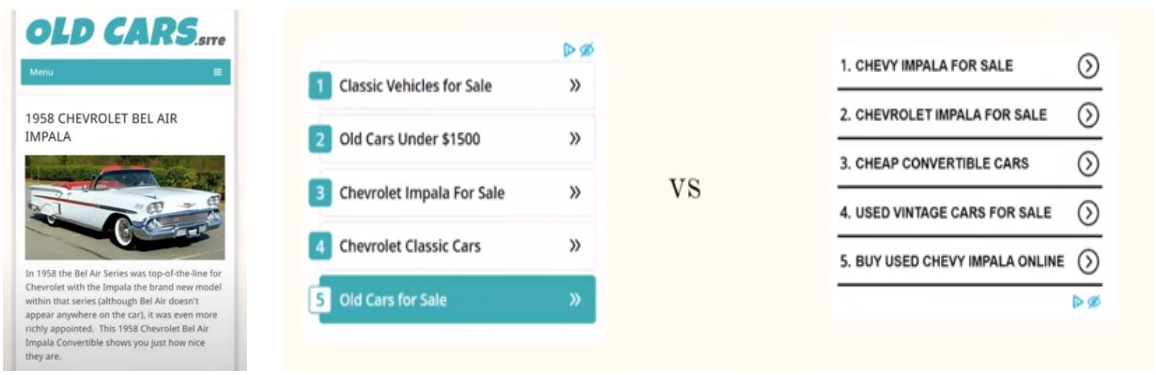
By intuition, you will prioritize Signup Flow as the metric is not performing well (20% success rate) and also it is bringing new users into the system, thus increasing the total lifetime value. Then you will start building features to improve the workflow based on your own ideas and feedbacks from your peers and users. But what you don't realize is that you might be chasing a metric that is actually incomplete. Suppose here the data says the Signup conversion rate for real users is 60%, in that case the success metric used in the workflow is not that terrible if you just account for valid signups! And hence improving it would just account for a minor upgradation. Thus, data can be very relevant in such cases to make the right decisions and prioritizations:

- Prioritized Feature – A simple robotic activity check feature during signup
- Prioritized Focus – To shift to Course Completion Flow
- Resolve metric ambiguity – Signup conversion to Valid signup conversion

<u>Example 2:</u> Ambiguity about outcomes

Let's say you have 2 ads and you need to decide which one of them gives better CTR or Click Through Rate:

Intuition says to go with the first ad as the structure is friendly and might perform better because of the following reasons: Native look and feel, better user experience, seemingly logical continuation of the content.



Using data to take decisions:

- Iterations can be numerous but small and measurable
- Adapt with each iteration
- Tend towards more favorable outcomes
- Do not hesitate to go against the user's opinion if data/experiments do not backup the same. Instead use user's opinion as the starting hypothesis

What to measure: Deciding KPIs/ vital metrics

1. Understand user journey:

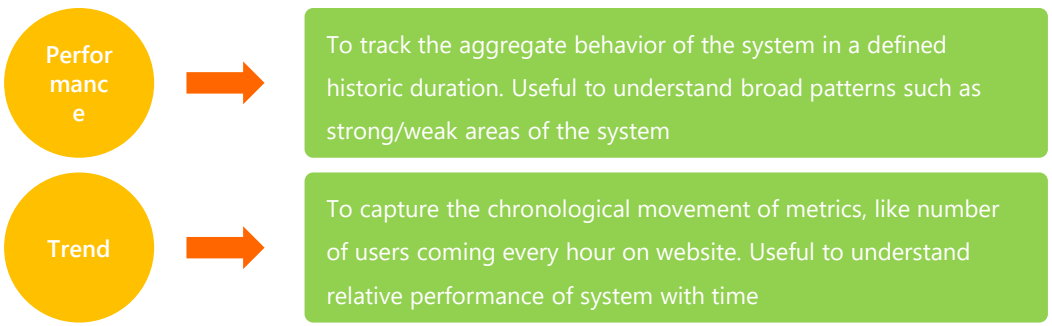
- 5 E framework – Entice, Enter, Engage, Exit, Extend
- Understand objective in each
- Map objectives to metrics

1. Feedback loops which can impact performance:

Example – While finding the cheapest supply path to a good:

- Cheap supply leads to more successful transactions
- More successful transactions lead to better performance
- Better performance leads to higher paying propensity by buyers
- Suppliers increase prices attributing to increased paying propensity
- Fewer successful transactions due to increased price, resulting in poorer performance
- Paying propensity decreases

What to measure: Trends vs Performance



Tools that can be used when doing data analysis in a product management environment are as follows:

- Pixel based logging – Google Analytics
- Data management and visualization – Tableau, Power BI, Google Data Studio
- Alert Management – PagerDuty (also Opsgenie, VictorOps)

A/B Testing in Product Development

What is A/B Testing?

A/B testing is a method of comparing two versions of a variant, which can be screen, CTA, product, process flow, webpage or app, against each other to determine which one performs better for the users. It tests one experience with another and can be performed for small or medium changes or even big redesigns.

How to do A/B Testing?

- Pickup similar audience cohort for both the variants and distribute them randomly across the versions.
- More than 2 variants or multiple experiences testing is also possible and known as A/B..N Testing.
- Only one parameter is usually changed in 2-variant A/B Testing as compared to Multivariate Testing which involves multiple changes in parameters. In such tests, it sometimes becomes difficult to predict as to which particular change led to the version being preferred more than the others. But many-a-times due to paucity of time and lack of resources people do multivariate testing with a leap of faith.



Why do PMs do A/B Testing:

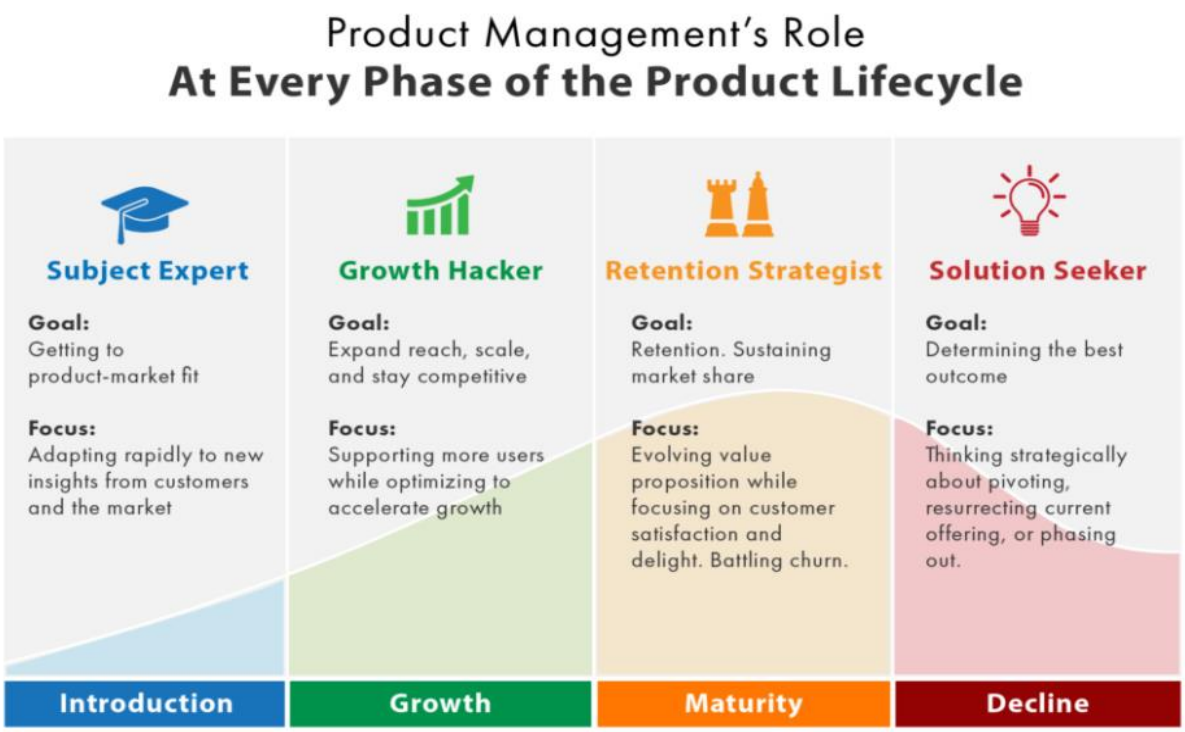
- To validate hypothesis
- For Iterative Product Development
- For Data driven decisions

A/B Testing Life Cycle:

- Define a goal that you are trying to optimize for
- Define a success metric of that goal
- Develop a hypothesis – Figure out where the friction in the flows for the users are. Variables are metrics need to be clearly defined
- Set up Experiment
- Run Experiment
- Analyze Results – Look for significance of the data (eg. T-test), consider possibility of burn-in effect (happens usually for new notifications which interests users for the first few times, but if the same notification appears repeatedly, the user may not be interested)

Example – Hypothesis is that a faster website will get more users in. You will have to design the experiment like for how long will you run the test, what will be your target population and how frequently that population comes back.

Depending on the state of the product, A/B Testing can run for a couple of days or a couple of weeks depending on how matured the product is already and what is the goal. It is a continuous process and should be done on a frequent basis as the number and type of users interacting with the product or visiting the website is not constant and changing regularly.



Limitations of A/B Testing:

- Does not tell why users preferred one version more than the other and not always conclusive
- In case the population is small and the users are not frequent users, we might have to keep running the test for a longer time in order to get significant result

Case Study – HealthifyMe SignUp:

1. Objective: Increase Signup rates
2. Metric: Signup Conversion
3. Hypothesis: Phone number signups may increase conversions
1. Setup: Make phone number signup as defult for new users
1. Run experiment
2. Results: 20% increase in signup converison

B2B Product Management & User Research

Why B2B User Studies are different?

1. End users of the product are completely different - Its difficult to put yourself, as a PM, in the shoes of a customer in B2B settings where there are multiple things happening at the same time. Also, you are not the one who is using the product on a day to day basis, so it becomes very difficult to empathize and put on the product manager's hat and understand the problem from customer perspective. Hence, talking to users becomes mandatory.
2. There are multiple stakeholders involved – A purchase decision in B2B setting is collectively made by different teams, unlike B2C where there are comparatively lesser stakeholders, hence user study becomes even more important.
3. Feature Requests – Easy to fall into the trap of asking the customer what they want.
4. Secondary Research – There's a lot of secondary research available in a B2B setting which is a miss in a B2C setting. Can help focus in asking the right questions to users.

Objectives of a good user study:

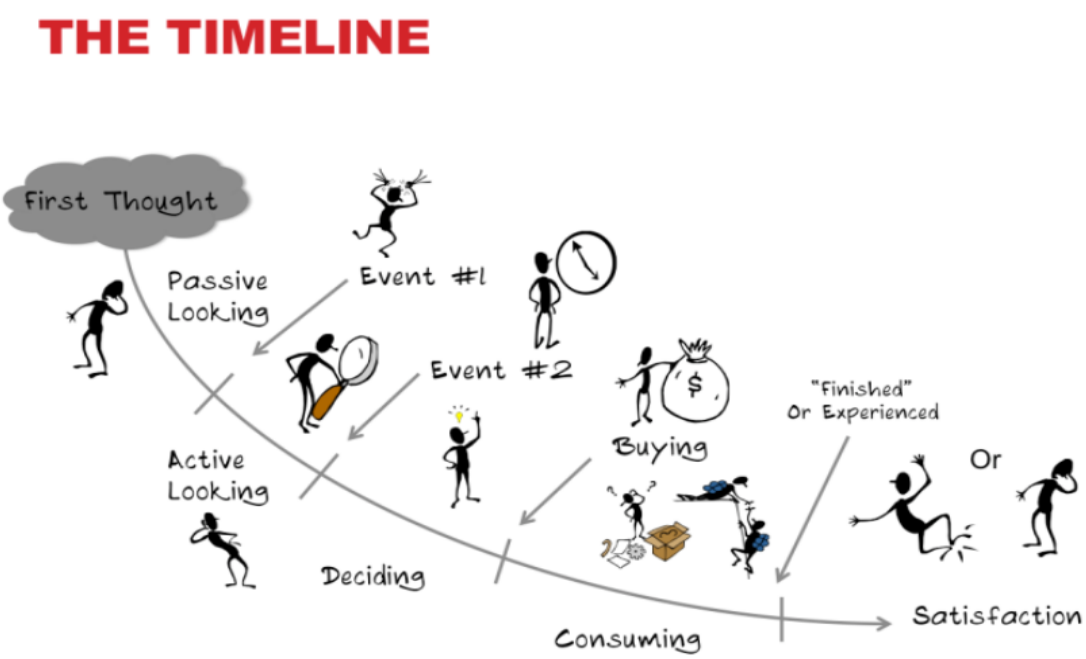
A good user study should be able to understand and implement the four key drivers of business:

1. **Market** – How many users are there in this market?
2. **Product** – What problems does my user have?
3. **Pricing** – How much will they be willing to pay?
4. **Channel** – Where can I find my users?

Principles of doing a good user study:

- **Research online before talking to users** – Start with secondary research. It will help you look for what end users search. It will also help you as a PM to map out the market and quickly drive the insights before even starting your customer study process. It can help you achieve a faster reach. Secondary research materials can include case studies, user reviews for similar products and competitors, forum discussions, sureys and results.

- **Identify and talk to the right users** – Key drivers of recruiting a user:
  - **Users who have recently started or stopped using my platform** – Objective is to analyze their switching behavior, why did they switch from one platform to another, for example, switch from field events to webinars
  - **Outlier users who use a lot** – for example, an user visiting my platform 4 times a month has recently started visiting it everyday. Such users are called outlier users or extreme users
  - **Users who don't use my platform** – The biggest example can be a person who is visiting my website, but is actually not buying any product
  - **Users of different types** – The different segments of people you would want to go after who might have varied purposes behind using your product
- **Talk about specific stories:**
  - A good story is about the user and not about the product. It should talk about an incident, and not inolve a generic discussion. For example, "I bought 3 books last month" is a good story and more specific compared to "I usually buy a lot of books".
- **Each story has a timeline:**



Example – Assume the problem statement is the organizers wanting to understand how to get more people to talk in CoLAB PM talk. Let's say they approach a speaker who has been giving this talk for the last 3 years, as he would be the right candidate to help the organizers identify more such speakers. The person would start with his reponse to the question "How did he end up as a speaker" as follows:

- First Thought – When he heard a colleague giving a similar talk
- Event 1 – During placement interviews, the coordinator told him that he should come back to campus and give the batch a talk
- Event 2 – The HR of his company coming to him and asking him whether he would be interested to give a talk in his MBA campus

So, knowing this information would help the organizers reach out to the right avenues like placement committee of colleges , company HRs, etc. who could help them in getting speakers across multiple domains. If the organizers would have instead asked the first speaker a generic question, they would have never been able to get this valuable information in the first place. That's the use of mapping story to a timeline.

- **Look for key decisions in the timeline:**

For the same example look for – How will the organizers get more people to talk to, what were the alternative talks that the speaker wanted to do, what did he like in each of the alternative and what was his preference for choice of topic.

Case Study:

BrowserStack is a cross-browser testing company for developers wanting to test their websites on multiple browsers like Chrome, Safari, etc. The company, few years back, had launched a new platform to test mobile devices.

Problem Statement for BrowserStack was – How to get more users to test on mobile devices?

How did they go about doing it:

1. Secondary Research – Market reports, Hacker News (a forum used by many developers), articles written by end users, engineering blogs of Airbnb and other companies. Lot of insights like how do people choose their devices, came from secondary research.
2. Identified the right users – Multiple segmentations were done as follows:

- Freelancers, Startups and Enterprises
- Users who had recently started testing on mobile devices
- Users who tested a lot on mobile devices
- Users who weren't earlier testing on mobile devices, but had suddenly started doing so

3. Prepared the right questions to ask – Why did they suddenly started testing on mobile devices? Why these specific devices? When did they first think of testing on mobile devices? Questions like these can be orchestrated and mapped over the timeline.

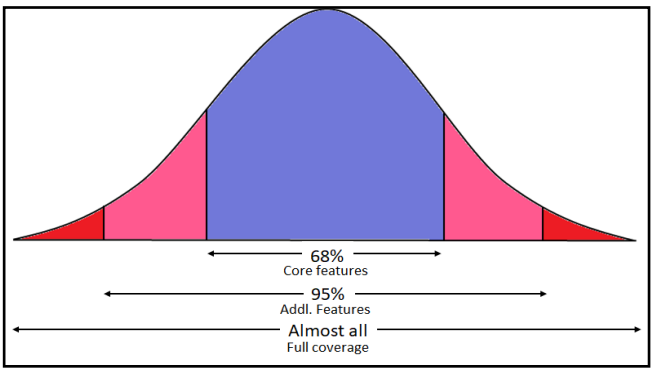
What was found:

- First thought – During development, while the user was showing it to a colleague some misalignments were observed
- Event 1 – Looking at the Google Analytics data, when he found out that a lot of his customers come from mobile devices
- Event 2 – When the Support team filed a bug from a top customer and asked for a deadline on when the bug would be fixed on an Iphone.

It was only after these events that the user decided to evaluate multiple tools and started testing mobile devices on BrowserStack. Hence, the user study helped BrowserStack to gain significant insights and find a solution to their problem statement.

MVP to Full product Rollout: Case Study at Ola

MVP vs Full Rollout



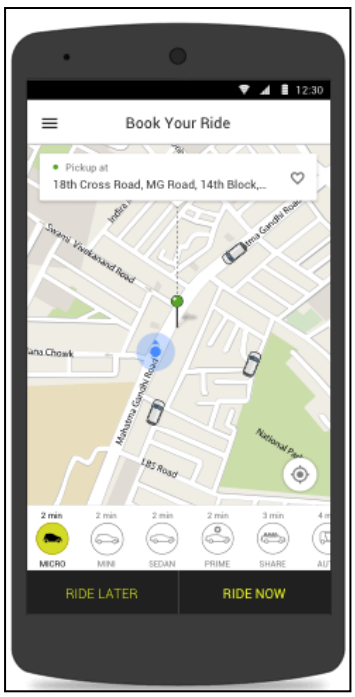
When it comes to planning of new features, MVP (most viable product) or the first rollout will be addressing the main set of features which comes around to roughly 68% and further use cases are addressed with subsequent phases of the rollout.

Feature: Ola Queued Allotment

What is this feature?

If you recall coming across a message that popped up on your screen that said *"**Your driver is completing a trip nearby and will reach you shortly**"*, then you probably have encountered that feature.

ontext (2016-17):



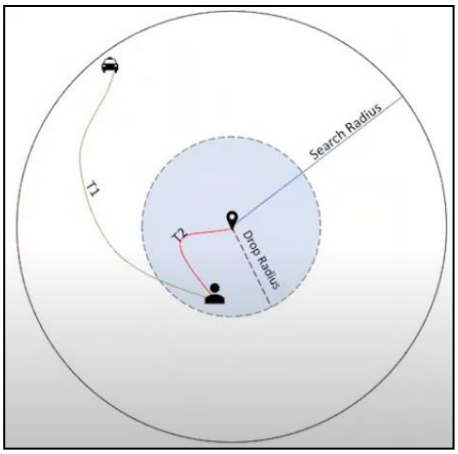
- No route marker indicating the route the cab is going to take
- Primitive Google maps and App technology

More importantly, if one booked a ride in Ola (or Uber) at the time, you would get an empty cab.

Objectives of feature introduction:

- Improve driver utilization
    - 46% utilized: Time spent ferrying the rider
    - 46% idle: Drivers did not get a ride and they were waiting or cruising around
    - 7% "Dry-Run": Driver is allotted a ride, but the cab is traveling empty to the location. On an average, this distance was 2-2.1km

- 1% utilization improvement -> ~2% reduction in cash burn
- Reduce Dry-Run



Queued Allotment Algorithm

- Larger circle is the "Search Radius" (~2km) and the smaller one is called the "Drop Radius" (~0.5km)
- The center of the circles represents the location of the new user who is requesting a ride
- The objective of the algorithm is to search for a cab in "Search Radius" (which is currently with a passenger) that will be dropping off a customer in "Drop radius"

- The dry run in this case is reduced significantly and will be somewhere close to the drop radius
- The total time promised to the customer would be T1+T2+ buffer (to account for cash payments)
- Metrics:
- Success metrics: Dry run should reduce and subsequently, utilization should improve
- Control metric: Cancellations should not increase significantly as compared to the general trend

Phase I: Apply algorithm without any UX for customer/driver

- Started in 2 locations in Bangalore with high demand
- The hypothesis that a car is dropping off a customer in the vicinity of the new passenger was verified since several allotments per day

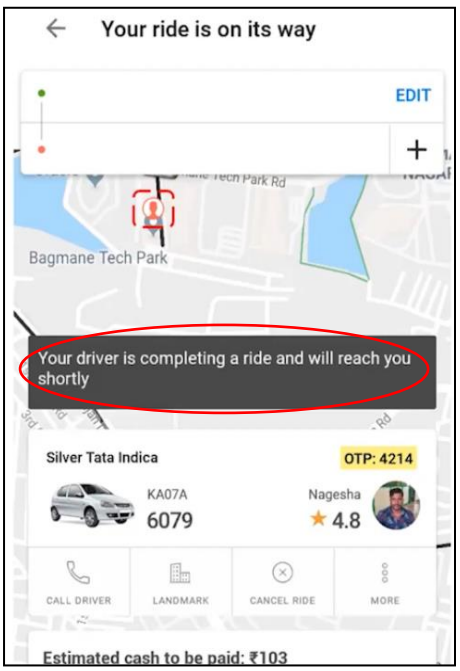
Problems faced:

- Customers would book the cab and notice that the cab is not moving in their direction ultimately resulting in cancellation.
- Since this feature was simply turned on in the backend, the customers would be shown only the updated ETA. However, customers would notice the unusually high ETA for such a short distance and would end up cancelling.
- The drivers were also new to this experience and would inform the customers that they are already carrying another customer. This would again result in cancellation either by the driver or the customer
- A few peculiar cases were observed where in the driver abandoned the current customer since he received the information of the next ride

Takeaways from Phase I: The underlying algorithm worked well although the control metric failed

Phase II: Add UX for customer

Initially, a Toast Message (a message or a notification that pops up for some time and then disappears) was added for customer as soon as the ride gets allotted



Issues:

Customers do not monitor screen after selecting the "book ride" option. As a result, they would miss the message and situation reverts to phase I

Modification: A route marker to be added along with the Toast Message

- Adding a polyline at that point in time (2016-17) posed an engineering problem
- As an alternative to that, a marker was added that

- would indicate the intermediate drop location. The Toast Message would still appear for 3 seconds and then disappear.
- The customers could tap on the marker which would re-display the Toast Message
- The customers *__assumed__* that the cab would take longer to reach and end up cancelling the ride (Ola had found that any ETA more than 9 minutes would cause a cancellation, and this was based on the historic trend of Ola under-committing the ETA)
- Polyline was added later to display the route that the cab will be taking indicating an intermediate stop

Phase III: Add UX for driver

Problems addressed:

- The driver was not cognizant of the queued allotment and the display screen would only bear the details of the current ride.
- Drivers would log off after their ongoing trip was completed and would not be aware of the trip that was allotted after that.

Feature modification:

- Toast notification added notifying the driver regarding the newly allotted ride
    - **Issue**: Likely that the driver missed the toast notification
- Persistent push notification that the next ride has been allotted. This would appear at the top of the screen so that the driver is always notified about the next ride

Phase IV: Bugs and extensibility

- There was no way for the driver to decline the ride. Ola had a ride decline mechanism and the drivers would be compelled to accept the ride. The drivers were allowed only 2-3 declines a day

    - Contrary to this, Uber had an "accept" model; drivers who are willing would accept the ride and declining would not result in penalty

- No downtime between rides
    - Drivers were unable to take any break (lunch break, etc.) or refuel

Further modifications:

- Added a mechanism of "Go offline" which would allow drivers to take a break or be done for the day once the quota for the day is done.
    - Drivers tried to game the system by cancelling rides in order to get rides that would get them closer to their home location.
- Extended the mechanism to incorporate "Go Home" functionality giving the drivers the option to conclude their day
    - This functionality would help drivers cover 75%-90% of the distance to their home
    - Later extended the feature to include the "Go To" feature.
- The design was reused for Ola Share with colors indicating pairs of pick-ups and drops

Conclusion

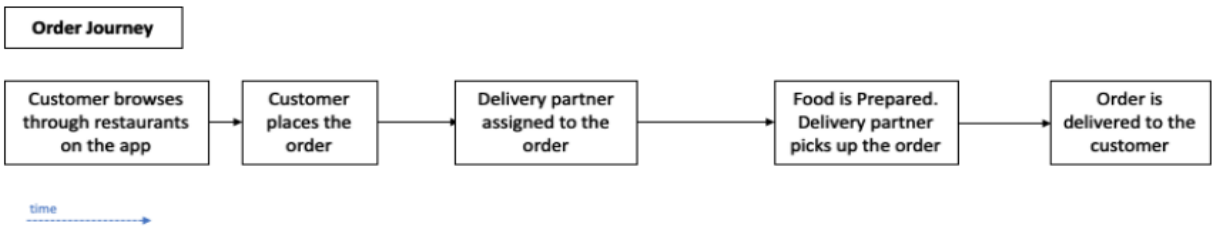
- Scaling requires devising new solutions to old problems
- Scaling and Agile go hand-in-hand
    - Define metrics for success and monitor regularly
- Fix bugs at each step to make solution robust
- Do NOT go for full rollout right after MVP

Relevant articles on popular products



Swiggy's effort to counter COVID-19 threat

Typical Order journey at Swiggy



Before order is placed to the customer

Ensuring restaurant staff remain healthy	Ensuring hygiene within restaurant premises	Sensitizing delivery partners about social distancing
• Checking temperature of their staff everyday • Anyone with temp above 99.1F or associated symptoms are advised to stay at home until tested negative for post mandated quarantine	• Complete scrub down of all the possible touch points - chair, table, door knob, counter tops, sinks in every 4 hrs • 3 Ply face masks are provided • Hand wash stations are maintained at pick up	• Constant communication in form of gifs and videos • **Deep Learning based algorithm to ensure delivery partners wear mask regularly** • Every delivery partner is required to click a selfie

period

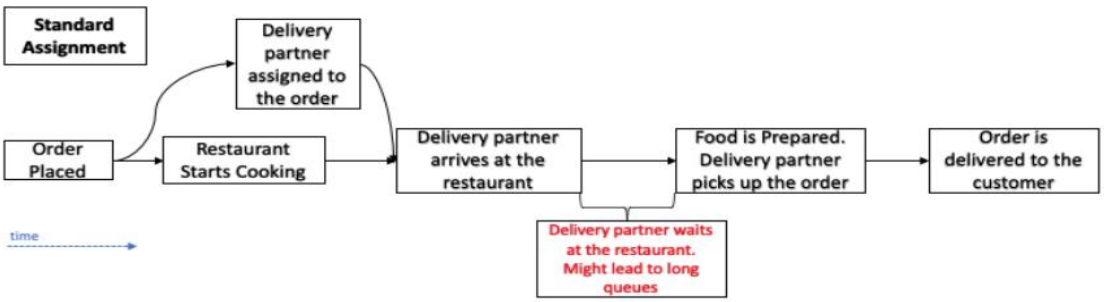
points

- Orders packed in additional bag
- Restaurants following these are **categorized as "Best Safety Standards" on the app**

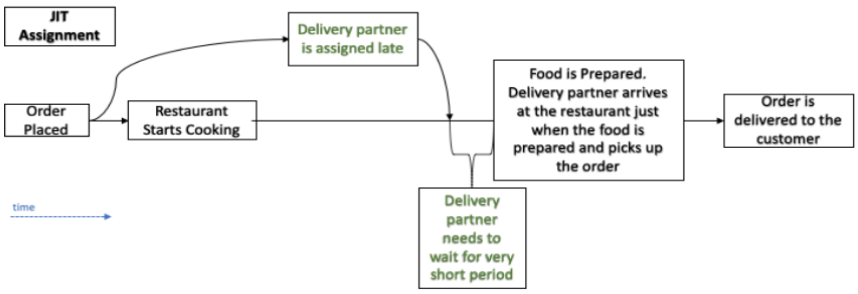
and post it on their app when they log-in to the system and start receiving orders. The algorithm checks if the delivery partner is wearing a mask and does not allow partners without masks to log-in.

When a delivery partner is assigned to an Order

Standard Procedure: Swiggy delivery algorithm assigns delivery partner a few minutes after order is placed. The work flow is as mentioned below.



Pain Point	Solution
Delivery partner to wait at restaurant for as significant amount of time till the food is prepared leading to crowd in times of high demand	Just in time delivery partner arrival- delivery partner assigned to particular order arrives at restaurant right when the food is about to be ready It reduces wait time and long queues at restaurant



Food Ready- Delivery Partner Assignment

Standard Procedure: Swiggy delivery algorithm assigns delivery partner a few minutes after order is placed. The work flow is as mentioned below.

Pain Point	Solution
Scenario where there are some prepared orders at restaurant whose assigned delivery partners have not yet arrived, while there are some delivery partners waiting at the restaurant whose orders have not been prepared yet	Delivery partners are assigned to restaurants instead of specific orders. As soon as food is ready for a particular order, it is assigned to delivery partner who is already present at restaurant

Workflow

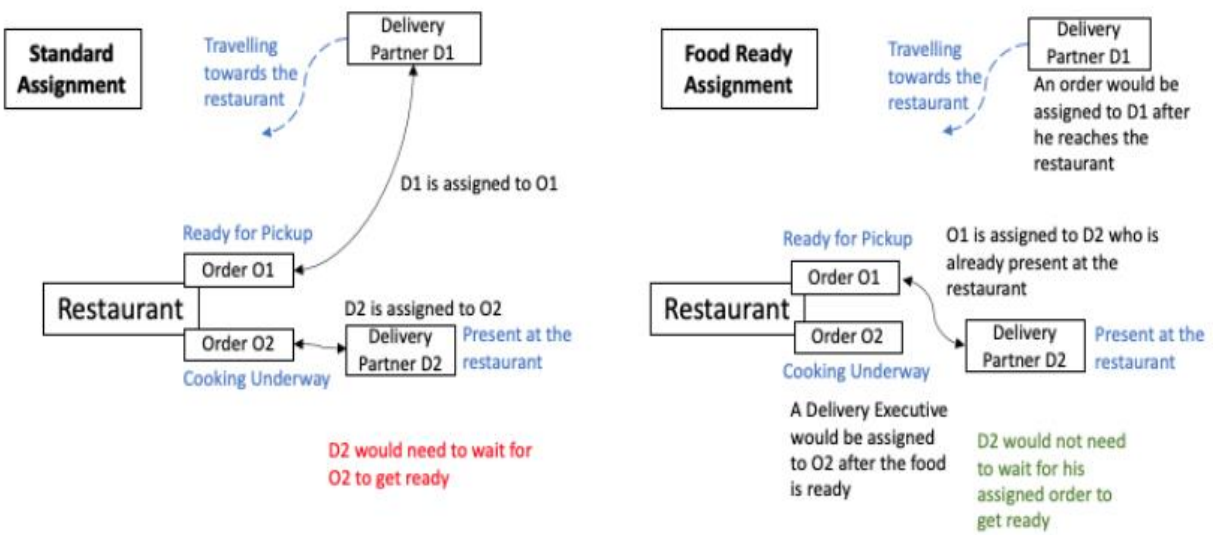
An algorithm predicting time taken for a delivery partner to travel to the restaurant
An algorithm to predict the time taken to prepare the order

→

Based on these predictions, right time to assign a delivery partner is picked up so that he reaches the restaurant when food is ready

→

This has led to 25% reduction in average wait time for delivery partners



Contactless delivery

The delivery partner delivers food outside customer's doorstep and then leaves. → Delivery partner submits a picture of delivery every time he delivers a contactless delivery → Customer gets notification on app saying food has been delivered

UI and UX changes in Big Tech due to COVID-19

Facebook	• It pushed out a number of informational resources within their app to help get people to COVID-19 information faster • They have a COVID-19 dedicated page showing data on number of cases, educating people about facts of COVID-19, Prevention tips, Promoting accredited institutions, showing latest posts from government, civic and health organisations
Instagram	• Instead of solely promoting informational resources, Instagram decided to go a more community-based route in order to foster socializing and togetherness. • A `Stay Home` story was added to the Story bar, which congregates all stories by people you follow who have used the `Stay Home` sticker. • In addition, they've added group video chatting. Users can all browse through Instagram together, looking at whatever content they want.
Microsoft	• Collaborative project with CDC – a healthcare symptom checker chatbot (Meant to corral users to the type of healthcare they may need to pursue. It should be noted that it is not intended to diagnose or providing information on testing facilities)

	• Bing's COVID-19 data live map tracker for cases globally. There are a lot of tools, graphs, statistics that can be broken down for accessing information
Google	• It has added numerous COVID-19 portals throughout their search pages • It has been promoting "DO THE FIVE" -5 guidelines to help slow spread of corona virus • Google's corona virus information site has compiled news articles and tweets related to the virus, including health info for those looking for symptoms and treatments • The map on this page brings out very simplified global case tracking map
Apple	• It has created a chatbot style "COVID-19 Symptoms checker" asking questions step by step to determine if they are aligned with COVID-19. • Users can also get latest COVID-19 news, learn about preventive techniques and testing information

Source: https://masterdesignblog.com/ui-and-ux-changes-in-big-tech-due-to-covid-19/

The Psychology behind TikTok's addictive feed

Popular videos on TikTok have the below mentioned things in common.

1. Simple (Very short and basic)
2. Unexpected (Curiosity Gap)
3. Concrete (e.g. relevant to COVID-19)
4. Emotional (Fun/fear/music driven)
5. Story telling

These are the key elements that make an idea stick hence the sticky content on TikTok.

A typical User journey on TikTok

- ✓ TikTok allows the feed to adapt to the watching habits of users
- ✓ Once the user is hooked, sponsored ads start sneaking in
- ✓ The addictive "swipe up" process goes on and on
- ✓ Few moments later, Tiktok's "digital wellbeing" notification pops up with message "you have watched xxx videos in 60 minutes" suggesting for a break.

(Providing Exit Points on TikTok- It allows users to disengage from your product with a sense of completion. If it is not done people will associate your product to a never ending lit of tasks. Not only it helps in long term user retention, but most importantly it is a humane thing to do.)

TikTok videos are more addictive than other social media platforms for the below mentioned reasons

1. Short- 16 seconds in average (9x shorter than Facebook's average). The feed hence adapts faster
2. Snapped- There is no aim needed, just flick your finger and the video will be immersive every time
3. Surprising- You either learn , laugh or be weirded out but you always get something personalized

This mix of very low cognitive task and high variability makes TikTok a textbook example of "addiction-forming design"

Source: https://growth.design/case-studies/

Which apps showed higher engagement during the lockdown period as an impact of COVID-19?

Key Findings:

- Some app industries are seeing a surge in user engagement — some as high as a 177% average increase in daily active users (DAUs).
- For struggling industries, app launches are down as much as 60-80%.
- Some app verticals are seeing prolonged spikes in engagement (Content & Media), while others are experiencing temporary peaks before returning to normal (Grocery)

Which apps are seeing higher engagement?

Grocery Apps	• Unlike Food & Delivery apps, Grocery apps have seen a dramatic increase in DAUs during the month of March as governments around the world issued lockdown orders. • App launches were up 177% on March 24 as consumers scrambled to secure supplies and prepare to shelter in place
Content and Media Apps	• With one-third of the globe confined to their homes, streaming media apps are becoming an essential source of breaking news. • Content & Media apps are holding steady at a 40% increase in DAUs since early March.

Gaming Apps	• Gaming apps saw a constant increase in app launches throughout the month of March, signalling that users aren't experiencing app fatigue and are in fact turning to mobile games more and more as a way to connect with friends and entertain themselves as they self-isolate.
Education Apps	• Online learning is undergoing a significant transformation as COVID-19 forces students around the world off-campus and into virtual classrooms. • Parents are overseeing their children's home schooling, people are taking the opportunity to pick up new skills and hobbies, and learners are looking for new resources and ideas. • It's no surprise that Education apps are seeing a significant rise in app launches — up nearly 20% before seeing a decline at the end of March.
Medical and Medicine delivery Apps	• Throughout the course of this pandemic, essential services like access to healthcare and pharmacies have been in high demand. • People are both more interested in their overall health and are turning to video visits and emails rather than go to hospitals for non-urgent care. • Medical & Medicine Delivery apps have seen spikes in DAUs as high as 40% before appearing to normalize as the month of March ended.
Health and Fitness Apps	• With gyms closed, people are turning to their Health & Fitness apps to help them stay active indoors (and log all their quarantine snacks). • Health and fitness brands finding ways to encourage people to access their library of digital workouts and connect with the whole community via their mobile apps.

Apps showing drop in engagement during the lock down period as an impact of COVID-19

➢ Given widespread bans on public gatherings and severe restrictions that have brought

- nearly all travel to a standstill, it's no surprise that **Event Ticketing, Travel, Hospitality, and Ride Hailing apps** are especially hard hit as this pandemic stretches on. The World Travel and Tourism Council projects a global loss of $2.1 trillion in revenue.*

  While things may seem dire, there are initial signs of recovery in China. As restrictions are being lifted, hotel bookings increased by 40%, flights rose 230%, and the Chinese domestic tourism market says it's preparing to recover 70% of losses over the next six months.

- **Food & Delivery apps** have seen a precipitous drop in engagement, but have since shown signs of recovery. After dropping over 40% on March 25, app launches have since risen — down just 4.7% from 15 days previous on March 28.

  One possibility: customers who were previously unsure about the safety of eating prepared food were reassured as more information came out about viral transmission, resuming take out/delivery orders. Another possibility: customers either ran out of groceries or got tired of cooking for themselves as shelter-in-place orders were extended.

- Bans on public gatherings have forced most leagues to either indefinitely postpone or outright cancel their seasons. As a result**, Sports Content and Sports Games (Fantasy) apps** are facing an extended off-season with little content to lure users and keep them engaged. App launches are down over 50% for both categories.

Source: https://clevertap.com/blog/q1-data-impact-of-covid-19/

Interview experiences - Summers 2019



Product Manager role interviews

Interview 1 - Amazon

Suggestion/advice about preparation for interview

Keep it data driven and do not be too stuck on using frameworks

Interview experience

Case: Should Amazon launch a new feature prior to Great Indian Shopping Festival (GISF).

Response: I started with understanding the feature and what were the apprehensions of Amazon. The interviewer told me that the system could handle only some X number of requests per second of the new feature currently. I further asked data about the expected number of requests per second and the probability of exceeding X requests per second. The three feasible options were: a. Launch for selected group b. Launch for all c. No launch for all Now to select one option, I used factors such as probability of exceeding, criticality of feature, alternate options available, pros and cons of launching and delaying among others.

Interview 2 – Amazon

Suggestion/advice about preparation for interview

- Be thorough with you CV

- Make your approach customer centric
- Have a favorite app or digital product and your suggestions to improve it further

Interview experience:

Cases & approach:

Questions from CV: My workex was in Product Management. So, they asked questions around the projects I did and how that can help company like Amazon.

Other: How to make FireTV compete with existing cable and dish network. (This was basically a product feature and UX design case, so I focused on customer pain point and tried to provide solutions)

Behavioral: How will you resolve conflict between two teams at Amazon. If there is conflict on which team should get the banner space to promote their products (This was behavioral question. I approached by understanding both teams view point, then moved to business goals and customer benefits. It was open ended question. They were looking for your conflict resolution approach)

Interview 3 – Optum

Interview experience

Case & approach:

Problem statement: You are from Uber. After IPL match ends everyone wants a cab and often, we see a shortage in cabs - how to address? Solve it as a product manager and not as a consultant.

(**Inference** - Solve using Product Management frameworks and go through customer pain points and customer journey. Only ask domain specific questions to the interviewer and not repeated questions like in a consulting case) **Candidate**: Followed **CIRCLES** framework

- **C- Comprehend the situation** and ask basic questions to the interviewer)
    - **Candidate**: Is the issue only with Uber or its competitors also? What is the goal behind the case - customer satisfaction only or Uber profitability or both?
    - **Interviewer**: For all. But only consider Uber. Profitability as well as customer satisfaction.
    - **Candidate**: So only a smaller number of cabs are able to reach the stadium even though many customers are searching for cabs or are trying to book Uber. Then Identify the target customers.
- **I- Identify customers** - All customers present in the stadium – adult to old people, singles, couples, a 2-3-member group and a joint family; also consider UberGo and UberPool in case of Uber segmentation. Mostly couples and adult people (without family) face issues.
- **R-Report Customer Needs** - to get seat in cab to reach their destination. Sometimes customers book the whole cab even he/she is the only passenger. (Uber cabs go with one passenger only and seats for 3 passengers remain vacant. (UberGo))
- **C - Cut through prioritization** – booking based on number of seats solution to be given highest priority
- **L - List all the solutions** –
    - Booking of UberPool to be given more priority than UberGo
    - UberCommute where people coming with their own vehicles can lend a ride to others
    - Using google analytics or any analytics algorithm on Uber App to track traffic and eyeballing. People usually open the app and close it again during the closure of the match if they see a cab 5-10 mins away. They believe that they can get the cabs even if they book outside the stadium and normally fail to do so. So, redirecting cabs where there is more traffic rather than having more cabs where there is less requirement

    - Making drivers take trips of people going to places near the stadium so that more cabs can reach the stadium
    - Surge Pricing to lure drivers near the stadium
- **E- Evaluate trade-offs** - none
- **S - Summarize recommendations** based on Uber's profitability and customer satisfaction. Select only 2 feasible solutions at first as a MVP feature and then you can give more solutions as long-term strategy

- **Questions from CV:**

- Explain your work and role in your previous company.
- How will your previous work experience help in Product Management role in Optum?
- Tell me what does a product manager do? And how is it different from a project manager?
- Are you a mobile person or a desktop person?
- Name 5 apps that you use frequently, and 3 features in one of them that you can improve. (You can use competitive analysis here)
- What are sentiment analytics?
- What is marketing myopia? Give an example.
- What is the difference between sales and marketing?
- If asked to launch a new product, who all will be its stakeholders?
- What is MVP?

Interview 4 – HUL IT

Suggestion/advice about preparation for interview

Stick to the basis of Supply chain and Use of IT in enhancing it. Learn the role of data in a firm like HUL

Interview experience

The interview in summers was mainly related to my projects and about myself. There were no HR QUESTIONS but they were asking questions related to project and HR mix.

Case: Use of IT in improving the distribution in Rural Areas

Program Manager role interviews

Interview 1 – Amazon

Suggestion/advice about preparation for interview

For Amazon interviews, try and integrate some of the leadership principles in your answers, be it work ex related or HR. They place a lot of emphasis for these principles.

Interview experience

Q: Some questions on my work experience - the way I contributed to the projects, arrival at a particular quantification

Q: **Guesstimate**: Suppose we launch Prime video in India during Sale, estimate the number of grievance-calls that Amazon would receive

Ans: Broke down the problem to only Tier 1 cities and picked 1 city - Hyderabad - population -> Educated -> Adopters & extended the same to other cities Break it down and show the structural thinking as much as possible. You should have a big tree structure to show this.

Q: App improvement suggestions

Ans: Search improvement on Amazon app

Q: Case: How will you leverage your work experience to bring about an improvement in any of the Amazon products?

Ans: I'd suggested that I'd leverage my experience with Audio algorithms to lead a team to improve customer engagement on Amazon Prime video while watching movies or web series.

Interview 2 – Amazon

Suggestion/advice about preparation for interview

Be thorough about all your CV points and read about the company details (their specialties, functions, principles, etc.) before going to the interview

Interview experience

Behavioral

- My interview started with the recruiters asking me about my intro including my hobbies. The initial conversation is mostly an ice-breaking discussion with the recruiters asking about my family, my college, etc.
- **Significant Achievements in Life**: After this, they asked about my most significant achievement in college and I answered relating to the project I have done in my college committee.

CV: Then, they started asking about my CV points particularly my work-ex related. After asking a few random points from my CV, they started drilling on a particular point. The point is related to the record time handover of a building I have done in 4 months instead of 6 months. They asked questions like:

- 'How did you achieve this milestone?'
- 'What resources you have received from your superiors?'
- 'What problems you have faced while achieving this milestone?'
- 'If you were asked to do this now, what would you do differently?'
- 'If more resources were given, would you be able to reduce this time further? If so, which resources? If not, why?', etc.

- They also asked me about my biggest failure in life till now and what I learned from that.

Case: I was also given a **guesstimate** regarding the number of delivery boys required to do 1-day delivery in Lucknow. In the end, to conclude the interview they asked me regarding the leadership principles.

Interview 3 – Amazon

Suggestion/advice about preparation for interview

- Prepare each point of your CV well. You should be able to explain and justify the interviewer about how you reached a particular number quoted in the CV.
- Also, for Amazon's interview, do read about the 14 Leadership Principles and frame instances (you may use the STAR framework) from your previous experience, where you had demonstrated.

Interview experience

Case: Come up with the metrics to be tested before certifying an app to publish on an Android App store

Other questions: Drilling questions on CV Specific points, Questions on Amazon's Leadership Principles

Interview 4 – Microsoft

Suggestion/advice about preparation for interview

Be upfront but polite. It's okay to ask questions and say you don't understand. Give reasons for assumptions. Develop stories around CV points and solutions

Interview experience

Cases: Single runway airplane scheduling, social media app for terminally ill, remote control for elderly, new feature for MS Word

HR questions:

- Who am I in a team?
- Why was the background diverse?
- Why MBA?
- Why the stream shifts?
- About myself.

General suggestions for Summers for IT / Analytics / Prodman roles

- Try to come up with a structure and list all possible options and then start eliminating or narrowing down, i.e. first cover breadth and then depth.
- Be well prepared with your CV points and also understand how the company you are interviewing for works.
- Favorite apps and their features. Why do you like those apps?
- Apps you don't like and why?
- CV work-ex preps. Long-term and short-term goals
- Go through the frameworks (CIRCLES, AARRR, Customer Journey).
- Go through the list of ProdMan companies and:
    - Gather insights about two of the company's products
    - Look at their values and frame your answers accordingly.
    - Talk to the seniors who have interned in that particular company.
- Basic terminologies like MVP, User persona, customer journey, agile, digital transformation
- Go through 'Decode and Conquer' book by Lewis Lin

- Refer to Materials provided by Disha and BizTech but in the end your basic understanding and problem-solving skills will help you sail through this process.
- Attend BizTech sessions to know more about these frameworks. Practice cases using these frameworks.

Interview Experiences - Summers 2020



Amazon

Vidushi Shahu - AMAZON (ProdMan)

Education Background - Bachelors of Engineering in Electrical & Electronics, Shri Vaishnav Institute of technology and science Indore

Work Experience - Tata Consultancy Services (40 months)

ROUND 1

Q1) Tell me about yourself?

Q2) A case on Customer Journey (Mentioned this in my introduction)

Q3) Favourite product? Why is it your favourite? Suggest some improvements. (detailed discussion about this too).

Q4) Any questions for the panel?

ROUND 2 (HR)

Q1) One time that you have to take ownership that went out of the way of your work?

Why do you consider this as an ownership outside your area of work?

Q2) One time when you have to convince someone on your same level? Like a peer or someone?

Q3) One thing that you would do differently if given a chance in your life?

Q4) One solution that you proposed that made a difference? Detailed steps of the solution.

Q5) How would you say that you are customer centric?

Q6) About my hobby?

Q7) Questions for the panel?

TIPS/ DO'S AND DON'TS

Try to be as enthusiastic about Product Management as you can. They judge whether you really want it or not. Try and answer keeping the LPs in mind.

Swapnesh Wagh - AMAZON (ProdMan)

Education Background -BTech., Electrical, IIT Hyderabad

Work Experience - Polycom (4 years 7 months)

ROUND 1 (CV/Case Based)

Q1) Tell me about yourself.

Q2) Tell us about 1 project where you have taken feedback from the customer. What did you do after taking feedback?

Q3) Have you used Alexa ? Suggest improvements in Alexa.

Approach: 1. Accent recognition

2. Can integrate Alexa with android phones to control different apps.

Q4) Give an example to integrate Alexa with an app.

Approach: Uber

Q5) Work-ex related question.

ROUND 2 (CV/HR Based)

Q1) Give us an example of a project that you did. How did you do it ? What role did you play ? Follow up questions.

Q2) Give us an example where you went beyond your limits to solve a problem.

TIPS/ DO'S AND DON'TS

Create stories around 14 LPs.

Himani Gomra - AMAZON (ProdMan)

Education Background -BE., Computer Engineering, NSIT

Work Experience - SharedReach (2 years 10 months)

ROUND 1 (CV/HR Based)

Q1) Tell me about yourself.

Q2) How would you solve a problem.

Q3) Would you add a new feature or improve an existing one. How would your approach change according to timeline in first 3, 6, 12 months

Approach: Would do POC, Prototype, testing in 3, 6 and 12 months respectively.

Q3) Tell me about a situation where you had to solve a very complex problem that took you months. How did you approach it ? What was the outcome ?.

ROUND 2 (CV/HR Based)

Q1) Give us an overview of yourself.

Q2) Was there an instance when you deep dived into a problem? How did you go about it? What was the outcome ?

Q3) Tell us a time when you collaborated with different departments to achieve something.

Q4) Talk about any recent technological advancement.

Approach: The candidate talked about applied filters getting cleared on Myntra shopping app after closing the app.

TIPS/ DO'S AND DON'TS

Be thorough with your CV and 14 LPs.

Vyom Wadhwana - AMAZON (Marketing Management)

Education Background - B. Tech, Mechanical Engineering, 2017, Pandit Deendayal Petroleum University

Work Experience - Segment Lead (CNG), IndianOil - Adani Gas (3 years)

ROUND 1

1. Tell me something about yourself
2. Any situation wherein you used Data for making a strategic decision
3. The sales of books are down on Amazon. How will you drive the demand?
4. Any questions for us?
5. How is online mode going?

ROUND 2 (HR)

1. Explain your role in Alumni committee
2. If you had to pick up one concept from Kotler and apply it in your present POR of Alcom,

what would it be and how will you execute it

3. Explain the challenges you faced while doing business development and how did you approach them (Work Ex)

4. Why Marketing?

5. Any questions for us

Rutul Shah - AMAZON (Program Manager)

Education Background - B.Tech (Mech. Engg), Pandit Deendayal Petroleum University

Work Experience - John Deere India (36 months)

CV/ HR QUESTIONS

Q - Introduction

Q - Tell me three important projects that you were a part of and what was your contribution?

ROUND 1

Q1) Follow up questions on CV

Guesstimate: Estimate the market size of shoes in Bangalore

ROUND 2 (HR)

Q1) Introduction

Q2) Explain your role briefly?

Q3) What according to you is most important for success of the project

Q4) Tell me about a time when you missed on the deadline of the project

Q5) Tell me about a time when you failed in a project? How differently would you do it now?

Q6) Any questions for the interviewer?

TIPS/ DO'S AND DON'TS

Know about the qualities that the role demands. Read about 14 leadership principles of Amazon

Amazon preparation in general

	Questions	Tips
CV Specific Questions	Work-ex projects	Know your projects inside out
Behavioural / HR questions	- Tell me about yourself. - Questions around Amazon's 14 LPs. Ex- • 1 project where you have taken feedback from the customer. What you did after taking feedback. • Example where you went beyond your limits to solve a problem. • Tell me about a situation where you had to solve a very complex problem that took you months. How did you approach it ? • Deep dive into a problem. How did you go about it? Outcome ? • Why do you consider this as an ownership outside your area of work?	Diligently prepare stories around the 14 LPs. They "make-or-break" your interview.
Domain knowledge	Product Development Lifecycle About AWS (for AWS role)	Show enthusiasm about PM role

Company knowledge	- Questions around different Amazon products - new feature & improvements suggestions	
PM Cases	- Improvements in Alexa - Favorite product. Any improvements? - Case on Amazon Prime customer journey - Guesstimate: Total mobile phone sales on Amazon	

Adobe

	Questions	Tips
CV Specific Questions	Questions around work ex and hobbies	Prepare your CV well
Behavioural / HR questions	- Tell me about yourself - Why PM? - Why Adobe?	Make your introduction interesting
Domain knowledge expected	- What is PM? - Why PM?	Don't expect anything technical. Just be logical.
Company knowledge expected	Prepare at least one of Adobe's product and improvement suggestions for it. Eg: Identify threats to Adobe PDF reader	
PM Cases	- Design a regional music app - features, 4Ps - Design Uber for women - features, 4Ps - Design a model for groceries app	Be structured in your response

DRUTI CHANDRA - ADOBE

Education Background - Management Studies, SSCBS

Work Experience - Zomato (Marketing)

CV/ HR QUESTIONS

Q - Introduce yourself.

Q - Why Product Management?

Q - Why Adobe?

Q - What metrics did you use to measure growth/ success at Zomato?

ROUND 1 (CASE BASED)

Case 1:

Design an app for regional music - how would you promote regional music of the place of your choice? What features would you add? What metrics would you evaluate?

- Target Audience?

- What are the different costs involved?

- How would you price it? *(Tip - Adobe seeks only value based pricing for all products.)*

- How would you monetize it?

- How would you make the app relevant to the COVID times?

- How would you promote it?

- How can you make the person listen to more songs? - By giving them better suggestions without being monotonous and varied vocals/ beats.

- How would you differentiate it from Spotify, Apple Music, Soundcloud?

Approach:

The candidate used the CPCC and the 4P's framework to approach the questions.

TIPS/ DO'S AND DON'TS

Be honest. Have a proper reasoning for every suggestion - interviewers question each and every point. Accept your mistakes.

They have loads to ask. Try to keep them engaged in one particular section.

Be thorough with some of their products.

Deloitte

	Questions	Tips
CV Specific Questions	Past work experiences, roles and responsibilities	Prepare your CV well
Behavioural / HR questions	- Introduce yourself - Why consulting	Make your introduction interesting
Domain knowledge expected	- What is cloud? - Benefits / challenges of cloud	Not looking for jargons, but basic technical cloud knowledge is expected
Company knowledge expected	- Why Deloitte	Always asked, prepare this question well
PM Cases	- Cloud migration strategy - Should the company move, what factors need to be looked at, costing.	Be structured, convey your thought process

APEKSHA GUPTA - DELOITTE (Tech Strategy)

Education Background - B. Tech. (CSE), JIIT Noida

Work Experience - 3.5 years across two startups

CV/ HR QUESTIONS

Introduce yourself.

What was your role in the last organisation?

How do you think working in Deloitte would be different from the startup culture?

ROUND 1 (TELEPHONIC - GUESSTIMATE ROUND)

- Asked the DevOps cycle - Explained what was followed in my last workplace.

Guesstimate: Estimate the market demand of iPhone12 in India.

Approach: Divided the population of India on the basis of income level & age group. Assumed a certain % of people in each category and calculated the total demand of premium phones. Finally, factored the market share of Apple to reach the final value.

ROUND 2 (CASE BASED)

Case (Cloud Migration):

There is a retail company that is growing tremendously for years. Recently, there has been an increase in the data centre costs. As the data centres are old, there have been a lot of outages lately. The owner has heard a lot about the cloud from his friends and is thinking of moving to the cloud. Should he move to the cloud? What will be the cost of moving to cloud?

Approach:

Asked a few clarifying questions -

C - Where is the company located? - *India*

C - What is the size of the company and the user base? - *Similar to Flipkart*

C - About outages - if it is a trend or a seasonal issue? - *Seasonal/ Promotional*

C - Since it is not a consistent issue, he can consider moving to cloud as cloud allows the flexibility to upscale/ downscale storage as per requirement. If he increases the data storage at the current data centre, that might become redundant at other times of the year, hence incurring him extra costs.

C - What is the infrastructure/ platform of the data centre?

I - Why do you want to know that?

C - So this would have a huge impact on my decision to move to cloud. Certain cloud platforms are not compatible with some technologies. For eg - AWS does not support Mainframe applications and Unix based applications. They need to be *refactored and replatformed* respectively in order to perform the migration.

Since you mentioned the data centres are quite old, they might be using one of these (obsolete) technologies.

Other considerations -

- cost-benefit analysis of moving to cloud vs having in-house setup with regular maintenance,
- current employee expertise,
- data security and compliance issues,
- number of applications that we are looking to move to the cloud - *do a suitability and a financial analysis for all apps*

Costs -

- Capital costs - Hardware costs, labour costs,

Operating costs - Maintenance costs

ROUND 2 (HR)

The interviewer asked questions about one of my work experiences and then wanted to know more about that work industry.

TIPS/ DO'S AND DON'TS

Have a smile, and keep the conversation engaging till the end. State all your thoughts and assumptions. Be honest.

QUESTIONS ASKED AT THE END

- What is expected from interns who have no prior experience in cloud tech?
- Where should I see myself in the complete cloud strategy chain?

Electronic Arts (EA)

Akarsh Jain - EA

Education Background - Aerospace Engineering, IIT Kanpur

Work Experience - Nucleus Software exports (3yrs)

CV/ HR QUESTIONS

- Why gaming?
- Which are your favourite games?

CV based questions

ROUND 1 (Case Based - case was provided 3-4 days earlier and solution submitted)

1. Questions around "D2 retention"

2. Difficulty level spike post D2

3. How will you check if there was a difficulty level spike?

4. Was the last activity a loss?

5.How will you price Plants vs Zombies

6. Why gaming?

ROUND 2

Which game with the best UX?

-Mass effect 3

Guesstimate:

Q1. How many photos are uploaded on Instagram in a day?

-stories/posts?

-do we know the number of accounts?

-meme pages, business, celebrity, personal- high, med, low usage

Q2. Rev. drop 5% Week on Week

-which questions will you ask?

-other competitors

-patterns of customers?

-change in app which has led to decrease in paying customers

Q3. Covid has boosted the gaming industry, what will happen post Covid?

-casual vs social gamers

-casual will have lower retention

Q4. Unlimited funds and resources, which game will you make for India?

-Culture is important, mythology could be leveraged. Mythology strategy

Q5. Plants vs Zombies is famous in NA but not in India

- Diwali theme

- Smaller SKUs in terms of payable services

Q6. What is freemium vs premium?

Google

Anjali - Google (Capability Development)

Education Background - BA (H), English, DU

Work Experience - Silverpush (1 year)

CV/ HR QUESTIONS

How do you deal with different kinds of people in your team ?

Suppose a teammate of yours takes a route which is unethical but gets the work done, how will you handle it? Tell me some resources that you would refer to, in this situation.

An instance where you took a risky project and how did you get other members on the team on-board. What was the outcome?

ROUND 1 (Google product case/HR Based)

Q1: Introduce Yourself.

Q2: How do you engage with cross-functional teams?

Q3: What did you like the most about your previous organisation?

Q4: How do you deal with different kinds of people in your team ?

Q5: What is your favourite google product ? What do you like about it ? What do you don't like about it ? What are the changes that you would recommend for the same ?

ROUND 2 (Case Based/HR Based)

Q1: Tell me about yourself.

Q2. Is there any app that you'd like to introduce? How will you go about it? Why did you select that particular app? What will be it's USP? How will it be different from its competitors? Lay out a GTM strategy for the app. (A lot of questions on the app, the market that the candidate chose and how will she pitch the app to investors)

Q3. How has digital marketing changed in the recent years? Some new trends that you have observed? (Give examples of some recent trends/ads) What do you think will be the future of digital marketing?

Q4. Suppose Airtel is launching a product targeting teenagers, how will you convince them to use YouTube as their medium, instead of TV?

Q5. What are some of the challenges that you have faced while starting an MBA programme, online? How did you cope with it? The internship might also be remote, what kind of challenges do you think you'll face? How will you build connections in 2 months in a virtual setup?

Q6. Suppose a teammate of yours takes a route which is unethical but gets the work done, how will you handle it? Tell me some resources that you would refer to, in this situation. (Resources on Environmental, company ethics etc)

Q7. Suppose a new intern is joining the team, what will be your suggestions for them?

Q8. Suppose a person in your team, don't believe in hearing out diverse viewpoints, how will you convince them?

Q9. Think of any product of your choice and suggest ways to increase its sales. Focus on digital marketing aspects as well.

ROUND 3 (Case Based/HR Based)

Q1: Tell me about yourself.

Q2. Do you have questions related to the new role?

Approach: Asked a couple of questions on the role and what would it entail)

Q3. What according to you is the most striking point in your CV?

Q4. How to monetise Google chrome? Further questions on the measures that the candidate suggested.

Q5. Do you think Chrome should be monetised?

Answer: No.

Q6. Follow up question on why do you think Google hasn't done it yet, why would Google make a product and not earn money from it?

Q7. Showed some data with 5 different segments (A, B, C, D, E) with their customer reach, google's revenue and total market size, then asked questions on where do you think google is weak, where are they strong and which segment provides maximum growth opportunity. (Asked further questions on how the candidate arrived at the answer and my approach)

Q8. Another dataset with a company's CPC, CTR and number of impressions from a particular digital campaign and asked questions on how to calculate the revenue generated. In which case is the ROI better? How?

Q9. An instance where you took a risky project and how did you get other members on the team on-board. What was the outcome?

Q10. Suppose you have a list of publishing websites (Websites which use Google ads and Adsense) ex: TOI, what will be some of the metrics that you will judge these websites on? How will you convince them to use google products? Lay out your strategy for the pitch.

TIPS/ DO'S AND DON'TS

1. Major part of the interview revolved around situation based HR, hence preparing HR answers using STAR method, is really important.

2. Knowledge about Google products (very important) and basic digital marketing terms helps for case based/technical questions.

3. Each interviewer introduced themselves in great detail at the start of the interview and asked if I had any questions for them at the end of the interview. So, apart from the questions that I had in mind, I also asked specific questions based on what they mentioned in their introduction (about working in Google and their role) which they seemed really keen on answering and showed my interest in the company and the role, as well.

4. They don't pick up points directly from CV, but the kind of questions they probed into required me to include stories from relevant CV points. So if you've prepared stories/ talking points around CV pointers, it helps.

QUESTIONS ASKED AT THE END

How has the experience of WFH been like as the interviewer had started his career at Google entirely remote.

What are the kind of projects that I would get to work in, during the internship?

Akanksha Mehta - Google (Product Marketing Manager)

Education Background - BTech. Biotechnology, Amity University

Work Experience - PCL Technologies (1 year 1 month)

CV/ HR QUESTIONS

Tell me about yourself.

A lot of behavioural and situational questions.

ROUND 1 (CV/Situational/HR Based)

Q1: Tell us something about yourself

Q2: CV Based questions.

Q3: Tell me about a situation where you dealt with a diverse team.

Q4: Tell us about a situation where you confronted someone. How did that pan out?

Q5: Suppose you were part of a team and all of you contributed equally but 1 person ended up taking the credit. How would you deal with that?

Q6: Suppose there was a project you noticed was not being executed to the best of its potential. You felt like you could contribute to the project more. How would you approach the team without offending them?

Q7: What does achievement mean to you?

Q8: What has been your biggest achievement in life ?

ROUND 2 (Case Based/HR Based)

Q1: Introduce yourself.

Q2: Who was your most difficult client and how did you deal with them?

Q3: What was the most important thing when you were dealing with B2B and what was your approach when you approached your lead for the 1st time?

Q4: What is your long term goal?

TIPS/ DO'S AND DON'TS

Be thorough with your CVs.

Kubric

	Questions	Tips
CV Specific Questions	- Internship related question: Give the reasoning to a particular UX design that candidate chose in his internship	Thorough CV prep is a must
PM Cases	PM Role • Myntra's users have been dropping for the past month. Can you think of why this is happening • What is your Favorite app? What is the feature that you don't like about it and how can you improve that? what metric would you choose to evaluate its performance ? PMM Role Email Writing Task As a Kubric PMM, write an email to a free user in order to upgrade them to a premium pricing model. You are free to assume any user/vendor	Following framework whole answering questions helps a lot

Gaurav Meena - Kubric (PM)

Education Background - Mechanical Engineering, IIT Roorkee

Work Experience – NA

CV/ HR QUESTIONS

Tell me about yourself

Internship experience related to UI/UX design during college

ROUND 1

Question 1:

Internship experience related to UI/UX design during college
Candidate mentioned how his work had led to an increase in website/app traffic and no. of leads. Towards that, how he

1. collected the Data
2. Made design changes based on the insights from data analysis

Interviewer probed about candidate's thought process behind his work on UI design for a new service,

Question 2:

What is your Favorite app? What is the feature that you don't like about it and how can you improve that? what metric would you choose to evaluate its performance ?

Case 1:

Myntra's users have been dropping for the past month. Can you think of why this is happening

Approach:

Clarifying questions

- App or Website

- Problem among new users or current users

User journey

Login >> Landing Screen >> Product Listing Page >> Product Page >> Cart >> Payment Options

Issues in customer journey

- Discussed issues in each stage of the User Journey that might be causing for the new user to drop out.
- Candidate engaged with the interviewer as and when he identified a prospective issue at any point

Interviewer asked to take up any two problems and think of possible solutions

Candidate followed a similar approach as the case example in Disha Product management Case book (Page: 33/38)

Saket Narayane - PMM

Education Background - BTech Computer Science, National Institute of Technology, Warangal

Work Experience - Mu Sigma, 3+ years

CV/ HR QUESTIONS

- Introduction

- Questions around work ex

ROUND 1

Basic interaction that lasted for 20 minutes about the candidate and his work ex

ROUND 2 (Email Writing)

Question 1: Exercise on email writing: As a Kubric PMM, write an email to a free user in order to upgrade them to a premium pricing model. You are free to assume any user/vendor. Candidate was given 10 minutes to write this email and he considered a sweet shop owner using Kubric's free services.

Approach:

- Catchy subject line
- Focus on personalization
- Discounts in form of cashbacks/flat 50% sale
- Exclusivity of the offer made

Customer centric: a call to the vendor to reach out in case of clarity, etc by directly replying to this email or contacting on phone

TIPS/ DO'S AND DON'TS

Keep smiling, interviewers appreciate a pleasant disposition

Media.net

	Questions	Tips
CV Specific Questions	- Work ex problem area based PM case	
Company knowledge expected	A basic idea of Ad-tech – the bidding system in selling ad spaces	
PM Cases	Around 4-6 PM cases per person in two rounds. Examples: • You are a PM of YouTube, average viewing time has come down without any effect on daily active users. Investigate. • You are a PM at google search. How will you tackle the problem of spam clicks on google native search ads. • Design a P2P (person to person) vehicle renting app • Your budget is $100,000. How will you spend it on the marketing of a premium category of chocolates by cadbury?	Engage with the intervieweer and give a logic reasoning to your assumptions

Sumit Kumar Pal - Media.Net

Education Background - Computer Science Engineering, Jadavpur University

Work Experience - Samsung, 21 months

CV/ HR QUESTIONS

A case based question was asked related to an area of candidate's work ex

ROUND 1 (CASE BASED + Favourite App Analysis)

Case 1:

You are a PM of YouTube, average viewing time has come down without any effect on daily active users. Investigate.

Approach:

Clarifying/Scoping questions:

Web app or mobile app?

Customer personas

- Entertainment seekers
- Educational/DIY type content consumers
- Users watching whatever is trending

Customer journey to do a root cause analysis

- Problem in downloading the app (no, pre-installed)
- Login issues
- Change in type of recommended videos
- Intrusiveness/relevance of Ads
- Buffering of videos

Candidate was asked to explore every possible touch point in the customer journey and also

give reasons to candidate's hypothesis on issues and non-issues

Metrics to reflect average viewing time

Viewing time/DAU

Favorite app:

1) Quora

2) Candidate's favorite messaging app - 10 things he likes about it and 8 things he hates

Case 2:

How would you filter out spam messages on email and how will you test the effectiveness of this spam filtering feature?

Approach:

- Cluster content
- Allow for users to mark messages as spam and train the algorithm to identify such messages
- Metrics (+ve and -ve)
    - proportion of times candidate moved back a message from spam to main folder
    - Number of times user disabled the spam feature
    - Number of times user opened a spam message

Case 3:

You are a PM at Facebook, what metrics would you use to judge the effectiveness of Facebook Like feature?

ROUND 2 (Case Based)

Case 1:

You are a PM at Swiggy, design the home page and the restaurant recommendation engine.

Approach:

Candidate approached the question using the perspectives of these key stakeholders and their interactions with the customer using the app:

- Customer
- Restaurant
- Delivery agent
- Payment gateway
- Swiggy as an aggregator

Case 2:

Design a P2P (person to person) vehicle renting app

Approach:

Clarifying questions

- **Type of vehicles: 2 wheeler/4 wheeler**
- **Short term/long term type renting**

Key stakeholders: Renter & Rentee

Concerns that both types of customers face

- Profiles of the rentee, the type of usage, experience in driving etc
- Profile of the renter, vehicle condition

Customer journey approach to designing the app : Right from the preliminary step of verification of profiles to showing listings on the app based on key filters like type of use

Case 3:

You are a PM at google search. How will you tackle the problem of spam clicks on google native search ads.

Approach:

Deep dive into following sections:

- **Why would spam clicks happen in the first place?**
- **Impact of spam clicks?**
- **How to detect these spam clicks?**
    - Track IP addresses and flag any repeated clicks from the same IP address
    - Track any unnatural rate of clicks
    - Swap/randomize the sequence of ads for similar keyword searches-
    - Time stamps: Detect frequency of clicks on ads from a specific IP.
    - Use IP block listing for IP exclusion.(punitive actions once detected)

Gradual reduction of incentives for every clicks from a single user (from the ad provider's side)

Dipin Kumar Kerketta - Media.net

Education Background - BTech (CSE), Bharati Vidyapeeth College of Engineering

Work Experience - Software Engineer, iNICU Medical Private Limited

CV/ HR QUESTIONS

Tell me about yourself: Candidate talked about his work, interests and hobbies.

ROUND 1 (Guesstimate Based)

Question 1

Calculate the number of badminton racquets sold in Delhi in a day.

Approach:

- population of delhi-> subdivided into age groups and demand from each.

- Number of stores -> premium and normal ones catering to each age group
- Personalized the guesstimate by accounting for sale at sports complex shops by DDA

ROUND 2 (Case Based)

Case 1:

Book my show bookings are declining. Investigate.

Approach

Scoping question: is the decline happening among existing/active users or new users?
Evaluating

1. Price
2. Competitor offerings
3. App specific:
    1. Are the advertisements wrongly targeted?
    2. Placing of the events on app

The interviewer just wanted to get a sense on how well the candidate is able to deep dive into the problem without expecting him to necessarily diagnose the problem

Case 2:

Book my show bookings are declining. Investigate.

Approach:

Scoping question: is the decline happening among existing/active users or new users?

Evaluating

- Price
- Competitor offerings
- App specific:

- Are the advertisements wrongly targeted?
- Placing of the events on app

The interviewer just wanted to get a sense on how well the candidate is able to deep dive into the problem without expecting him to necessarily diagnose the problem

Case 3 (work ex based case):

Candidate worked in a healthcare product based startup and started to talk about the solution that his company offered. To this, the interviewer asked him to focus on articulating only the 'problem' or business objective that the candidate's company was trying to offer.

Approach:

Candidate worked on a product aiming to simplifying child delivery care in ICUs in India

The case was then moulded around a new problem in this domain and the candidate was asked to go step by step through the solution and give an MVP. He was asked to go beyond the existing offerings that his company had provided.

Case 4: Your budget is $100,000. How will you spend it on the marketing of a premium category of chocolates by cadbury?

- Candidate started with the STP analysis of this chocolate and identified the touch points where a prospective customer can be targeted for eg. Kiosks at the airport
- Interviewer asked the candidate to focus on the online advertisements (because, ofcourse!)

Approach:

The following factors would affect what price should be bidded for an online ad:

- Cost of one chocolate and the profit margin
- Expected number of impressions through the ad
- Expected conversion rate

- What the max amount you are willing to pay vs what the competitor is bidding (Game Theory reference)

Since this was a fairly new territory for the candidate the interviewer helped him by citing this example:
If, for person A, a painting outside his dining hall seems worth rs 1000$ and for person B, the same one outside his gallery seems worth 5000$. What would be the bid amount that both these persons would be willing to pay?

Candidate's answer: it's prudent for only the person B to bid i.e. 1001$ which is just above the maximum amount that A is willing to bid.

The candidate was encouraged to think on similar lines, competition in bidding in new chocolate vs competitor say Lynt chocolates.

Microsoft

	Questions	Tips
CV Specific Questions	Discussions on CV projects	
Behavioural / HR questions	- Tell me about yourself - About hobbies - Describe a problem that you solved at work	
Company knowledge expected	- Discussion on recent Microsoft acquisition of Tiktok and value proposition around same	
PM Cases	- Amusement park for 2-7 year olds, facing complaints of high waiting time for a ride. How will you address it? - Youtube for elderly - How can you add value due to changing scenario of COVID in classroom - You are the PM of a website. Traffic is expected to be increase during festivals, 3 ways in which you will manage - Swiggy app exists for end-users. Should there be a Swiggy app for suppliers as well. What features would you consider?	Think properly before saying anything, in depth cross questioning is done on every word you say

PM Cases	- Design an alarm clock. Pick your own customer segment. - Design a security system for Satya's new house. - Design refrigerator for blind - You are a budding entrepreneur. You need to design a raincoat. - Redesign privacy settings in WhatsApp. - Create an app to track COVID cases and to ensure media news authenticity. - How would you go about integrating payments on IRCTC? - Design an enterprise level chatbot to help employees deal with stress. - Metrics to analyse COVID situation in the country.	

Debapriya Chhotray - Microsoft (PPI from PM challenge)

Education Background - B.Tech (Mechanical Engineering), NIT Rourkela

Work Experience - Capgemini (1 year 10 months)

CV/ HR QUESTIONS

Tell me about yourself.

Some basic follow up questions on my hobbies.

ROUND 1 (CASE BASED)

Q1: Tell me about yourself.

A1: Told the interviewer about family, educational and professional background and about my hobbies.

Q2: Some basic follow up questions on my hobbies.

Q3: Considering the current Covid scenario where people have shifted to a work from home/study from home model, How can you as a product manager, improve the customer experience?

A3: Considering the two models, I chose the study from home model where I would improve Microsoft teams in enhancing the classroom experience. I took two customer segments that are teachers and students. After which, I figured out pain points around notes, exams/quizzes, attendance and doubts. I moved forward with the doubts pain point and decided to bring some 5-6 features around it after which I had to prioritize some and decide on some metrics that would help me gauge the success of my proposed features.

ROUND 2 (CASE BASED)

Q1: How would you redesign the privacy settings in WhatsApp?

A1: I did not get the problem statement at the first go. I talked about encryption initially. The interviewer nudged me to rethink the privacy features. Then I picked up “Last seen”/”Blue

ticks"/"Online status". I talked about how I can customize all this in a manner where a user would be able to apply these features for some people selectively. Example: This would help a user to show his/her last seen to the people he/she is comfortable with and it is not shown to the rest of the people from his/her WhatsApp contact book.

TIPS/ DO'S AND DON'TS

1. Understanding about the interview experience/roles from the seniors.
2. They are looking for a structured approach to a problem.
3. Take a sufficient number of Mock PIs/Cases.
4. The features/solutions need not be extraordinary. They would like to hear your rationale behind your decisions, and that should be convincing.

Gaurav Rajan - Microsoft

Education Background - B.Tech. (Computer Science), IIIT Delhi

Work Experience - Microsoft (3 years 8 months)

CV/ HR QUESTIONS

- Introduction

- About Work experience? Hobbies.

- Why did you leave Microsoft to join a B-school?

ROUND 1 (Group Case Round)

Case:

In the growing age of WFH (Work from Home), what should Microsoft (known for its wide variety of Productivity software and cloud services) invest in to make this work from home more productive than it is today! This could be a new product line or enhancements to an

existing product line for Microsoft to win this new age of remote working.

Approach:

I thought of improving the existing product teams instead of going for a new product line as it has lots of scope for improvement and also we could leverage its existing features to tap in additional customer base. I considered three target groups- working professionals(user personas- managers/developers), college/school people(user personas- teachers/students), vocational workers. For each of them, I tried to identify a primary need which might get fulfilled with the enhancement of MS Teams.

- For working professionals, I thought of improvement by adding plugins for developers and bookmarking by allowing chats to save in Microsoft one note.

- For college students/teachers, I considered a platform similar to google classroom- collaborate, discuss, share lecture notes. Provision to conduct quizzes for profs, faster access via Microsoft forms- since there are bugs in google forms used currently.

- For vocational workers, I thought of means of monetization through a hands-on tutorial for DIY sessions to customers having issues at home. They could be paid by integrating digital wallets like GPay, Paytm, Bhim, PhonePe etc.

- I also suggested multilingual bots to provide guided sessions for first time users.

- Finally considered trade-offs and took college students/teachers as the highest priority group.

ROUND 2 (CASE BASED)

Q1: Tell me about yourself

A1: Talked briefly about my educational background, past work experience and hobbies. Since I had prior work-ex at Microsoft only, discussed projects briefly (no in-depth discussions) and few HR regarding leaving MS to join B School.

Case:

Design a security system for Satya Nadella's house. What all features would you consider,

trade offs etc.?

Approach:

I was asked to design the product from scratch. I had to start with need gap analysis, existing systems in place, enhancements on previous models. Consider the target groups here (since I knew who all are there in his family- tried to build upon their needs), considering creating an IOT based system. Connecting all devices together through a mobile app. Features thought of- intrusion detection, temperature control, digital protection by encrypting laptops/personal computers at home. Gestures and scanning features. A lot of questions were built upon the responses I was giving. We also talked about Satya's son who is differently-abled and how the system can be made smarter to aid his needs- I talked about the UI of the app, gesture controls, alarms and signals through just a button click.

ROUND 3 (CASE BASED)

Q1: Questions on past experience at Microsoft.

A1: I worked in Microsoft Consultancy services. The interviewer had knowledge about the vertical. Asked about the team I worked with, projects I undertook, managers, leadership principles- tried to know how much I knew about MS :)

Case:

Design an enterprise-level chatbot to help employees deal with stress/anxiety during COVID.

Approach:

I had worked with chatbots in past projects. Had to design an end to end chatbot to help users cope with stress levels. I was asked to share my screen and design on the go. Used Think Aloud strategy- asked about the kind of users who might use it, was free to make assumptions. I considered two TG- employees/managers. Considered it as a conversational bot- had to be interactive and funny- tried to include funny quotes during user conversation, the interviewer smiled at it. I considered a user's journey while using the bot. I further identified the user's problem - categorized personal/professional, the bot can recommend

solutions using AI/ML models - send happy thoughts, recommendations, links to motivational websites, videos etc. Interviewer finally asked to share the doc where I wrote the solution for the bot.

TIPS/ DO'S AND DON'TS

- Be confident; interviewers are very friendly. Try to build upon the conversation; don't just stick to the solution. See how it can be improved further.

- Do not necessarily follow a particular framework in each case. Sometimes the interviewer is guiding the solution; try to follow what he/she wants instead of a predefined format-they might/might not appreciate it.

- During GD case discussion, be thorough and to the point. They can cut you in between and not allow you to complete the solution. Focus on the main issues you are addressing through your solution.

- Try to make the interview more interactive. Asks doubts/clarifications wherever required, they are very helpful in this regard. Do not directly jump to the solution- take some good 5-7 mins to think through the solution, I tried doing so in the 2nd round, and he asked me to take time before writing anything since I was sharing the solution on teams, so any mistakes they can easily catch then and there.

Megha Priya - Microsoft

Education Background - B.Tech. (Computer Science), Birla Institute of Technology

Work Experience - Tata Communications (3 years 2 months)

ROUND 1 (Group Case Round)

Case: In the growing age of WFH (Work from Home), what should Microsoft (known for its wide variety of Productivity software and cloud services) invest in to make this work from Home more productive than it is today! This could be a new product line or enhancements to

an existing product line for Microsoft to win this new age of remote working.

Approach: steps followed -

Identified the users - Students, Professionals, teachers

Identified the need of each user segment- like what they would require to focus more on their work and become more productive. For example, students, to focus more they would need some kind of interactive platform, as online classes get extremely boring and difficult to focus.

According to the needs suggested solutions. Of the several solutions, one which he picked was adding a quiz/poll icon in MS-Teams.

Interviewer cross questioned the solution and asked how I would actually implement it and why not just use other quizzing websites link in the chat to conduct a quiz or a poll to check if people are active or not. Answer to that was to get an edge over meet & zoom and include it as an icon for the host to create a real time quiz to gauge the interest of the people attending a meeting or a class. Posting a link to other websites would only increase their visibility, causing no good to Microsoft and also not many would know about the websites and how to create such quizzes.

ROUND 2 (CASE BASED)

Q1: Introduction, Explain a project from the CV.

Case: Consider an e-commerce website which has the highest number of visitors but has lowest sales among all e-commerce platforms. What could be the reasons behind this ?

Approach: Using customer journey tried to identify the issues at each step.

ROUND 3 (CASE BASED)

Q1: Introduction and Project based questions.

Case 1: Design swiggy app for sellers. what features should it have ?

Case 2: How will you analyse the current Covid-19 situation in a country as a government

official

ROUND 4 (CASE BASED)

Case: Due to WFH, employees are feeling frustrated and losing motivation. MS is planning to introduce a new product for providing counselling/mental wellness services. Design this new product. What feature should it have and explain end to end customer experience and customer journey. Use word/paint to draw the prototype or process flow of the product.

TIPS/ DO'S AND DON'TS

Follow a structured approach to start with the case. I followed two basic approaches -

MX Player

MAYANK BHARATI - MX PLAYER

Education Background - B.Tech, NSIT

Work Experience - AutoNinja, Swiggy, Meesho

CV/ HR QUESTIONS

Which was the best organization to work for you out and why?

ROUND 1 (GENERAL HR BASED)

1. Introduce yourself?
2. Which team would you like to join at MX Player and why ?
3. Why product management?
4. Why MBA?

TIPS/ DO'S AND DON'TS

Candidate was well prepared on below points though these were not explicitly asked in his interview:

- Read about MX Player's job profile and requirement
- Prepare about the product and offerings and areas that you think require improvement
- Which product do you like the most and why?

Mohit Sehra- MX PLAYER

Education Background - B. Tech. (Chemical Engg), IIT Kharagpur

Work Experience - 2 years in analytics at OYO and SG Analytics

ROUND 1 (GENERAL HR BASED)

Q1) Tell us something about yourself? Follow up – fav project at OYO and learnings?

Q2) Fav. project that you did while working for SG analytics in digital marketing? (Increase in average viewership by analysing poll results)

Q3) Asked about one of the projects mentioned in my SGA work ex- related to digital marketing? (Incremental revenue by optimizing brand merchandising)

Q4) Why do you want to do an MBA, what motivated you to do pursue an MBA?

TIPS/ DO'S AND DON'TS

- Be very thorough through your whole CV- Each and every point can be discussed in the interview
- Try to read about the company beforehand and make notes to remember the crucial things
- If you have analytics BG, be very thorough with each and every number that you have mentioned in the CV. For example – you have written 10% revenue growth, so you should know what was the base and the value post the increment and obviously your contribution towards it.
- Do prod man cases with seniors- as many as you can. 5-6 decent sessions would suffice
- Try to go through analytics and prod man prep related course books (at least 2-3), try following the material and frameworks provided by Biz tech and Disha.
    - Read about MX Player's job profile and requirement

- Prepare about the product and offerings and areas that you think require improvement
- Which product do you like the most and why?

Optum

Komal Vaity - Optum

Education Background - BTech. EXTC, VJTI

Work Experience - Wipro (2 year 7 months)

CV/ HR QUESTIONS

Tell us 3 things you would like to talk about in your CV.

What is automation, innovation, invention?

What will you do if your teammate is not performing?

ROUND 1 (CV/HR Based)

Q1: Tell us 3 things you would like to talk about in your CV. Tell us more about it.

Q2: Why do you want to join Optum ? What is it's vision ?

Q3: What is automation ? (mentioned in Candidate's CV)

Q4: What is innovation ? What is invention? Difference between the two ?

ROUND 2 (Case Based/HR Based)

Q1: Tell me about your work-experience.

Q2: Why Optum ? What do you know about US Healthcare?

Q3: What will you do if your teammate is not performing?

Q4: More questions around the US healthcare industry.

Q5: Who can be the potential customers of Optum.

Q6: What are the different products that Optum can offer.

Case 1: Design an elevator for a building in COVID Times.

Case 2: A user of apple watch is not using the feature to check heart rate after using it for 2 months. What could be the reason? How will you solve it ?

TIPS/ DO'S AND DON'TS

They ask a lot about company and US healthcare. It is good to read about it.

Jyothi Tangella - Optum

Education Background - BTech. EXTC, Vasavi College of Engineering

Work Experience - Accenture (2 year 11 months)

CV/ HR QUESTIONS

1. Why do you want to join Optum and work in the health industry?
2. What are the top 3 achievements of your life?
3. What is your best achievement in work-ex?
4. One of your team members is not working properly. What would you do in this situation?

ROUND 1 (HR Based)

Q1: Introduce yourself in brief

Q2: Why do you want to do prodman ?

Q3: Why do you want to join Optum and work in the health industry?

Q4: Given the COVID condition, what do you think Optum could have done immediately? What should have been considered in US healthcare.

ROUND 2 (Case Based/CV/HR Based)

Q1: What are the top 3 achievements of your life ?

Q2: What is the thing that made you inclined towards the health industry?

Q3: What is your best achievement in work-ex?

Q4: Follow up work-ex based questions.

Q5: Tell us about anything that had a great impact on your life.

Case: Improvise an elevator in COVID Scenario.

Approach:

The major factor that has to be taken care of is social distancing. So when a person wants to enter the elevator, instead of opening the door, voice commands can be used to open and close the door. Virtual circles could be made around people inside the elevator so as to ensure social distancing is maintained while they are standing. A permanent sanitiser could also be fixed in the lift and AI could be used to detect if people inside the elevator are wearing masks. If not, then an alarm could be rung and entry could be restricted. Another feature could be keeping a check that only one person can enter the elevator at a time.

Q7: Why do you want to pursue prodman? Why do you want to join the health industry?

Q8: Name any 3 challenges that you faced at your workplace?

ROUND 3 (Case Based/HR Based)

Q1) What is the difference between prodman and project management ?

Q2) What is important to you - solution or the problem?

Approach: The candidate said the problem was important because if you don't have a clear understanding of the problem then one cannot design an effective solution.

Q3) If you were given a problem. how would you solve it ? Strategically or tactically ?

Q4) Given the US health condition, what is the potential problem that you would pick and what are the measures?

Approach: The candidate picked up diabetes because she had a good knowledge about it.

Q5) What are the different stages in design/development of a product.

Q6) One of your team members is not working properly. What would you do in this situation?

TIPS/ DO'S AND DON'TS

It is good to be aware of the US healthcare industry, medications etc.

Read JD and responsibilities.

Focus on improving the personalisation experience of patients.

Read Casebooks, BizTech instagram posts, and have a good number of case slots.

Interview Experiences - Summers 2021



Adobe

Abhigna Parikipandla- Adobe

Education Background - Integrated degree - BTech in Engineering Design and MTech in Automotive designing, IIT Madras

Work Experience - Programmer Analyst, Boeing India (21 months)

CV/ HR QUESTIONS

Q - Introduce yourself.

Q- CV specific questions

ROUND 1 (CASE BASED)

Q1. What are your favourite applications?

C - Webtoon, Spotify

Q2. What is webtoon

C - Explained about the application

Case: How can you improve the total number of users using adobe acrobat reader for opening the pdf files

Approach: I segregated the users.

Pain points - There is no together viewing experience to discuss, single page viewing people use web browsers only when highlighting is needed then use adobe. If we have comments on pdf added other people reading can have references for the book. We can also have a platform for group study - like google doc platform.

They liked user segregation. They really helped me throughout the case.

ROUND 2 (CV/CASE BASED)

Q1. Brief introduction and about one project I had in my CV on automation tools - 5 to 10 min

Q2. Favourite Apps?

C - Webtoon and Spotify

Case: Spotify PM is trying to increase premium count. Help him

Approach: I used CIRCLES framework. Customer segregation - Spotify users who listen, artists who want to promote themselves. They were interested in the second segment.

Q3. What should Spotify provide them?

Approach: I segregated as discovery, promotion, retention. If you are not getting approach, you can always go back and discuss with them

Q4. Discuss all the pain points, success metrics

Approach: Went through customer journey. They were looking more for approach

Q5. Give solutions

Approach: Interactive chat option with fans, Having artist schedules

Quick to catch on what they told. My ideas were not that great but they made me comfortable.

TIPS/ DO'S AND DON'TS

For PM, have as many cases with seniors to be comfortable.

Have at least 2 favourite applications - top to down

Amazon

Akanksha - AMAZON (Program Manager)

Education Background - B.Tech (IT), BIT Mesra

Work Experience – Tata Steel (36 months)

ROUND 1 (CV/HR)

Q1. Introduction

Q2. Tell me a situation when you deep dived to solve some issue in a project, how did you go about it?

Approach: (Question based on one LP) Took one project mentioned in my CV, explained the issue raised and how I collaborated with different stakeholders involved and steps taken to solve the problems and what was the outcome.

Q3. Mention a time when you had conflict with your senior and how did you go about it?

Approach: I took one of the projects from my workex where I had faced some issue about my approach, framed my answer around it explaining the reason of the conflict. Mentioned the steps I took and the outcome (He was more interested on the data driven method that I took to present my side of approach to senior to resolve issue).

ROUND 2 (HR)

Q1) Introduction

Q2) Explain your role briefly?

Q3) What according to you is most important for success of the project

Q4) Tell me about a time when you missed on the deadline of the project

Q5) Tell me about a time when you failed in a project? How differently would you do it now?

TIPS/ DO'S AND DON'TS

Go through Amazon Leadership Principles, all answers should have some elements of these principles.
Be thorough with the CV points written, they may pick up any point and ask question based on it.
Follow framework provided by Team Disha to answer the HR questions.

Do ask some question when interviewer ask for it, try to pick up some details from his introduction, shows your interest

QUESTIONS ASKED AT THE END

Q. Asked about the project that the Interviewer mentioned in his introduction and had 5 min conversation on it, also asked about challenges faced by program manager in day-to-day work.

Atlassian

Shreyash Krishna - Atlassian (Product Management)

Education Background - B.Tech Ashoka University

Work Experience – Product Manager, IIFL

ROUND 1 (CV/CASE BASED)

Q1. Introduction

Q2. Case Submission summary

Q3. How do you prioritise suggested features?

Q4. Develop an Alarm clock for the blind

ROUND 2 (HR)

Q1. A situation where-in you had conflict with senior management

Q2. A situation where you self-initiated a project

Q3. A challenging experience from workex/college

ROUND 3 (HR)

Q1. A situation where you had to self-initiate a project

Q2. A project wherein you had to manage the prioritisation of multiple stakeholders

Q3. A situation where you had to lead a team, what were the obstacles faced and how you overcame

TIPS/ DO'S AND DON'TS

Practice all types of cases- Design, Improvement, RCAs, Industry specific, etc.

Always use positive and negative metrics while approaching case solutions

American Express

Raghu Kumar Reddy – American Express

Education Background - BTech. NIT Calicut

Work Experience - Latentview Analytics, Paytm

CV/ HR QUESTIONS

Q1. Tell me about yourself.

Q2. What do you know about AMEX?

Q3. What are the revenue sources of AMEX?

Ans. American Express earns most of its money through discount revenue, primarily represented by earnings on transactions that take place with partner merchants. The company also generates revenue from cardholders through annual membership fees, interest on outstanding balances, conversion fees, and more. American Express has a "spend-centric" model aimed at growing the number of overall transactions on its cards through special offers and relatively low fees.

ROUND 1 (CV Based/Puzzle)

Q1. You've got someone working for you for seven days and a gold bar to pay him. The gold bar is segmented into seven connected pieces. You must give them a piece of gold at the end of every day. What and where are the fewest number of cuts to the bar of gold that will allow you to pay him 1/7th each day?

Ans. Cut the bar in 3 pieces or segments of size 1/7,2/7 and 4/7. Give piece 1 at the end of first day. Give piece 2 at the end of second day and take piece 1. Give piece 1 at the end of third day. give piece 4 at the end of 4th day and take pieces 1 and 2. Give piece 1 at the end of day 5. Give piece 2 at the end of day 6 and take piece 1. Give piece 1 at the end of day 7. So with 3 cuts we can achieve this.

Q2. How do you flag a fraudulent transaction in corporate credit cards which are used for personal purposes?

Q3. Asked about whether I have an idea about SQL, excel and given a short case study asked me how do I plot the graph in Pivot Table using excel.

Q4. Asked to explain key projects at Latentview Analytics and Paytm I was part of.

ROUND 2 (CV/HR Based/Case)

This round was longer than first round and taken by a senior person. The interview was more of a conversation than of an interview.

Q1. Started with Introduction and major projects I have worked on at Latentview Analytics and Paytm. Lots of cross questions involved while I was explaining these projects which involved working with credit card company and loyalty analytics.

Q2. I was given a short case study on how to flag users who would stop using AMEX cards and plans to counter that.

Q3. Further discussion was based on culture fit and assessing my workstyle.

Q4. What are your Strengths and Weaknesses?

Q5. How would you handle a failure?

Q6. How would you convince your manager? What do you expect from a workplace?

Q7. What motivates you?

TIPS/ DO'S AND DON'TS

Get to know more about AMEX, their business model and how is it different from VISA and Mastercard before interview. Play on your strengths and it's okay to take time to answer questions. Maintain calm composure and smile, it would help a lot.

Emeritus

Rishabh Virulkar – Emeritus

Education Background - BTech. D.Y. Patil College of Engineering & Savtribai Phule Pune University

Work Experience – Fresher

CV/ HR QUESTIONS

Q1. Tell me about yourself

Q2. What is product management?

Q3. What do you know about Emeritus?

Q4. Have you visited our website? What did you see there?

Ans. Started with the company's history, founder's names, culture, vision and mission statements.

Q5. Can you tell me about a situation where you overcame a challenge?

ROUND 1 (Number skills)

I: Let's do a some small numerical.
C: Sure!
I: Your target is 100 enrolments. The successful conversion rate is 5%. How many leads would you approach?
C: (Basic maths) 5% of X is 100 => X = 2000 leads (Interviewer didn't give much time for answers)
I: That's correct. Now, let's consider if the price of one product is Rs5000. Assume no profit/no loss situation. What will be the cost per lead to the company?
C: Ans= 5,00,000/2000 = Rs 250 per lead. It took 60 seconds to answer. The interviewer expected a little faster.

I: Anything you want to ask?

C: Emeritus has a global presence, and it recently entered the Indian market in 2019. What are the short-term plans of the company? What are the expectations from the intern?

I: Emeritus is planning to increase its product portfolio. The intern is expected to discover market opportunities, develop new courses to address them, bring them to market, drive enrolment, pricing and ensure learning outcomes for learners.

C: Thanks! That was a detailed answer.

TIPS/ DO'S AND DON'TS

Prepare your HR answers well. Review HR answers with multiple persons and go with advice you feel the best. Check the company websites; JD and RG will be enough. Above all, most important is confidence and a smile. Do not panic and take a deep breath. Take as many case slots as possible

GamesKraft

Diptanshu– GamesKraft

Education Background - BTech BIT Mesra

Work Experience - Infosys Ltd

CV/ HR QUESTIONS

Q1. Tell me about yourself

Q2. 2-3 questions related to cv?

ROUND 1 (CV Based/Case)

Q1. Interviewee directly jumped to CV point where I mentioned e-sports competition winner during college fest. It revolved around Why particular game (CS 1.6), Why not other games (DOTA). Follow up questions were on improvement in the game and how will you measure it.

Q2. In the end one case study was given - Counter Strike planning to launch mobile based application. Should it do or not. Suggestions - Special focus on KPIs.

ROUND 2 (Case Based)

Q1. After settling interviewer asked my favourite app. I mentioned telegram. He then asked 2nd favourite app to which I said Swiggy.
Questions on Swiggy -
Q2. Why this app? (What I like)
Q3. What improvement required.?
Q4. Success Metrics on the solution suggested.
Q5. Discussion on success metrics.
Q6. Which e-commerce app do you use for which I mentioned Amazon.
A case was given -

I: You are a manager of electronics category and CEO wants to increase the profitability by 10% for the same category.
C: Started with basic profitability framework and interviewer asked to work on both revenue and cost. I started with value chain, suggesting areas where costs can reduce or revenue can increase finally moving to app improvements and giving success metrics. Interviewer mainly focused approach and ideas given.

TIPS/ DO'S AND DON'TS

Taking prod man case slots is a must. Interviewer was asking follow up questions on different success metrics. Interviewer might ask any games you played. It will be good if you keep an analysis of popular games.

Google

Akash Gupta– Google

Education Background - BTech. IIT Roorkee

Work Experience - Samsung R&D Bangalore, Optum India

CV/ HR QUESTIONS

Q1. Tell me about yourself

Q2. Why google?

Q3. Why PM?

Q4. 2-3 questions related to cv?

ROUND 1 (CV Based/Guesstimate)

Q1. How URL works (In detail)

Q2. Web-app scalability (How communication happen between backend & UI) (Since I have developed a Web-app in workex)

Q3. How to fetch image/video from server and load on website. How does it work on different devices (laptop, mobile)

Guesstimate:

I: Number of people in a single day on airport?

I: Choose airport of your choice.

C: Delhi (near to my hometown) I used Supply side approach to give the answer Interviewer was happy with the approach then he asked me to use demand side approach also. Apply demand side approach also, interviewer was happy with the approach

I: Anything you wanted to ask?

C: I asked few questions related to interviewer workex, had 10 min discussion with him. He was impressed with the questions (Brownie points)

ROUND 2 (Case Based)

Product Improvement:

Q1. Name your 3 favourite products (Also give reason why you like the product)?

Q2. Improve google calendar (my 3rd fav product) for different customer segment (40 min intense discussion) Some in-between PM questions like what will do as a PM to solve this problem, how will you convince your tech team to work on the features your suggested.

ROUND 3 (Case Based)

RCA:

I: 15% decrease in Facebook groups usage. Find out the reason and recommend solution

C: Is it specific to web or app?

I: Both

C: Other few clarifying questions i.e. from how long, geography

I: 6 months, All over world

C: Made user journey and pin-point possible issues at each page. Pointed out around 10-12 issues. Then Interviewer selected 3 out of those and asked me to provide solution with success metric Intense discussion on all point Some in-between PM questions like what will do as a PM to solve this problem, how will you convince others.

TIPS/ DO'S AND DON'TS

They are looking for Structured approach. Cases are open ended so make sure to touch all the pain points. Success Metric are very important.

Sindhu Srinath - Google (Product Management)

Education Background – B.E. ECE, PES Institute of Technology

Work Experience – Soroco India (36 months)

CV/ HR QUESTIONS

Introduction

CV specific questions.

Why Google?

ROUND 1 (CV/Case Based)

Case 1: If your users are not able to access your webpage, what could be the possible issues? How would you prevent this from happening again?
Approach: Went through the entire journey of a web request within the network as well as how it is serviced covering possibility of failures at DNS, Network, Load Balancer, Servers (Are down), Database
Case 2: How many drones would be required to deliver all the ecommerce packages in <your city>?
Approach: Asked Interviewer roughly how long does each drone take to deliver a package. Used Supply side approach to calculate number of warehouses in my city (based on time taken by drone to deliver), number of packages/warehouse to be delivered per day to get number of drones.

ROUND 2 (Case/HR Based)

Q1. List top 3 favourite apps and why? Need not be google apps
Q2. Which do you use the most? How would you design this for the elderly?

Interviewer cross questioned every point the candidate was suggesting here
Q3. If you had unlimited budget, what change would you make in this app?

Candidate was pushed to suggest more and think more

Q4. If you promised customers a bunch of features and your engineering team is not able to deliver it, how will you go about it?

ROUND 3 (Case/HR Based)

Case. You are the PM of a product and you just changed the signup flow- you are now taking in additional information. When you did A/B testing for this, you found that there was an 8% increase in filling details and a 20% decrease in 7-day retention rate. What could be the issue?

Approach: Laid down rough reasons I'd want to explore - Change in the way we analyse data, characteristics of the population chosen for A/B testing, Possible Macro issues. Was guided by the interviewer to arrive at their point.

Q1. You are the CEO of a company and you just invented teleportation. What would you do with it?

Approach: Was asked to state my approach only. So mentioned just the headers as follows:

a. Identify the objective the CEO had in mind - Increase Revenue or use product for common good

b. Identified possible Customers and prioritise based on objective

c. Based on customer stated the value proposition for the product

d. Discuss the Promotion and launch strategy for the product

e. Discuss the Evaluation Metrics to track how well product is doing

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TIPS/ DO'S AND DON'TS

Be thorough with your CVs.

Practice RCA and Prod Design/Improvement questions

Practice at least 1 guess estimate per day

Microsoft

Akarsh Kandpal – Microsoft

Education Background – BTech. IIT (BHU), Varanasi

Work Experience - Toppr, Zilingo

CV/ HR QUESTIONS

Q1. Tell me about yourself

Q2. Why Product Management?

ROUND 1 (Group Case Round)

Group Case Round:

Case: Design an effective disaster management system for India. It can be a new product or build upon any Microsoft product. A group of 20+ candidates was given a case to solve within 30 minutes and submit it on a word document.

Approach: I chose pandemic as a disaster and tried to come up with user personas and pain points that were faced during Covid. I gave basic but feasible solutions—covered points like hospital bed availability and booking and a platform for getting medical supplies and equipment. For hospital bed availability, I mentioned it to cover this solution in 2 iterations. In the first basic version, we can release a platform for checking the bed availability, and in the next iteration, we can go for hospital bed booking as well.

I also provided with 2 metrics for judging the success of the features.

After the submission had a small 1-1 discussion where the interviewer majorly focussed on metrics. Questions like why you chose these metrics and what information you were trying to get out of them. The interviewer also asked me to look for other metrics specifically for my 2nd iteration of the 1st solution. The metrics I suggested were No. of hospital bed bookings/ No. of Hospital searches. I believe the interviewer just wanted to gauge whether I do have

some idea about funnel analysis.

ROUND 2 (Case Based)

Case: Design an Air traffic controller

Approach: Asked 2-3 clarification questions in order to get idea about what we were looking for in this question.

I: I am looking for 2 things: first, metrics you would consider for scheduling landing and take-off, and second, drawing out the final console.

C: My basic algorithm to decide which flight will take off or land will be based on First-come-fist-serve, where whichever flight had the earliest timing will be scheduled for its arrival or departure. Then I mentioned basic principles under them I tried to cover a few points. First, one is Loss of human life, under which I covered points critical fuel level, medical emergency, bad weather. The second principle was Loss of value/money (Cost) under which I mentioned A flight that has already been delayed a lot or flight that has not been given clearance for landing for a long time (basically the waiting time) and also gave preference to international flights over domestic ones.

Finally, I had to draw the console and explain how it would look like. After this, I was asked to keep any top 3 features for my console, stating why I felt those were important.

ROUND 3 (Case Based)

Case: Design an experience for preventing information leak through emails in organizations

Approach: Asked a few clarification questions to gauge more into the problem. We would want to prevent information leaks externally as well as internally. Furthermore, the information leak can happen either through email content or through attachments. This feature can now be part of the email client itself or an extension on a browser.

For preventing external leaks, a simple block on sending emails to external domains can be used. For internal leaks, a feature can be provided which will allow keeping the email thread within the group. Tagging other employees and forwarding mail will be disabled. To prevent

leaks through attachments, the files will not be allowed to download and can only be viewed.

This was a small 15-20mins round where the interviewer was just trying to understand the through process and structure of the candidate.

Yatharth Yadav- Microsoft

Education Background – Btech, NSIT, Delhi

Work Experience – Zomato

CV/ HR QUESTIONS

Q1. Tell me about yourself

Q2. Discussion on anime for 2-3 min from the introduction?

Q3. Why did you leave your job as Product Manager to join MBA only to sit for Product companies again?

Ans: I had already prepared an answer. I included few product management terms like user obsession, curiosity and passion with examples to support my story.

Q4. 2-3 questions related to cv?

ROUND 1 (CV Based/Case)

Case: Improve Zomato app for senior citizens (I had Zomato in my workex)

Approach: Followed CIRCLES framework. Defined user persona, for example, senior citizens are not tech savvy. They have weak eyesight. They forget things quickly. They have very picky taste, so generally eat the same food from particular restaurants (I was not sure of this point so I mentioned that I have seen it with my grandmother and stated the assumption that every senior citizen does this). Apart from this, I presented 2-3 more characteristics. After persona characteristics, I went for the pain points by combining persona and flow on the app. For example, senior citizens are picky about taste but particular about restaurants so we can

show them previously ordered food rail in search and other flows. We can even show dishes on the homepage under restaurant tiles. I also suggested bigger text and images in UI (got a cross question on this in Round3). Apart from this, I presented 2 more features.

Q1. Which of the features would you prioritize?

Ans: Applied RICE framework and prioritized 2 features. Was stopped in between as we ran out of time.

Q2. Any questions for me?

Ans: I want to know even after 14 years of product management experience, that too in a company like Microsoft, what keeps you going as a product person. For example, at this starting point in career I get excited by solving user problems and building solutions to help them. I want to know what excites you at this stage of your career so I can relate and see PM role for long term.

It was a 45 mins interview.

ROUND 2 (Case Based)

Case: Design Zomato App for restaurants from scratch (FYI - I had Zomato in my workex)

Approach: Mentioned all the stakeholders involved with the restaurants like Zomato app, delivery guys, their own kitchen (updating status of the food), Zomato salesmen etc

Divided the restaurant journey into Onboarding (further divided into Pre, During and Post onboarding), Info update - like updating menu time to time, and journey after food is ordered like order acceptance, sending order to kitchen, keeping track of number of orders (helps in updating expected delivery time), marking order complete, handing over food to - waiter (in case of dining), delivery guy (in case of online ordering) and customer (in case of takeaway order). Based on stakeholders and journey I gave all the features covering all the above-mentioned aspects. It was a 35 mins interview

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ROUND 3 (CV/HR Based)

Q1. I have seen your CV and read your previous 2 interviews, so let's jump straight to

questions. You have mentioned PM experience in your CV, tell me about an incident where you solved customer pain point and also kept business metrics in mind.

Ans: Mentioned one of my experiences. I mentioned that business metrics and revenue was not our goal in that case but ultimately solving user problem helped in increasing revenue too.

Q2. Cross question from my solution in Round1 and asked for its implementation. Feature in question was UI improvement for senior citizen.

Ans: I asked for 2 mins to think of an implementation. Suggested age as input in signup flow and accordingly set a default UI for senior citizens. Then dived deep into each aspect of implementation. (Interviewer was very well prepared and had really good questions to each part of implementation)

It was a 20 mins interview

TIPS/ DO'S AND DON'TS

If you have Product mentioned in your CV then be thorough with that product. Be prepared for questions like "Mention some incident where you showcased customer obsession or where you displayed prioritization or stakeholder management". Microsoft is famous for questions based on CIRCLES which have a certain style in the form of Design or Improvement case. Make sure to solve or read a few.

Nitin Singh- Microsoft (Program Management)

Education Background - B.Tech. IIT Guwahati

Work Experience – Indian Oil Corporation Limited (47 months)

ROUND 1 (Group Case Round)

Case: Design an intelligent assistant for the elderly population in India to empower every person and every organization on the plant to achieve more.

Approach: Submitted the case as per format, tried to keep it as structured as possible. Next was individual interactions post submission. Panelist went straight into case discussion. Added few new points which came to my mind at the spot during interaction.

ROUND 2 (CV/HR/CASE BASED)

Q1. Panelist asked the candidate to introduce himself.

Q2. Candidate was asked about his work experience and highlights in his work experience.

Case: Design an OTT Platform for Kid

Approach: Tried to keep it as structured and interactive as possible. Panelist asked the candidate to delve deeper into few points he mentioned.

ROUND 3 (CV/HR/CASE BASED)

Q1: Panelist was very senior level employee, wanted to know about the candidate's work experience and he tried to make this a stress interview by asking tricky HR questions.

A1: Candidate kept his energy level high, and answered HR questions through STAR framework

Case 1: Design an experience to prevent information leak over email in any enterprise

Case 2: How will you improve WhatsApp privacy features, when you hide your last seen it hides for everyone how will you do it for selective people.

TIPS/ DO'S AND DON'TS

1. Take Case Slots for maximum domains possible. Don't keep a rigid mind towards summers.
2. Take lots of mock interviews with seniors of different domains to get the best idea of your preparations.

Optum

Vijaya Aditya Errabati - Optum

Education Background -

Work Experience - Systems Engineer in Infosys (24 months), Technical consultant in Adjoint, a start-up (21 months)

CV/ HR QUESTIONS

Q Questions on Background, work ex, projects worked on

- Explain architecture of the projects worked and how it impacted the business
- As you have worked with clients, Did you work with health care clients, and if you did, what were certain differences between the normal clients and the health care clients?
- What was your role in the projects you worked on?
- How would you design this(relating the project in CV) for a health care user

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ROUND 1 (CV/CASE BASED)

Q1. Which health care app do you use?

C - Pharmeasy

Q2. Improve the app, think about the customer journey and go through pain points and how would you solve it?

Q3. How would you introduce the app to customers?

C - Did STP for the product and listed features, issues for the product to be introduced.

(Quite exhaustive went about for 10 min)

Q4. What do you know about Optum?

Approach: Told about revenue it holds, on parent company, headquarters, what kind of customers do they cater to. The interviewer then explained in detail about what Optum does and his role

ROUND 2 (CV/HR BASED)

Q1. Tell about the Payment product you worked on

Q2. What role do you want to work in Optum? Insurance or payments or what sector?

Q3. Can you tell subtle differences between Indian and US health care

Approach: Didn't know either of the structures, but the interviewer told to think about it and helped me. Then I came up with some differences. One was people here don't mostly take general insurance. There are many that the government implements here.

Q4. Talk about your role in payments, How did you interact with clients. Can you give the names of the clients you worked with?

C - I didn't say the names and he was fine with it

Q5. What do you think the role of HR would be in Optum

Q6. Why PM? What do you expect from Optum?

Q7. What health app would you develop such that it helps Indian Citizens now?

Approach: I gave the best feature, pain points of already existing apps. Then solutions to the pain points

ROUND 3 (HR BASED)

Q1. How Would you introduce a product?

Q2. What would you do to introduce a product if the market is collapsing?

Q3. Told me about compensation and asked if I have any location constraints

TIPS/ DO'S AND DON'TS

Don't try to force fit the framework.

Know about the company, role being offered to you.

QUESTIONS ASKED AT THE END

Q. How long has the interviewer worked?

Q. Roles and expectations from my role, Do all projects have the same responsibility?

PayPal (Risk)

Anup Ghagre – PayPal (Risk)

Education Background - BTech. Yeshwantrao Chavan College of Engineering

Work Experience - CETKING

CV/ HR QUESTIONS

Q1. Tell me about yourself

Q2. What do you think are the important qualities to be a product manager?

Q3. Suppose you are a product manager and there are two people in your team who have conflict. How would you go about solving the situation?

ROUND 1 (Case Based)

Case: There is an account of Bill Gates on LinkedIn. How would you identify if it is a fake account or a genuine account?

Approach: Firstly, we can compare the frequency of posts on his LinkedIn profile with the ones on other social media platforms. Secondly, we can look at the quality of content posted by this account to identify if it is fake. The next step will be to check if Melinda Gates is there on LinkedIn then whether she's following this account. Another point will be to check if the post is demanding any PII data from his followers, then it may be a fake account.

ROUND 2 (Case Based)

Q1. Tell me about five products which you like and why?

Case 1: Suppose you are a Product Manager of the Amazon app. What changes will you make in the Amazon app that will help you increase your revenue in the Apparels category without burning any cash in terms of marketing? (as the candidate mentioned in the previous question for buying apparel, he prefers Myntra)

Approach: If the customer is a regular user of Amazon, then the size he ordered in the past should be shown as the recommended size when the customer is placing a new order. Another measure that can be taken is to collaborate with influencers to market the product.

Case 2: Suppose there is a vending machine with two products – Milkshake and chocolate and it is meant for the children. How would you go about building this product?

Approach: Given the product is meant for the kids; its design should be kept simple. There would be just two buttons- one for each product. The milkshake button would be colored white, and the chocolate button would be brown colored so that the kids can quickly identify. Given there are just two products, the size of the vending machine will be compact.

For the payment system interviewer pointed out that cash cannot be used. Hence, the candidate said coupons could be used for payment. The interviewer says here that coupons can get misplaced by the kids. The candidate then answered that an image recognition system can be created to recognize the kids' faces and have a database with one-to-one mapping. The parents can top up the kids' accounts and keep a close watch on how much the kids are spending.

TIPS/ DO'S AND DON'TS

Be confident. Another important thing is having the presence of mind. Both these things helped the candidate a lot.

QUESTIONS ASKED AT THE END

Q1. What are the skills that the candidate should learn in order to be better prepared for the role?

Q2. How is the work culture at PayPal?

Salesforce

Shraddha Singh– Salesforce

Education Background - BTech. Thapar Institute of Engineering & Technology

Work Experience – Capgemini

CV/ HR QUESTIONS

Q1. Tell me about yourself

Q2. Talk about any one project/task from your workex form beginning to end, the approach and ideas behind it. (Questions around my take on the whole activity)

Q3. Why product management

Q4. Tell about a task in your workex that was challenging and how did you overcome it. (Technical Questions around it)

Q5. Tell me about your Undergrad final year project. (Basic questions around whatever I told)

Q6. Asked about the application I developed on Blockchain from my workex.

ROUND 1 (CV Based/Roleplay)

Single round of Interview of 1 hour

30 min by a product manager:

Q1. A roleplay situation: You are a Hyderabad car seller, you have built a site for your shop, identify how you will define a good visitor and a bad visitor on your site. (On clarification: good visitor is someone who will be worth the effort and bad is someone who is just a passerby not seriously looking to buy anything and would be a waste of time)

Q2. Your favourite product, its best features and any improvement.

30 min by senior product manager(joined in later):

Q3. What subjects are you planning to take up in the senior year. Have you discussed any

with your seniors

Q4. what do you prefer B2B or B2C? (question because saleforce is B2B) (kind of a trick question, be honest with your answer)

Q5. You must have read up about Salesforce and its products. Do you have any questions about it. (asked 2-3 questions around it)

TIPS/ DO'S AND DON'TS

Prepare for why product management, why technology? CV based questions should also be prepared thoroughly

Interview experiences - Summers 2022



Media.net

Utkarsh Agrawal – Media.net

ROUND 1

1. Introduce yourself.
2. Questions about CV.
3. Why Product Management?
4. Guesstimate : annual stadium revenue for Allianz arena (I had mentioned I'm a bayern Munich fan)
5. Metrics for YouTube CEO Dashboard
6. Explain like I'm 5 : API
7. 3rd party cookies
8. Cookie matching
9. Google is phasing out cookies what are the alternatives
10. User identification in contextual advertising

ROUND 2

1. Intro
2. Why Product Management
3. RCA Case : Average revenue per ride has gone down in Uber
4. PD Case : design a system to counter fake news on FB

Ankur Mishra – Media.net

UG: B.Tech. in IT from NIT Allahabad

Work ex: Software Engineer at Fidelity Investments (almost 3 years)

Summers Interview: Media.net

BUDDY ROUND

1. Tell me about yourself
2. What do you know about API, how would you explain it to a 5 yr old
3. Discussed on the submission
4. RCA on drop in revenue of amazon
5. What factors will Netflix consider for building movie recommender

ROUND 1

1. Puzzle - Two people are selecting a number from 1-9. First one to reach the cumulative sum of 49 wins the game. Find the winning strategy
2. What do you know about Media.net?
3. What do you know about Contextual Ads?
4. CV based questions

ROUND 2

1. You are the PM of Spotify. Should Spotify launch audio books? If yes, then should they do it in the same app or different one? Product Development case for the same.

Atlassian

Nithin Adari – Atlassian

ROUND 1: PM ROUND

1. Tell me about yourself
2. Tell me about a hardware product which you used recently and liked it or hated it
3. Why PM?
4. Why did you choose this app (in submission)?
5. Who are you solving for? - Talk about the customer persona
6. What other factors did you consider before choosing this pain point and why did you choose this particular pain point over those? - Clearly establish your objective
7. You're the PM and pitch this feature to your engineering team in 20-30 seconds - Talk about both customer and business impact
8. What metrics do you want to look at?
9. What would be the metrics from drivers' POV?
10. Why Atlassian?
11. Talk about any critical situation you faced during work and how you overcame it
12. Any questions for me?

ROUND 2: LEADS & INSPIRES ROUND

1. Tell me about yourself
2. Tell me something not on your CV
3. What was one situation where you had to confront someone and how did you do it?

4. What was one negative feedback that you received and how did you act on it?
5. What was the best project that you worked on?
6. What was one instance where you put somebody else's goals ahead of yours?
7. Any questions for me?

ROUND 3: VALUES ROUND

1. How would you describe yourself?
2. What do you do in your free time?
3. What was one criticism that you received and how did you act on it?
4. What keeps you motivated to perform better at workplace?
5. Tell me about one instance where you committed a mistake and how you rectified it
6. What was one project for which you alone were responsible from start to end? Would you like to change anything about how it went?
7. Any questions for me?

Soumyadeep Chatterjee - Atlassian

UG: B.Tech (CSE), KIIT, 2019

Workex: Temenos & Kony Labs, 34 months

ROUND 1

Case Submission - Improve any of the ride-hailing apps. Provide details about 1 feature that you like in the app. (Deadline - 2 days)

ROUND 2

• Interviewer introduced herself

• Introduction

Details about work ex, what and how

• Case (I got an abstract case, relating to payments that Atlassian can accept from clients. Reason -> I had worked in a banking software company)

• Case had no proper conclusion for me, the interviewer wanted to see structure and questions mainly.

ROUND 3 - LEADERSHIP ROUND

• Detailed out about the round

• introduction

• Tell me about the biggest challenge you faced, how you overcame it. Learnings (I took an example from personal life)

• Tell me about a time where you received feedback and how did you receive it? Was the feedback useful to you in life?

• Tell me about a time where you faced criticism and how did you handle it

• Time when you put someone else's needs before yours. (I got stuck with this, asked to circle back to this a little later)

In each question, the interviewer gave a structure in which he wanted me to answer and asked me to be precise. Use STAR framework, give a little context

ROUND 4 - VALUES ROUND

• Started with small talk

• Asked me about how I interpret the company values (I answered with interpretation from my own life instances, some work ex related, some personally related)

Can you tell me about a time when you had to give negative feedback to someone, how did they receive it and outcome (something around handling an uncomfortable situation)

• Tell me about the most efficient team you were part of (Answered with a team that I got to lead, what I did different, and what was the outcome)

• Time when you had to manage a difficult stakeholder, how did you go about it, what was the outcome. Follow up - Any other way you could have handled it, and why not go that way.

Each round ended with Questions for the interviewer. Have some questions ready, to show interest (Could be generic, but the interviewer should be able to talk at length about it and make them feel good talking about it).

Kimberly Clark

Abhishek Hobalidar - Kimberly Clark IT

UG: B.E in Mechanical Engineering

Work ex & Company: L&T Technology Services, 46 months

ROUND-1

GD on Crypto currency future

Made two entries including conclusion

ROUND-2

Personal Interview

Q1.Tell me about yourself.

A: I told that i was sponsorship head of college official EV racing team

Q2. Relate term 1 subject to your learnings as sponsorship head

A: Told about Halo effect, Horns effect, MBTI personality, pygmalion effect, etc. I also ended by saying i have joined a premier B school

Q3. Since you are in a premier b school tell me how is it possible to have losses and positive cash flow. Tell me an example in Indian scenario

A: Told that if they are getting lots of investment and burning cash to sustain they are in losses. Told about Byjus as example

Q4. What happened with Byjus?

A: Told about interview with Byju Raveendran in business standard published on 14 September and they had published their numbers one year late.

Q5. Do you have any questions?

A: What will be my work since I don't have any experience?

Optum

Tanumoy Ghosh - Optum

UG: B.Tech in Electrical Engineering from IIEST Shibpur

Work ex & Company: 3 years at PwC India

ROUND-1: 50 MINS

Q1. Tell me about yourself

Q2. Why do you want to join Product management? What is there in the future of Product management and long term goals?

Q3. What do you know about the healthcare industry?

Q4. How do you link Product Management and the healthcare industry together?

Q5. CV based question on past experience - What have you done in the healthcare sector?

Q6. Pick up any device around you and tell me 3 points of improvement in that device? Prioritize the points you have mentioned for a customer segment for the first launch. Define metrics for the improvements mentioned. - *Picked health band and cross questioning about all the above points in depth with clarification in each stage*

(Justification in each answer)

ROUND-2: 45 MINS

Q1.Tell me about yourself?

Q2. Why Product Management? Why the healthcare industry? Why Optum?

Q3. Situation where you were not able to stand up to your potential

Q4. Situation where you persuaded your team to success despite getting too much pushback

Q5. In a business scenario with a product launch, one launch in 4 months and one just started

3 months back with a timeline of 14 months, which one would you prioritize?

Q5. Another business scenario of 1) 100mn market - product launch with 20% market share and 8% growth rate 2) 50mn market - product launch with 30% market share and 16% growth rate. Which one would you prioritize

Q6. What would be your primary and secondary sources for the initial data of a product demand survey?

Q7. Where do you see yourself in the next 15 years?

(Justification in each answer)

Shashank Lipate Optum

A major discussion in my interview was about work ex projects and experience. It was a stressful interview, he asked all aspects and discussed different approaches from PM POV.

ROUND 1

1. Work ex
2. PM roles and responsibilities
3. Guesstimate: Total water consumption in Nagpur
4. Design methodology
5. Most imp skill of PM
6. Experience of MBA life
7. Goals and life vision

ROUND 2

1. Smooth discussion about my aspirations

1. Why Optum
2. Knowledge of HealthCare industry
3. Any healthcare tech product you use, why?
4. Challenges in Healthcare and future scope
5. Market analysis concept and process
6. Top skill of PM
7. Why PM
8. Expectation from MBA and Optum

SUGGESTIONS

1. Focus on CV and relate your work with JD
2. Study about Optum values, vision, mission and some facts
3. Difference between healthcare industry of US and India

Google

Kaushal Chhowala - Google

UG: BE(CS) - 2018 from RGIT, Mumbai

Total prior Work Ex: 47 Months

Software Engineer @ Majesco(32 months)

Consultant @ Deloitte USI (15 months)

ROUND 1:

Questions asked:

1) There was no tell me about yourself question, Casual greetings then started with guesstimate

3) I- (Guesstimate) - Marginal expenditure for google when a new user signs up (Went for 10-15 mins)

4) Case: Gmail sees a spike in signups , because other services like AWS are down. Google is running out of storage and they don't want their new sign ups to fail. What google can do in next 24 hrs

- Lot of questions on every point I suggested

5) Technical questions:

Horizontal scaling

How do you create data duplication system, eg backups.

High level design questions

Db questions ACID

Last 5 mins

Any questions?

ROUND 2:

Started with introduction

I mentioned in my intro that I Emcee/ host wedding and corporate events. Owing to that the interviewer dwindled onto the following case:

1) Design a wedding planner app

45 minutes thorough discussion on the following parameters:

- Goals

- Personas,

- User needs,

- Features,

- Prioritizations,

- MOSCOW framework

- Revenue Streams

2) Any questions (last 5-6 minutes)

ROUND 3:

Initial 5-7mins there were some audio & network issues from his end

While he was setting this up, we had our introductions exchanged

Questions

1. Typical google PM execution question

   - Imagine you are a google PM and a google Vice President tells you to come up with a vacation planning app. Do whatever come to your mind and keep monetisation as a key aspect while developing your MVP, timeline given was of 6 months.

   (This discussion went for like good 20-25 minutes, we discussed all aspects of a PD - Goals, personas, Success Metric, North star etc)

2. Guesstimate the revenue that would be generated from this app?
3. How would you make money of this product?
4. GTM, which all inventory will u think of ?
5. How will external collab lookalike?
6. Explain how you would talk to internal stakeholders.

Last few mins q&a

Adobe

Rishabh Rastogi - Adobe

Questions:

1. Introduction

2. Explain the heads given in priority Matrix

3. Justify a few scores given

4. Explain the metrics and what insights can be drawn (cross questions)

5. GTM for my top 2 features

6. Any questions for me

Disney+Hotstar

Case: Increase retention and engagement of freemium customers visiting Hotstar to watch live cricket matches

BUDDY ROUND (CASE BASED)

~30 mins telephonic interview

1. Can you walk me through your approach?
2. Why these user personas?
3. Why did you select the particular user persona?
4. Do you know about the plans hotstar offers?
5. Do you think the selected personal will not buy or lowest plan?
6. Explain the success metrics stated
7. What is the meaning of threshold mentioned in success metrics
8. How did you come up with threshold
9. What will be the guard rail metric for our features
10. Any questions for me

ON-CAMPUS INTERVIEW:

~40 mins HR-CV-fav app based

1. TMSAY
2. Any situation where you had conflict with your manager/team mate
3. How did you resolve it
4. What do you think are the qualities that have helped you excel? (Strengths)

1. What are the areas which you are currently improving? (Working)
2. What was your role in forecasting tool that was developed?
3. How did you manage the client expectations? Any instances of conflict with client? How did you handle it?
4. Which is your favourite app?
5. What do you like the most about it?
6. What are some of the competitors?
7. Why don't you prefer competitor apps?
8. How would you improve the app?
9. Why PM?
10. Any questions for us?

Mihir Jadhav – Disney+Hotstar

ROUND-1

Case Submission- Increase engagement for freemium users (Deadline- 2 days)

ROUND-2

Telephonic Interview- Case discussion

1) My understanding of the problem statement

2) Asked me to explain the prioritized features in brief

3) Reasons for prioritization (components of RICE framework)

4) Metrics- Success metrics and Guard rail metrics

5) Why Disney+Hotstar?

6) Do you have any questions?

ROUND-3

Personal Interview- Case discussion + CV/HR

1) Introduce yourself

2) Why get back to product management when you were already a PM?

3) Why PM at Disney+Hotstar?

4) Questions on case submission focusing on feature description (Revenue perspective)

5) Reasons for de-prioritizing features (RICE Framework- focusing on the reach and impact)

5) One achievement from my work-ex

6) Biggest challenge you faced as a PM

7) Do you have any questions?

Vaishnavi Pendke - Disney+ Hotstar PM

Education Details – Electronics Engineering

Work experience – L&T Technology Services (27 months)

ROUND 1- BUDDY CALL

Questions asked-

1. Tell me about yourself
2. Questions related to case submission – User persona, Feature prioritization, success metrics, a trade-off for the suggested feature.
3. Implementation strategy for the suggested feature
4. Questions if any to the interviewer (would recommend keeping 1-2 questions ready)

ROUND 2- FINAL INTERVIEW

Questions asked-

1. Tell me about yourself
2. Tell me more about your start-up (mentioned this in the introduction), your role in it, and the current status of the start-up.
3. Asked about my favourite app. best features, improvement points, and how will you implement the suggested features.
4. Can you tell me about any situation where you resolved a conflict within your team?
5. Have you ever led a team? What challenges did you face? How did you overcome those challenges?
6. Any situation when you were short of resources and time and had to deliver the task on time. What did you do?
7. Questions if any to the interviewer

Amazon

Chandana Sreenivasa Murthy - Amazon Prodman

UG: B.Tech (CSE), NMIT '19

ROUND 1

The interviewer asked me to introduce myself.

After that he looked at my CV. There was a bucket called "Product Development" in my CV with 3 points. He asked me to choose any point that was close to my heart and asked me to explain what I did for that. All 3 of my points were different(2 were website development and 1 was a dashboard using Python and Django).

I chose the dashboard one and explained in brief what I did. One point I would like to mention here is that I had written "I led a team of 4 in building a dashboard", so I started with the background and went on to say my team did this. He told me to tell him what I did in the project, not the entire team. After that it was all questions based on that single point on how I saved man-hours and where did the time saved go to etc.

I would say go through your entire CV thoroughly. They will ask you regarding every word in the CV which is related to Prod. Especially points where numbers are mentioned(eg 600+ man hours, he asked me how I got 600. Asked me to do the calculations. And then I had written a reduced error rate by 5%, he asked me how I got that and what was the use of reducing it. So questions like this)

The interview went on for about 20 minutes

ROUND 2

Round 2 was more of a HR based interview. He asked me to introduce myself and asked a few questions on my CV. Then he asked some questions like "What do you think is the one constructive feedback you received and how did you react to it", "Tell me something you did

that was not defined in your job description", "Tell me one instance where you had a disagreement with your teammate" and a few more along the same lines.

This round went on for 30+ minutes

Amazon - Tiyasha Das

UG: B.TECH in Computer Science, IEM Kolkata

Work ex & Company: 2 yrs Cognizant, 6 months Vedic Maths Forum India

ROUND-1: 40 MINS CV & CASE

Q1. Tell me about yourself

Q2. One situation when you used Data extensively to solve a problem

Q3. Why do you think you are well-suited for Product management in Amazon?

Q4. RCA: Identify the reason for drop in orders by 50% in Swiggy on a particular day. (External and Internal situation analysed, User journey, Touch point where cancellations can happen)- Add to Cart page issue, recent update shifted the apply promo code option to Choose payment page.

Q5.What metric would you check if you were told at the beginning that the issue is with the 'Add to Cart Page'?

Q6. What data points will you check to identify the issue if you have access to all Swiggy's data points?

ROUND-2: 30 MINS CV BASED

Q1. Situation when you took initiative to solve a persistent problem. (Asked to explain and calculate the percentages mentioned in the CV)

Q2. A situation when you could not fulfil your commitment

Q3. If given a chance, would you handle the situation differently?(Follow up on previous question)

Anmol Jain

UG: B.Tech, IT, USICT

Summers: Amazon PM

ROUND 1

Introduction round and some basic flow of interview (like answer slowly so he could write)

Asked about a situation I persuaded my team to follow non-traditional approach.

Ans: Talked about my entrepreneurship/business expansion stint. It was related to e-commerce so was asked in-depth questions.

Q2. If you would launch a new product in food segment would you choose Amazon which is marketplace with variety of items or new apps like zepto who's fmcg and food focussed

Ans: Talked about the logistic help Amazon provides and the supply chain network and assistance.

Grilled over numbers as to how much could I save and if it was even beneficial for a new and small brand to go on Amazon. Number crunching over advertisement logistic salary etc given.

Ended with if you have any question.

ROUND 2:

The panellist looked bit distant as he was writing on another screen and making pointers throughout

Q1. A project you took apart from you responsibility

Ans: Talked about a new market analysis and product research picked at work.

Grilled on timeline and asked to split days the project went

Q2. Why did you take it and why were you the only one to take.

Ans: used Amazon values of learning new things and customer centric to explain

Q3. Why did your manager choose you?

Ans: talked about multi-task and how handled other projects at the same time without affecting performance.

Ended with any question?

HUL

Ramya Agaldiviti - HUL IT

UG: B.Tech (IT), G.Narayanamma Institute Of Technology & Science,, 2018

Workex: Adobe Systems India Pvt. Ltd,

Summers:

ROUND 1

Form submission - The form had normal details + HR questions

HR Questions:

1. Please share highlights of your academic achievements including any external certifications or scholarships that you would like us to note. Tell us about moments in your academic history that give you a sense of accomplishment
2. Let us know about any niche skills or advanced knowledge in a subject that would be relevant to this internship
3. Please share details about any formal work experience including organization name(s), your designation, responsibilities, and tenure. (Please do not include internships here)
4. Please share details about any internships including organization name(s), project details and duration.
5. Please share highlights about your professional achievements and in your assessment, what differentiated you amongst your peers.
6. What is the only value that trumps honesty for you and why?
7. Tell us about one strength of yours that distinguishes you.
8. Describe the most important piece of constructive feedback that you have received. What was it and how did you act on it?

9. How have you grown as a person in the last 2 or 3 years?

10. Please share anything else about your candidature, which is not yet represented, that you would like us to note.

ROUND 2- ONLINE VIDEO INTERVIEW

3 questions were asked in the video interview. 1 min to think and 3 mins to record.

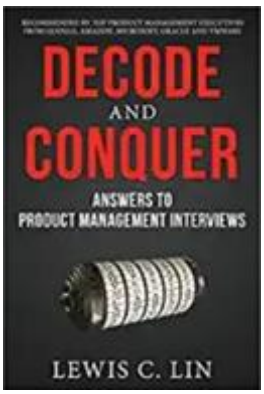
1. How has the adoption of Cloud impacted the approach towards application development. Give some examples
2. You are an IT lead for Logistics Function for a particular Unilever cluster (set of 7-8 countries) which spends 7% of their turnover as distribution costs. You are expected to put together technology interventions to help business function reduce costs by 20% and improve customer experience by improving servicing. Make assumptions and share how would you go about the transformation program.
3. With new Digital Capabilities, what are relevant ones for Unilever to consider to help reduce the cost of Technology to the organization.

ROUND 3 - INTERVIEW ROUND (TECHNICAL)

1. Asked to introduce myself and talk more on the work experience.
2. How can digital marketing help HUL gain customers? How can technology help HUL improve its SCM?
3. Difference between normal online purchases vs a kirana store owner ordering through Shikhar app? What can be improved in the Shikhar app? How can HUL use the data that it has to capture demand and fulfil it?
4. What are the emerging competitors of HUL? Are Mama-Earth and mCaffeine competition for HUL?
5. Question: Do you have any questions for us?

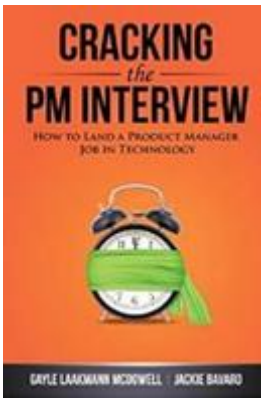
PM Resources

Books for reference



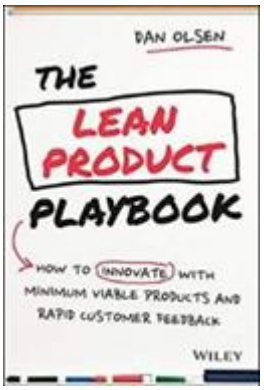
Decode & Conquer *by Lewis C. Lin*

Decode and Conquer is known for its easy-to-use frameworks — including CIRCLES, AARM, and DIGS — that help candidates tackle an extensive range of case interview scenarios. It is the first book every PM enthusiast must read to grasp the types of questions asked in the PM interview. The book also offers a dialogue- style format for answering the example questions. It is worded simply and easy to understand even if you are a complete novice in the field.



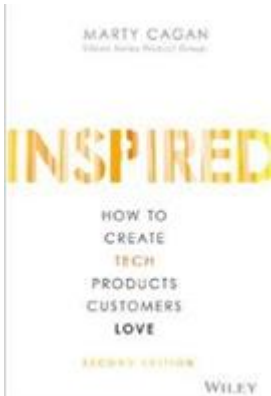
Cracking the PM Interview *by Gayle Laakmann McDowell*

Learn how the ambiguously named "PM" (product manager / program manager) role varies between companies, what experiences you need, how to make your existing experience translate, how to master the PM interview questions (estimation questions, behavioral questions, estimation questions, product questions, technical questions, and the super-important "pitch")



The Lean Product Playbook *by Dan Olsen*

The Lean Product Playbook walks you through the steps required to achieve product- market fit- determine your target customers, identify underserved customer needs, create a winning product strategy, define your minimum viable product (MVP)



Inspired *by Marty Cagan*

The book covers how product management fits in with other functions- engineering, interaction design, visual design, project management, product marketing. The difference between product management and product marketing is often misunderstood but Inspired explains it simply.